

Functional genetic report

Genetic Health Report

Sérgio F · Report date: 02-10-2026

Categories: **NUTRITION · HEART & BLOOD VESSELS · INFLAMMATION & AUTOIMMUNITY · BODY FAT & APPETITE · BLOOD SUGAR CONTROL · FITNESS**

This report repeats, with DNA of Sérgio F, the study of an example health report (72 traits in 4 sections, the same rsIDs and labels, equivalent methods). It is informative and educational, based only on genes. It is neither a diagnosis nor a prescription. All recommendations are marked as suggestion (not applied) and crossed with the v2 genetic plan.

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Personal information

Name

Sérgio F

This report does not diagnose any condition. If a condition exists in the family, or if there are doubts about the results, they should be discussed with a health professional.

Summary

72 genetic results	50 recommendations of the example (crossing with the plan)	931 rsIDs of the example report	585 in the chip	241 imputed (DR2≥0.8)	105 SNP not found in the data
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Of the unique 931 rsIDs of the example report (1362 lines, because some SNPs enter several traits), 585 come directly from the genotyping chip and 241 from the local imputation with quality DR2≥0.8. 105 were not found: 69 were imputed with low quality (DR2<0.8, treated as not found) and the remaining are indels, rare variants or do not exist in the data. The 11 polygenic scores (PGS Catalog) used 8 805 111 variants in total.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of the chip, 593 directly agree with the alleles of the dbSNP, 0 needed conversion to the complementary strand, 23 are palindromic (A/T or C/G; orientation confirmed by the position and by the reference allele) and 1 (rs688, cast rsID) was solved by the position. The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from the example report when the example uses the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Overall result: **17** favourable traits, unfavourable **10** and the others typical or without good / bad judgment. 44 of the 72 labels coincide with those of the client of the example report (different person).

Main results

Trait	Result	Interpretation
LDL Cholesterol	LOWER LEVELS	Cholesterol LDL: lower levels
HDL Cholesterol	LOWER LEVELS	Cholesterol HDL: lower levels
CETP (Cholesterol/ Longevity)	HIGHER ACTIVITY	CETP (cholesterol/longevity): higher activity
APOC3 (Blood Lipids/ Longevity)	LOWER ACTIVITY	APOC3 (lipids/longevity): lower activity
TCF7L2 (Blood Sugar/ Diet)	WORSE GENETICS	TCF7L2 (glycemia/diet): less favourable genetics; rs7903146 CT, rs12255372 GT: 2 allele(s) of risk
Type 2 Diabetes	TYPICAL LIKELIHOOD	2 type diabetes: typical probability
MTHFR	LOWER ACTIVITY	MTHFR: lower activity; C677T rs1801133 GA, A1298C rs1801131 TG — compound heterozygous, ~50–60% activity
Methylation	HIGHER ABILITY	Methylation: higher capacity
Iron Overload	TYPICAL LIKELIHOOD	Iron overload: typical probability; C282Y rs1800562 GG, H63D rs1799945 CG — H63D heterozygous; target ferritin <100
Vitamin D	DECREASED NEED	Vitamin D: decreased need
Calcium	INCREASED NEED	Calcium: increased need
Inflammation (CRP)	TYPICAL LEVELS	Inflammation (CRP): typical levels
IL1B (Inflammation/ Fatigue)	TYPICAL ACTIVITY	IL1B (inflammation/fatigue): typical activity
ADRB2 (Body Composition)	WORSE GENETICS	ADRB2 (body composition): less favourable genetics; Gln27Glu rs1042714 GC
UCP1 (Body Composition)	LOWER ACTIVITY	UCP1 (thermogenesis): lower activity -3826 rs1800592 TC
FTO (Body Composition)	TYPICAL	FTO (body composition): typical; rs9939609 TA

MC4R (Appetite / Blood Sugar)	HIGHER ACTIVITY	MC4R (appetite/glycemia): higher activity; rs17782313 TT, rs12970134 GG: 0 allele(s) of risk
Endurance	HIGHER	Aerobic resistance: higher

General view of the results

Colors: **unfavourable** **favourable** **typical or without judgment good/bad** **not reproducible**. The labels in English keep the categories of the example report.

Processes

Methylation

HIGHER ABILITY

Methylation: higher capacity

TCF7L2 (Blood Sugar/Diet)

WORSE GENETICS

TCF7L2 (glycemia/diet): less favourable genetics; rs7903146 CT, rs12255372 GT: 2 allele(s) of risk

Triglycerides

TYPICAL LEVELS

Triglycerides: typical levels

High Cholesterol

TYPICAL LIKELIHOOD

High cholesterol: typical probability

LDL Cholesterol

LOWER LEVELS

Cholesterol LDL: lower levels

HDL Cholesterol

LOWER LEVELS

Cholesterol HDL: lower levels

Lipid Oxidation

HIGHER

Lipid oxidation: higher

Oxidative Stress

LOWER

Oxidative stress: lower

Detox

HIGHER ABILITY

Detoxification: higher capacity

Phase I Detox

TYPICAL ABILITY

Phase I detoxification: typical capacity

Phase II Detox

TYPICAL ABILITY

II phase detoxification: typical capacity

Inflammation (CRP)

TYPICAL LEVELS

Inflammation (CRP): typical levels

Bone Health (Functional)

TYPICAL

Bone health (functional): typical

Type 2 Diabetes

TYPICAL LIKELIHOOD

2 type diabetes: typical probability

APOC3 (Blood Lipids/ Longevity)

LOWER ACTIVITY

APOC3 (lipids/longevity): lower activity

IL6 Gene (Inflammation)

TYPICAL ACTIVITY

IL6 gene (inflammation): typical activity; rs1800795 (-174) GG

TNF Gene (Inflammation)

LOWER ACTIVITY

TNF gene (inflammation): lower activity

CETP (Cholesterol/ Longevity)

HIGHER ACTIVITY

CETP (cholesterol/longevity): higher activity

IL1B (Inflammation/ Fatigue)

TYPICAL ACTIVITY

IL1B (inflammation/fatigue): typical activity

Nutrition

MTHFR**LOWER ACTIVITY**

MTHFR: lower activity; C677T rs1801133 GA, A1298C rs1801131 TG — compound heterozygous, ~50–60% activity

MUFA Response**TYPICAL**

Response to monounsaturated fat: typical

Protein Metabolism**WORSE RESPONSE**

Protein metabolism: worse response

Riboflavin (Vitamin B2)**TYPICAL NEED**

Riboflavin (vitamin B2): typical need; C677T rs1801133 GA, A1298C rs1801131 TG; without 677TT

Vitamin B6 (Pyridoxine)**TYPICAL NEED**

Vitamin B6 (pyridoxine): a typical need

Folate**DECREASED NEED**

Folate (functional): decreased need

Alcohol Sensitivity**TYPICAL**

Alcohol sensitivity: typical; ADH1B rs1229984 CC, ALDH2 rs671 GG

Calcium**INCREASED NEED**

Calcium: increased need

FUT2 (Gut/vitamin B12)**SECRETOR**

FUT2 (dentestine/vitamin B12): secretor; rs601338 GA: secretor

Carbohydrate Metabolism**TYPICAL**

Carbon hydrate metabolism: typical

Saturated Fat**TYPICAL RESPONSE**

Saturated fat: typical response

Unsaturated Fat**TYPICAL RESPONSE**

Unsaturated fat: typical response

Vitamin A**TYPICAL NEED**

Vitamin A: typical need

Vitamin B12**TYPICAL NEED**

Vitamin B12: typical need

Vitamin C**TYPICAL NEED**

Vitamin C: typical need

Vitamin D**DECREASED NEED**

Vitamin D: decreased need

Choline**DECREASED NEED**

Hill: decreased need

Iron Overload**TYPICAL LIKELIHOOD**

Iron overload: typical probability; C282Y rs1800562 GG, H63D rs1799945 CG — H63D heterozygous; target ferritin <100

HLA-DQ (Gluten)**TYPICAL GENETICS**

HLA-DQ (gluten): typical genetics; rs2187668 CC (DQ2.5), rs7454108 TT (DQ8)

Coffee-Related Sleep Problems**LESS LIKELY**

Sleep problems related to caffeine: less probable

Salt Sensitivity**TYPICAL**

Salt sensitivity: typical

MTRR (Methylation)**TYPICAL ACTIVITY**

MTRR (methylation): typical activity; A66G rs1801394 AA

VDR (Vitamin D)**TYPICAL ACTIVITY**

VDR (vitamin D): typical activity

PUFA Response**TYPICAL**

Response to polyunsaturated fat: typical

Fat Metabolism**TYPICAL**

Fat metabolism: typical

Lactose Intolerance**LIKELY TOLERANT**

Lactose intolerance: probably tolerant; -13910 rs4988235 GA

Caffeine Sensitivity**LOWER**

Caffeine sensitivity: lower

Caffeine-Related Anxiety

LESS LIKELY

Caffeine-related anxiety: less likely;
ADORA2A rs5751876 CC

CYP1A2 (Detox)

TYPICAL ACTIVITY

CYP1A2 (detoxification): typical
activity; rs762551 CA

GC (Vitamin D)

TYPICAL ACTIVITY

GC (vitamin D): typical activity

Body Composition

Bitter Taste Sensitivity

TASTER

Sensitivity to bitterness: taster;
TAS2R38 rs713598 CG, rs1726866
GA, rs10246939 TC — PAV/AVI

Sleeping

LESS LIKELY

Snacking: less probable

ADRB2 (Body Composition)

WORSE GENETICS

ADRB2 (body composition): less
favourable genetics; Gln27Glu
rs1042714 GC

UCP1 (Body Composition)

LOWER ACTIVITY

UCP1 (thermogenesis): lower activity
-3826 rs1800592 TC

PPARG (Metabolism)

TYPICAL ACTIVITY

PPARG (metabolism): typical activity;
Pro12Ala rs1801282 CC

FABP2 (Blood Sugar/ Cardiovascular)

HIGHER ACTIVITY

FABP2 (glycemia/cardiovascular):
higher activity Ala54Thr rs1799883 TC

UCP2 (Body Composition)

TYPICAL ACTIVITY

UCP2 (energy metabolism): typical
activity; -866 rs659366 CT

UCP3 (Body Composition)

TYPICAL ACTIVITY

UCP3 (energy metabolism): typical
activity; -55 rs18084 GG

Tendency to Overeat

TYPICAL LIKELIHOOD

Trend to eat in excess: typical
probability

SLC2A2 (Sugar Intake)

TYPICAL

SLC2A2 (sugar consumption): typical;
Thr110Ile rs54400 GG

CLOCK (Sleep & Metabolism)

TYPICAL ACTIVITY

CLOCK (sleep and metabolism):
typical activity; 3111 rs1801260 AG

Excess Body Fat

TYPICAL LIKELIHOOD

Predisposition for excess body fat:
typical probability

MC4R (Appetite / Blood Sugar)

HIGHER ACTIVITY

MC4R (appetite/glycemia): higher
activity; rs17782313 TT, rs12970134
GG: 0 allele(s) of risk

Sugar Cravings

TYPICAL LIKELIHOOD

Sweetness: typical probability

FTO (Body Composition)

TYPICAL

FTO (body composition): typical;
rs9939609 TA

Exercise and Fitness

Power

TYPICAL

Power: typical

ACTN3 (Power)

TYPICAL ACTIVITY

ACTN3 (power): typical activity;
R577X rs1815739 TC

Response Exercise (Metabolic)

TYPICAL

Metabolic response to exercise:
typical

Muscle Cramps

LESS LIKELY

Muscle cramps: less probable

Injuries

LESS LIKELY

Functional lesions: less probable

COL5A1 (Collagen)

HIGHER ACTIVITY

COL5A1 (collagen): higher activity;
rs12722 TT

Exercise Recovery

TYPICAL

Recovery after exercise: typical

Endurance

HIGHER

Aerobic resistance: higher

Results by category: favourable and unfavourable

The 72 traits grouped in the 6 categories of the example report. In each category, the results are separated into favourable and unfavourable; the typical and the activity of genes (without judgment good / bad) come next.

NUTRITION (Nutrition)

Favourable

- **HIGHER ABILITY** **Methylation** Methylation: higher capacity
- **DECREASED NEED** **Folate**: Folate (functional): decreased need
- **DECREASED NEED** **Vitamin D** Vitamin D: decreased need
- **DECREASED NEED** **Choline**: Choline: decreased need
- **LESS LIKELY** **Coffee-Related Sleep Problems**: Sleep problems related to caffeine: less probable
- **LIKELY TOLERANT** **Lactose Intolerance**: Lactose intolerance: probably tolerant; -13910 rs4988235 GA
- **LESS LIKELY** **Caffeine-Related Anxiety**: Caffeine-related anxiety: less likely; ADORA2A rs5751876 CC

Unfavourable

- **LOWER ACTIVITY** **MTHFR**: MTHFR: lower activity; C677T rs1801133 GA, A1298C rs1801131 TG — compound heterozygous, ~50–60% activity
- **INCREASED NEED** **Calcium**: Calcium: increased need

Typical / non-judgmentable

- **TYPICAL NEED** **Riboflavin (Vitamin B2)**: Riboflavin (vitamin B2): typical need; C677T rs1801133 GA, A1298C rs1801131 TG; without 677TT
- **TYPICAL NEED** **Vitamin B6 (Pyridoxine)**: Vitamin B6 (pyridoxine): a typical need
- **TYPICAL** **Alcohol Sensitivity**: Alcohol sensitivity: typical; ADH1B rs1229984 CC, ALDH2 rs671 GG
- **SECRETOR** **FUT2 (Gut/vitamin B12)**: FUT2 (dentestatin/vitamin B12): secretor; rs601338 GA: secretor
- **TYPICAL NEED** **Vitamin A** Vitamin A: a typical need
- **TYPICAL NEED** **Vitamin B12**: Vitamin B12: typical need
- **TYPICAL NEED** **Vitamin C**: Vitamin C: typical need
- **TYPICAL LIKELIHOOD** **Iron Overload**: Iron overload: typical probability; C282Y rs1800562 GG, H63D rs1799945 CG — H63D heterozygous; target ferritin <100
- **TYPICAL ACTIVITY** **MTRR (Methylation)**: MTRR (methylation): typical activity; A66G rs1801394 AA
- **TYPICAL ACTIVITY** **VDR (Vitamin D)**: VDR (vitamin D): typical activity
- **LOWER** **Caffeine Sensitivity**: Caffeine sensitivity: lower
- **TYPICAL ACTIVITY** **CYP1A2 (Detox)**: CYP1A2 (detoxification): typical activity; rs762551 CA
- **TYPICAL ACTIVITY** **GC (Vitamin D)**: GC (vitamin D): typical activity

HEART & BLOOD VESSELS (heart and blood vessels)

Favourable

- **LOWER LEVELS** **LDL Cholesterol:** Cholesterol LDL: lower levels
- **LOWER ACTIVITY** **APOC3 (Blood Lipids/Longevity):** APOC3 (lipids/longevity): lower activity

Unfavourable

- **LOWER LEVELS** **HDL Cholesterol:** Cholesterol HDL: lower levels
- **HIGHER** **Lipid Oxidation** Lipid oxidation: higher
- **HIGHER ACTIVITY** **CETP (Cholesterol/Longevity):** CETP (cholesterol/longevity): higher activity

Typical / non-judgmentable

- **TYPICAL LEVELS** **Triglycerides:** Triglycerides: typical levels
- **TYPICAL LIKELIHOOD** **High Cholesterol** High cholesterol: typical probability
- **TYPICAL** **MUFA Response:** Response to monounsaturated fat: typical
- **TYPICAL RESPONSE** **Saturated Fat** Saturated fat: typical response
- **TYPICAL RESPONSE** **Unsaturated Fat:** Unsaturated fat: typical response
- **TYPICAL** **Salt Sensitivity:** Salt sensitivity: typical
- **TYPICAL** **PUFA Response:** Response to polyunsaturated fat: typical

INFLAMMATION & AUTOIMMUNITY (Inflammation and autoimmunity)

Favourable

- **LOWER** **Oxidative Stress:** Oxidative stress: lower
- **HIGHER ABILITY** **Detox:** Detoxification: higher capacity
- **LOWER ACTIVITY** **TNF Gene (Inflammation):** TNF gene (inflammation): lower activity
- **TYPICAL GENETICS** **HLA-DQ (Gluten):** HLA-DQ (gluten): typical genetics; rs2187668 CC (DQ2.5), rs7454108 TT (DQ8)

Unfavourable

None.

Typical / non-judgmentable

- **TYPICAL ABILITY** **Phase I Detox:** Phase I detoxification: typical capacity
- **TYPICAL ABILITY** **Phase II Detox:** II phase detoxification: typical capacity
- **TYPICAL LEVELS** **Inflammation (CRP):** Inflammation (CRP): typical levels
- **TYPICAL ACTIVITY** **IL6 Gene (Inflammation):** IL6 gene (inflammation): typical activity; rs1800795 (-174) GG
- **TYPICAL ACTIVITY** **IL1B (Inflammation/ Fatigue):** IL1B (inflammation/fatigue): typical activity

BODY FAT & APPETITE (Body fat and appetite)

Favourable

- **LESS LIKELY** **Sleeping:** Snacking: less probable

Unfavourable

- **WORSE RESPONSE** **Protein Metabolism:** Protein metabolism: worse response
- **WORSE GENETICS** **ADRB2 (Body Composition):** ADRB2 (body composition): less favourable genetics; Gln27Glu rs1042714 GC
- **LOWER ACTIVITY** **UCP1 (Body Composition):** UCP1 (thermogenesis): lower activity -3826 rs1800592 TC

Typical / non-judgmentable

- **TYPICAL** **Fat Metabolism** Metabolism of fats: typical
- **TASTER** **Bitter Taste Sensitivity:** Sensitivity to bitterness: taster; TAS2R38 rs713598 CG, rs1726866 GA, rs10246939 TC — PAV/AVI
- **TYPICAL ACTIVITY** **UCP2 (Body Composition):** UCP2 (energy metabolism): typical activity; -866 rs659366 CT
- **TYPICAL ACTIVITY** **UCP3 (Body Composition):** UCP3 (energy metabolism): typical activity; -55 rs18084 GG
- **TYPICAL LIKELIHOOD** **Tendency to Overeat:** Trend to eat in excess: typical probability
- **TYPICAL ACTIVITY** **CLOCK (Sleep & Metabolism):** CLOCK (sleep and metabolism): typical activity; 3111 rs1801260 AG
- **TYPICAL LIKELIHOOD** **Excess Body Fat:** Predisposition for excess body fat: typical probability
- **TYPICAL** **FTO (Body Composition):** FTO (body composition): typical; rs9939609 TA

BLOOD SUGAR CONTROL (Glycemia control)

Favourable

- **HIGHER ACTIVITY** **MC4R (Appetite / Blood Sugar):** MC4R (appetite/glycemia): higher activity; rs17782313 TT, rs12970134 GG: 0 allele(s) of risk

Unfavourable

- **WORSE GENETICS** **TCF7L2 (Blood Sugar/ Diet):** TCF7L2 (glycemia/diet): less favourable genetics; rs7903146 CT, rs12255372 GT: 2 allele(s) of risk
- **HIGHER ACTIVITY** **FABP2 (Blood Sugar/ Cardiovascular):** FABP2 (glycemia/ cardiovascular): higher activity Ala54Thr rs1799883 TC

Typical / non-judgmentable

- **TYPICAL LIKELIHOOD** **Type 2 Diabetes:** 2 type diabetes: typical probability
- **TYPICAL** **Carbohydrate Metabolism:** Carbon hydrate metabolism: typical
- **TYPICAL ACTIVITY** **PPARG (Metabolism):** PPARG (metabolism): typical activity; Pro12Ala rs1801282 CC
- **TYPICAL** **SLC2A2 (Sugar Intake):** SLC2A2 (sugar consumption): typical; Thr110Ile rs5400 GG
- **TYPICAL LIKELIHOOD** **Sugar Cravings** Desire for sweets: typical probability

FITNESS (Exercise and physical condition)

Favourable

- **LESS LIKELY Injuries:** Lesions (functional): less probable
- **HIGHER Endurance:** Aerobic resistance: higher

Unfavourable

None.

Typical / non-judgmentable

- **TYPICAL Bone Health (Functional):** Bone health (functional): typical
- **TYPICAL Power:** Power: typical
- **TYPICAL ACTIVITY ACTN3 (Power):** ACTN3 (power): typical activity; R577X rs1815739 TC
- **TYPICAL Response Exercise (Metabolic):** Metabolic response to exercise: typical
- **LESS LIKELY Muscle Cramps:** Muscle cramps: less probable
- **HIGHER ACTIVITY COL5A1 (Collagen):** COL5A1 (collagen): higher activity; rs12722 TT
- **TYPICAL Exercise Recovery**Recovery after exercise: typical

Mechanisms of ageing

Each trait was linked to the aging mechanisms of the target list. The table shows, by mechanism, the unfavourable traits (where genetics demands more attention) and favourable traits.

Mechanism	Unfavourable	Favourable	Typical
Vascular Aging	HDL Cholesterol, Lipid Oxidation, CETP (Cholesterol/ Longevity), MTHFR	Methylation, LDL Cholesterol, APOC3 (Blood Lipids/ Longevity)	Triglycerides, High Cholesterol, MUFA Response, Saturated Fat, Unsaturated Fat, Salt Sensitivity, PUFA Response
Thermogenesis decline	ADRB2 (Body Composition), UCP1 (Body Composition)	—	FTO (Body Composition)
Insulin resistance	TCF7L2 (Blood Sugar/Diet), FABP2 (Blood Sugar/ Cardiovascular)	—	Type 2 Diabetes, Carbohydrate Metabolism, PPARG (Metabolism), Exercise Response (Metabolic)
Epigenetic Changes	MTHFR	Methylation, Folate (Functional), Choline	Riboflavin (Vitamin B2), Vitamin B12, MTRR (Methylation)
Hyperglycemia and Intolerance to Glucose	TCF7L2 (Blood Sugar/Diet)	—	Type 2 Diabetes, Carbohydrate Metabolism, SLC2A2 (Sugar Intake), Sugar Cravings
Osteoporosis	Calcium	Vitamin D	Bone Health (Functional), VDR (Vitamin D), GC (Vitamin D)
Thin Mass Loss	Protein Metabolism	—	—
Sarcopenia	Protein Metabolism	—	Power, ACTN3 (Power), Exercise Recovery
Oxidative stress	Lipid Oxidation	Oxidative Stress, Detox	Vitamin C, Iron Overload
Accumulation of Advanced Glycation Final Products (AGEs)	—	—	Type 2 Diabetes
Autophagy / Deregulated Macroautophagy	—	—	Iron Overload
DNA Damage and DNA Repair Deficit	—	Detox, Folate (Functional)	Phase I Detox, Phase II Detox, Alcohol Sensitivity
Oxidative Phosphorylation Decline	—	Endurance	Riboflavin (Vitamin B2)
Metabolic Homeostasis Decline	—	APOC3 (Blood Lipids/ Longevity)	Triglycerides, Fat Metabolism, Excess Body Fat, Exercise Response (Metabolic)
Epigenetic Derivative	—	Methylation	—
Redox imbalance	—	Oxidative Stress	Phase II Detox
Desynchronization of Biological Clocks	—	—	CLOCK (Sleep & Metabolism)
Dysbiosis	—	Lactose Intolerance	FUT2 (Gut/ Vitamin B12), Bitter Taste Sensitivity
Mitochondrial dysfunction	—	Oxidative Stress, Endurance	Vitamin B12, UCP2 (Body Composition), UCP3 (Body Composition)
Intestinal Barrier dysfunction	—	HLA-DQ (Gluten)	FUT2 (Gut/vitamin B12)
Neuromuscular Junction Disorder	—	—	Muscle Cramps
Dysfunction of adipose tissue	—	Choline, Snacking, MC4R (Appetite / Blood Sugar)	MUFA Response, Saturated Fat, Fat Metabolism, PPARG (Metabolism), Tendency To Overeat, Excess Body Fat, FTO (Body Composition)
Fragility	—	—	Bone Health (Functional)
Immunosenescence	—	Vitamin D	Vitamin A, VDR (Vitamin D)
Chronic Inflammation	—	TNF Gene (Inflammation), HLA-DQ (Gluten)	Inflammation (CRP), IL6 Gene (Inflammation)
Inflammaging	—		

		LDL Cholesterol, TNF Gene (Inflammation)	Inflammation (CRP), IL6 Gene (Inflammation), IL1B (Inflammation/ Fatigue), Unsaturated Fat, PUFA Response, Exercise Recovery
Neuroinflammation	—	—	IL1B (Inflammation/ Fatigue), Vitamin B6 (Pyridoxine)
Loss of muscle strength	—	—	Power, ACTN3 (Power)
Sleep disturbance	—	Coffee-Related Sleep Problems, Caffeine-Related Anxiety	Caffeine Sensitivity, CYP1A2 (Detox)
Circadian disorders	—	Caffeine-Related Sleep Problems, Snogging	CLOCK (Sleep & Metabolism)
Remodelling of the extracellular matrix	—	Injuries	COL5A1 (Collagen)
Arterial stiffness	—	LDL Cholesterol	High Cholesterol, Salt Sensitivity
Stiffness of the Extracellular Matrix	—	Injuries	COL5A1 (Collagen)

Overview of the recommendations

The 50 recommendations of the example report were reordered for Sérgio F: the risks of DNA with unfavourable result weigh more, and the crossing with the current stack and the genetic plan v2 gives each item a state. Items in conflict with the stack descend. **All: suggestion (not applied).**

#	Recommendation	Dose (example)	Risks of targeted DNA	Priority	State (v2 plan)
1	Methylfolate Methylate	400 mcg	MTHFR ↑	29.0	in the stack/plan v2
2	Strength training Strength Training	1 hour	ADRB2 (Body Composition) ↑	29.0	in the stack/plan v2
3	Maintaining optimal levels of vitamin D Maintenance Optimal Vitamin D Levels	1000 iu	Calcium ↑	24.0	in the stack/plan v2
4	Mediterranean Diet Mediterranean Diet	—	TCF7L2 (Blood Sugar/Diet) ↑	20.0	in the stack/plan v2
5	Green leaf vegetables Leafy Green Vegetables	—	Methylation, Calcium ↑, MTHFR ↑	12.9	in the stack/plan v2
6	Betaine (TMG) Betaine (TMG)	500 mg	Methylation, MTHFR ↑	11.8	in the stack/plan v2
7	Cinnamon Cinnamon	1g	TCF7L2 (Blood Sugar/Diet) ↑	10.0	caution (interaction or context)
8	Coffee Caffeine	100 mg	ADRB2 (Body Composition) ↑	10.0	in the stack/plan v2
9	Zinc Zinc	15 mg	Methylation, MTHFR ↑	8.6	in the stack/plan v2
10	Glutamine Glutamine	5g	TCF7L2 (Blood Sugar/Diet) ↑	6.0	compatible with stack
11	Beak grain Chickpeas	—	Vitamin B6 (Pyridoxine)	4.7	in the stack/plan v2
12	Quinoa Quinoa	—	Riboflavin (Vitamin B2)	4.3	in the stack/plan v2
13	Fiber-rich foods Eat Fiber-Rich Foods	—	—	4.0	in the stack/plan v2
14	Berberine Berberine	1000 mg	TCF7L2 (Blood Sugar/Diet) ↑	4.0	not recommended in this context
15	Alpha-lipoic acid Alpha-Lipoic Acid	100 mg	—	4.0	in the stack/plan v2
16	Psyllium Psyllium	5g	—	4.0	in the stack/plan v2
17	Pine bark extract Pine bark extract	100 mg	—	4.0	in the stack/plan v2
18	Garlic Garlic	—	—	4.0	in the stack/plan v2
19	Limit saturated fat Limit Saturated Fat	—	—	4.0	in the stack/plan v2
20	Lactobacillus rhamnosus Lactobacillus Rhamnosus	10 Billion CFU	Alcohol Sensitivity	2.4	compatible with stack
21	Sleeping 7 or more hours Sleep for 7+ Hours	—	Methylation	2.3	compatible with stack
22	Yoga Yoga	30 minutes	Methylation	2.1	compatible with stack
23	Plant-based integral food diet Whole-Food Plant-Based Diet	—	—	2.0	compatible with stack
24		30 minutes	—	2.0	compatible with stack

High intensity interval training (HIIT)

High Intensity Interval Training (HIIT)

25	Glucomannan Glucomannan	3g	—	2.0	compatible with stack
26	Walk Walking	30 minutes	—	2.0	compatible with stack
27	Exercise snacks (1 minute) Practice Exercise Snacks	1 minutes	—	2.0	compatible with stack
28	Lactobacillus plantarum Lactobacillus Plantarum	10 Billion CFU	—	2.0	compatible with stack
29	Okra Okra	—	—	2.0	compatible with stack
30	Avoiding organochlorine pesticides Avoid Organochlorine Pesticide Exposure	—	—	2.0	compatible with stack
31	Avoid polycyclic aromatic hydrocarbons (PAH) Avoid PAHs Exposure	—	—	2.0	compatible with stack
32	Avoid PCB Avoid PCBs	—	—	2.0	compatible with stack
33	Vinegar apple cider Apple Cider Vinegar	—	—	2.0	compatible with stack
34	Drinking at least 8 water cups per day Drinking at Least 8 Glasses of Water a Day	—	—	2.0	compatible with stack
35	Avoid phthalates Avoid Phthalate Exposure	—	—	2.0	compatible with stack
36	Whole body vibration Whole-Body Vibration	15 minutes	—	2.0	compatible with stack
37	Mindfulness meditation Mindfulness Meditation	30 minutes	—	2.0	compatible with stack
38	Inositol Inositol	2g	—	2.0	compatible with stack
39	Myo-inositol Myo-inositol	4g	—	2.0	compatible with stack
40	Avoid PFAS Avoid PFAS Exposure	—	—	2.0	compatible with stack
41	Black cumin (Nigella sativa) Black Seeds (Black Cumin)	1000 mg	—	1.0	caution (interaction or context)
42	Green coffee extract Green Coffee Extract	400 mg	—	1.0	caution (interaction or context)
43	Seed extract of Irvingia gabonensis Irvingia gabonensis, seed extract	300 mg	—	1.0	caution (interaction or context)
44	Black cumin oil Black Seed Oil	2 tsp	—	1.0	caution (interaction or context)
45	Whiteberry White Mulberry	—	—	1.0	caution (interaction or context)
46	Banaba Banaba	32 mg	—	1.0	caution (interaction or context)
47	Jiaogulan (Gynostemma pentaphyllum) Jiaogulan (Gynostemma Pentaphyllum)	1350 mg	—	1.0	caution (interaction or context)
48	Pheno-Greek Phenugreek	500 mg	—	1.0	caution (interaction or context)
49	Ketogenic diet Keto Diet	—	—	0.0	not recommended in this context
50		—	—	0.0	

↑ : risk of DNA with unfavourable result in Sérgio F. Priority = $\Sigma(\text{impact} \times \text{result factor})$ + adjustment to the plan (already in the 4 plan, compatible 2, caution 1); items not recommended $\times 0.2$.

Relevant recommendations for the stack

Specific recommendations of this report: link genetic results to the current stack and v2 genetic plan. The current stack is taken as it is; nothing has been changed.

1. LDL and ApoB: as low as possible, keeping the current treatment

Genetics: LDL Cholesterol LOWER LEVELS (p24); High Cholesterol TYPICAL LIKELIHOOD (p39); HDL Cholesterol LOWER LEVELS (p18); CETP (Cholesterol/ Longevity) HIGHER ACTIVITY (p11) (rs708272 GG, rs5882 AA); APOC3 (Blood Lipids/ Longevity) LOWER ACTIVITY (p18); Lipid Oxidation HIGHER (p77)

A score of LDL below average does not change the target: for LDL and ApoB, the lower the better. The CETP of higher genetic activity and the low genetic HDL pull in the opposite direction and reinforce maintaining the hypolipemic therapy. The favourable APOC3 helps in triglycerides.

Plan v2: pitavastatin 4 mg, ezetimibe 10 mg and bempedoic acid 180 mg KEEP; inclisiran or anti-PCSK9: ADD (discuss); obicetrapib (CETP inhibitor): future candidate; saturated fat <7% of energy; extra virgin olive oil 40–50 mL/day. • **suggestion (not applied)**

2. More fiber and hydrates with low glycemic index

Genetics: TCF7L2 (Blood Sugar/Diet) WORSE GENETICS (rs7903146 CT, rs12255372 GT: 2 allele(s) of risk T); Type 2 Diabetes TYPICAL LIKELIHOOD (p59); Carbohydrate Metabolism TYPICAL (p61); FABP2 (Blood Sugar/ Cardiovascular) HIGHER ACTIVITY; SLC2A2 (Sugar Intake) TYPICAL

TCF7L2 is the main genetic signal of glycemia in this report. The viscous fiber (psyllium, PHGG, glucomannan, legumes) and low glycemic index hydrates are the measures with better evidence for this genotype; more fiber is always better.

Plan v2: fiber ≥50 g/day; psyllium 15–20 g + PHGG 10 g (ADJUST, up); hydrates 35–40% of low IG and fructose added ≈0; acarbose, imeglimin and dapagliflozin: KEEP; berberine: AVOID (CYP3A4/rapamycin). • **suggestion (not applied)**

3. Methylation cycle: maintain active folate and reinforce betaine and choline

Genetics: MTHFR LOWER ACTIVITY (C677T rs1801133 GA, A1298C rs1801131 TG — compound heterozygous, ~50–60% of activity); Methylation HIGHER ABILITY (p7); Folate (Functional) DECREASED NEED (p10); Choline DECREASED NEED (p5); MTRR (Methylation) TYPICAL ACTIVITY; Riboflavin (Vitamin B2) TYPICAL NEED; © 2011 Elsevier Ltd.

The compound heterozygous of MTHFR reduces the production of 5-metiltetra-hidrofolato. The counts of methylation, folate and choline are favourable because other common alleles (e.g. MTRR 66A/A and COMT Val/Val) are lacking, but do not cancel the effect of MTHFR. The active folate is already in the stack; betaine and choline provide alternative pathways for homocysteine methylation. The choline of the v2 plan rests on other genes (PEMT, PNPLA3, SERPINA1, FMO3) and remains as it is.

Plan v2: 5-MTHF 800 µg and B12 1000 µg: KEEP; TMG: rise to 3 g/day (maximum 3 g because of LDL); choline 250–500 mg/day: ADD; folic acid: AVOID; B2: 5–10 mg arrives. • **suggestion (not applied)**

4. Iron: without iron supplements; target ferritin below 100

Genetics: Iron Overload TYPICAL LIKELIHOOD (C282Y rs1800562 GG, H63D rs1799945 CG — H63D heterozygous; target ferritin <100)

An H63D allele alone rarely causes overload, but excess iron feeds oxidative stress. Target ferritin below 100.

Plan v2: iron and multivitamins with iron: AVOID; little heme iron and no fortified cereals; tea/coffee/dairy at meals; vitamin C between meals. • **suggestion (not applied)**

5. Vitamin D and calcium

Genetics: Vitamin D DECREASED NEED (p84); GC (Vitamin D) TYPICAL ACTIVITY (p27) (rs2282679 TT); VDR (Vitamin D) TYPICAL ACTIVITY (p62) (FokI rs2228570 AG); Calcium INCREASED NEED (p22); Bone Health (Functional) TYPICAL (p30)

The vitamin D PGS is favourable, but the current dose is decided by the protocol. Calcium PGS suggests increased need: calcium mainly from food (tolerable dairy products by persistent lactase, green leaves, legumes, sardine).

Plan v2: D3 10 000 IU: remains (coordination note); K2 MK-7 240 µg: KEEP; Ca-AKG 1 + 1 g at meals: KEEP; no additional calcium supplementation. • **suggestion (not applied)**

6. Omega-3: already in the stack

Genetics: PUFA Response TYPICAL (p52); FADS1 rs174546 CC; Unsaturated Fat TYPICAL RESPONSE (p68)

Fish oil is already part of the stack. Genetics does not require change.

Plan v2: fish oil: 3.2–4 g EPA+DHA/day (coordination note), with ≥ 1 –1.2 g of DHA, at meals; small fatty fish 3×/week. • suggestion (not applied)

7. Inflammation: maintaining what is, without reintroducing polyphenols whose pause is suggested

Genetics: Inflammation (CRP) TYPICAL LEVELS (p26); IL1B (Inflammation/ Fatigue) TYPICAL ACTIVITY (p55); IL6 Gene (Inflammation) TYPICAL ACTIVITY; TNF Gene (Inflammation) LOWER ACTIVITY (p20); Oxidative Stress LOWER (p5)

Inflammatory genetics are typical to favourable (TNF and oxidative stress on the favourable side). Pausing curcumin, quercetin, luteolin and apigenin is a suggestion (not applied; the real stack has not changed); this report is no reason to resume them. The pulse of fisetin + quercetin only when the current protocol allows it (coordination note), outside the rapamycin window.

Plan v2: Omega-3, fiber, sulforaphane, ginger and naltrexone in low dose: KEEP; curcumin, luteolin, apigenin: pause is a suggestion (not applied; the real stack has not changed). • suggestion (not applied)

8. Coffee only in the morning

Genetics: CYP1A2 (Detox) TYPICAL ACTIVITY (rs762551 CA); Caffeine-Related Anxiety LESS LIKELY; Caffeine-Related Sleep Problems LESS LIKELY (p13); Caffeine Sensitivity LOWER (p18)

Typical caffeine metabolism and little genetic anxiety with it; the limit protects sleep.

Plan v2: caffeine ≤ 200 mg/day, only until 12h, filtered coffee; filtered decaffeinated after. • suggestion (not applied)

9. Exercise: strength and intervals as a priority

Genetics: ADRB2 (Body Composition) WORSE GENETICS; UCP1 (Body Composition) LOWER ACTIVITY; FTO (Body Composition) TYPICAL; Exercise Response (Metabolic) TYPICAL (p57); Endurance HIGHER (p92); Power TYPICAL (p65); ACTN3 (Power) TYPICAL ACTIVITY; COL5A1 (Collagen) HIGHER ACTIVITY; Injuries (Functional) LESS LIKELY (p19); Exercise Recovery TYPICAL (p42)

ADRB2 Glu27 and UCP1 -3826G are associated with lower lipolysis and base thermogenesis; exercise compensates these effects. With COL5A1 TT (harder tendons), gradual load progression and heating.

Plan v2: strength training 2–3×/week + HIIT or short intervals; creatine 5 g, collagen 20 g and protein ~ 140 g/day: KEEP. • suggestion (not applied)

10. Detoxification: crucifers and less combustion compounds

Genetics: Detox HIGHER ABILITY (p21); Phase I Detox TYPICAL ABILITY (p49); Phase II Detox TYPICAL ABILITY (p56); NAT2 rs1801280 CC

Phase I and II genetics, together with slow NAT2 and probable null GSTM1 (non-visible chip) justify reducing heterocyclic amines and PAH.

Plan v2: crucifers ≥ 1 –2 portions/day; avoid processed, smoked and carbonized meat; sulforaphane, NAC and glycine: KEEP (glycine: up to 5 g). • suggestion (not applied)

11. Lactose and gluten without restriction; zero alcohol

Genetics: Lactose Intolerance LIKELY TOLERANT (rs4988235 GA); HLA-DQ (Gluten) TYPICAL GENETICS; Alcohol Sensitivity TYPICAL
Persistent lactase and without HLA-DQ2.5/DQ8 tags: no genetic reason to restrict lactose or gluten. Genetic sensitivity to alcohol is irrelevant: zero alcohol.

Plan v2: lactose and gluten: SEM RESTRICTION; alcohol: zero. • suggestion (not applied)

12. Salt and potassium by food

Genetics: Salt Sensitivity TYPICAL (p49)

The genetics of salt sensitivity weighs little before the diet.

Plan v2: salt < 5 g/day; potassium 3.5–4.7 g/day of food, without potassium chloride or tablets (telmisartan). • suggestion (not applied)

Results in detail: Processes

Methylation · Methylation

HIGHER ABILITY higher capacity

15 score of 16 variants (signal effect alleles). the 7th percentile (mean position) in the reference population (n=633); 6% of the reference population has exactly the same score.

The methylation cycle (folate, B12, choline/betaine) provides methyl groups to DNA, neurotransmitters and homocysteine elimination. Variants in MTHFR, MTRR, MTHFD1, PEMT, BHMT and COMT can reduce cycle efficiency.

Example report: LOWER ABILITY · Category: NUTRITION · Mechanisms: Epigenetic Changes; Epigenetic Derivative; Vascular Aging

Alleles that pull to «LOWER ABILITY» — in the opposite direction to the result (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTHFR	rs1801133	GA	chip	A (+)	1	MTHFR 677T: –35% activity by allele
BHMT	rs3733890	GA	chip	A (+)	1	BHMT R239Q
GNMT	rs9296404	TT	imputation DR2=1.00	T (+)	2	GWAS: Plasma homocysteine levels (post-methionine load test) (Williams SR; p=2e-63); unfavourable T allele
PEMT	rs7946	CT	chip	T (+)	1	PEMT V175M: less phosphatidylcholine
CHDH	rs9001	TT	chip	T (+)	2	CHDH (as in the example report)
TRDMT1	rs12780845	AG	chip	A (+)	1	GWAS: Homocysteine levels (van Meurs JB; p=8e-10); unfavourable A allele
CBS	rs2851391	TC	chip	T (+)	1	GWAS: Homocysteine levels (van Meurs JB; p=2e-12); unfavourable T allele
MTHFR	rs1801131	TG	chip	G (+)	1	MTHFR 1298C: slightly reduced activity

No copies of the effect allele (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SHMT1	rs1979277	GG	chip	A (+)	0	SHMT1 1420T
MTR	rs2275565	GG	chip	T (–)	0	GWAS: Homocysteine levels (van Meurs JB; p=2e-10); favourable T allele
COMT	rs4680	GG	chip	A (+)	0	COMT Met: less activity
MTRR	rs1801394	AA	chip	G (+)	0	MTRR 66G
MTHFD1	rs2236225	GG	chip	A (+)	0	MTHFD1 1958A: more need for choline/folate
CPS1	rs1047891	CC	chip	A (+)	0	GWAS: Plasma homocysteine levels (post-methionine load test) (Williams SR; p=9e-13); unfavourable A allele
MTRR	rs1532268	CC	chip	T (+)	0	MTRR S175L
PEMT	rs12325817	CC	chip	G (+)	0	PEMT -744: more choline need

Undefined effect allele (32)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTHFR	rs2066470	GG	chip	—	—	
JMY	rs3797546	TT	chip	—	—	
BHMT	rs651852	CT	chip	—	—	
CBS	rs234706	GA	chip	—	—	
MTR	rs1805087	AA	chip	—	—	
FOLH1	rs61886492	GA	imputation DR2=1.00	—	—	
MAT1A	rs3851059	GA	chip	—	—	
BHMT2	rs625879	AC	chip	—	—	
SLC19A1	rs1051266	TC	chip	—	—	

PEMT	rs4646343	GG	chip	—	—
PEMT	rs12936587	GG	chip	—	—
MAT1A	rs7087728	GG	chip	—	—
MS4A6A	rs558660	GG	chip	—	—
MAT1A	rs2993763	GA	imputation DR2=1.00	—	—
FOLR3	rs651933	GA	chip	—	—
MTRR	rs1802059	GG	chip	—	—
TCN1	rs526934	GA	chip	—	—
COMT	rs4633	CC	chip	—	—
BHMT	rs567754	CT	chip	—	—
MTHFD1L	rs17349743	TT	chip	—	—
MMAB	rs7134594	CT	chip	—	—
MAT1A	rs4934028	GA	imputation DR2=1.00	—	—
CHMP4B	rs819171	CT	chip	—	—
ITCH	rs819147	CT	chip	—	—
MTHFR	rs3737965	GG	imputation DR2=1.00	—	—
FOLH1	rs202676	AA	chip	—	—
TYMS	rs2853533	GC	imputation DR2=0.91	—	—
DHFR	rs1643649	TT	chip	—	—
GNMT	rs10948059	CC	chip	—	—
NQO1	rs1800566	GG	chip	—	—
OGG1	rs1052133	CC	chip	—	—
MTHFD1L	rs6922269	AA	chip	—	—

SNP not found in the data (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PDXK	rs147242481	—	SNP not found in the data (Imputed DR2=0.03, below the 0.8 threshold)	A (+)	—	GWAS: Homocysteine levels (Thareja G; p=2e-11); unfavourable A allele
AHCY	rs13043752	—	SNP not found in the data (Imputed DR2=0.75, below the 0.8 threshold)	—	—	

TCF7L2 (Blood Sugar/Diet) · TCF7L2 (blood glucose/diet)

WORSE GENETICS genetics less favourable

Genotype rule (as in the example report): rs7903146 CT, rs12255372 GT: 2 allele(s) of risk T.

TCF7L2 regulates insulin secretion and GLP-1. The T allele of rs7903146 is the strongest genetic signal for type 2 diabetes (~1.4× per allele) and is associated with a worse response to diets rich in refined hydrates.

Example report: WORSE GENETICS · Category: BLOOD SUGAR CONTROL · Mechanisms: Insulin resistance; Hyperglycemia and Intolerance to Glucose

With the effect allele — in the direction of the result (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TCF7L2	rs7903146	CT	chip	T (+)	1	2 diabetes risk allele
TCF7L2	rs12255372	GT	chip	T (+)	1	2 diabetes risk allele

Triglycerides · Triglycerides

TYPICAL LEVELS typical levels

PGS002719 (PGS Catalog): 959 485 of 981 472 variants with genotype (cover 98%). the 68th percentile in the reference population (n=633); 0% of the reference population has the same score.

Triglycerides depend heavily on the diet (sugar, alcohol, caloric excess), but about 30–50% of variation is genetic. Genomic polygenic score.

Example report: TYPICAL LEVELS (p16) · Category: HEART & BLOOD VESSELS · Mechanisms: Metabolic Homeostasis decline; Vascular Aging

Alleles that pull to «HIGHER LEVELS» (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ACACB	rs149793040	AA	chip	A (+)	2	GWAS: Triglyceride levels (Jee YH; p=6e-35); allele A increases the trait
LPL	rs17091905	GG	chip	G (+)	2	in PGS002719: G allele increases the score
APOB	rs533617	TT	chip	T (+)	2	in PGS002719: T allele increases the score
CLPTM1	rs483082	GT	imputation DR2=0.99	T (+)	1	GWAS: Triglyceride levels (Graham SE; p<1e-300); T allele increases the trait
NLRC5	rs1800775	CC	chip	C (+)	2	in PGS002719: C allele increases the score
FADS2	rs174546	CC	chip	C (+)	2	in PGS002719: C allele increases the score
BUD13	rs117794084	GT	chip	T (+)	1	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=8e-45); T allele increases the trait
APOE	rs141622900	GA	chip	A (+)	1	GWAS: Triglyceride levels (MTAG) (Kamiza AB; p=2e-249); allele A increases the trait

Alleles that pull to «LOWER LEVELS» (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
LPL	rs17489373	GA	imputation DR2=1.00	A (–)	1	GWAS: Triglyceride levels (Sinnott-Armstrong N; p<1e-300); allele A decreases the trait
SIDT2	rs964184	GC	chip	C (–)	1	GWAS: Triglyceride levels (Graham SE; p<1e-300); allele C decreases the trait
PCSK7	rs662799	AA	chip	A (–)	2	GWAS: Triglyceride levels (Graham SE; p<1e-300); allele A decreases the trait
LPL	rs268	AA	chip	A (–)	2	in PGS002719: allele A decreases the score
APOE	rs7254892	GG	chip	G (–)	2	in PGS002719: G allele decreases the score

No copies of the effect allele (26)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOC3	rs147210663	GG	chip	A (–)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=3e-22); allele A decreases the trait
CETP	rs5882	AA	chip	G (–)	0	in PGS002719: G allele decreases the score
ANGPTL4	rs116843064	GG	chip	A (–)	0	GWAS: Triglyceride levels (Graham SE; p<1e-300); allele A decreases the trait
APOA4	rs12721043	CC	imputation DR2=0.89	A (–)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=8e-83); allele A decreases the trait
PAFAH1B2	rs186808413	CC	chip	T (–)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=1e-80); T allele decreases the trait
LPL	rs75218485	CC	imputation DR2=0.89	T (–)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=9e-119); T allele decreases the trait
LPL	rs117604010	GG	imputation DR2=1.00	A (–)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=4e-70); allele A decreases the trait
MAU2	rs58542926	CC	chip	T (–)	0	GWAS: Triglyceride levels (Jee YH; p=9e-290); T allele decreases the trait
MLXIPL	rs113296769	AA	chip	G (–)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=5e-36); G allele decreases the trait
DNAJC30	rs13242693	CC	chip	T (–)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=9e-56); T allele decreases the trait
WSB2	rs7973260	GG	imputation DR2=0.81	A (–)	0	in PGS002719: allele A decreases the score

MLXIPL	rs71556711	CC	imputation DR2=0.93	T (−)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=1e-213); T allele decreases the trait
—	rs118204057	GG	chip	A (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=3e-36); allele A increases the trait
APOA5	rs3135506	GG	chip	C (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p<1e-300); C allele increases the trait
APOA4	rs12721041	CC	chip	T (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=2e-91); T allele increases the trait
BACE1	rs116987336	GG	imputation DR2=0.89	A (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=1e-140); allele A increases the trait
LPL	rs1801177	GG	chip	A (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=3e-82); allele A increases the trait
MAP1A	rs55707100	CC	chip	T (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=8e-82); T allele increases the trait
LPL	rs186868868	CC	imputation DR2=0.86	A (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=3e-46); allele A increases the trait
APOE	rs7412	CC	chip	T (+)	0	GWAS: Triglycerides to Total Lipids in Large HDL percentile (Zoodsma M; p<1e-300); T allele increases the trait
EIF3J	rs151291132	AA	chip	G (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=8e-47); G allele increases the trait
SOST	rs76868109	AA	chip	C (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=8e-67); C allele increases the trait
SLC30A3	rs116170113	GG	imputation DR2=0.81	A (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=5e-22); allele A increases the trait
BUD13	rs114594921	TT	chip	C (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=4e-28); C allele increases the trait
NECTIN2	rs138607350	TT	chip	G (+)	0	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=1e-11); G allele increases the trait
APOE	rs429358	TT	imputation DR2=0.99	C (−)	0	GWAS: Triglycerides to Total Lipids in Large LDL percentile (Zoodsma M; p=3e-259); C allele decreases the trait

Undefined effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PCSK7	rs5128	CC	chip	—	—	

SNP not found in the data (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
—	rs201079485	—	SNP not found in the data	A (+)	—	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=8e-39); allele A increases the trait
—	rs149808404	—	SNP not found in the data	A (+)	—	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=4e-13); allele A increases the trait
PLA2G12A	rs41278045	—	SNP not found in the data	A (−)	—	GWAS: Triglycerides (Klarin D; p=1e-12); allele A decreases the trait
A1CF	rs41274050	—	SNP not found in the data (Imputed DR2=0.62, below the 0.8 threshold)	T (+)	—	GWAS: Triglyceride levels (Jee YH; p=1e-32); T allele increases the trait
GTF3C2	rs149117895	—	SNP not found in the data (Imputed DR2=0.71, below the 0.8 threshold)	A (+)	—	GWAS: Triglyceride levels (Sinnott-Armstrong N; p=7e-13); allele A increases the trait

High Cholesterol High • Cholesterol

TYPICAL LIKELIHOOD typical probability

PGS002700 (PGS Catalog): 916 425 of 937 601 variants with genotype (cover 98%). the 39th percentile in the reference population (n=633); 0% of the reference population has the same score.

Genetic probability of high total cholesterol. About half of the variation in cholesterol is hereditary.

Example report: TYPICAL LIKELIHOOD (p46) · Category: HEART & BLOOD VESSELS · Mechanisms: Vascular Aging; Arterial stiffness

Alleles that pull to «MORE LIKELY» (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PCSK9	rs28362286	CC	chip	C (+)	2	GWAS: total cholesterol (mean, inv-norm transformed) (Verma A; p=1e-287); C allele increases the trait
APOE	rs7254892	GG	chip	G (+)	2	in PGS002700: G allele increases the score
APOE	rs4420638	AG	imputation DR2=1.00	G (+)	1	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p<1e-300); G allele increases the trait

Alleles that pull to «LESS LIKELY» (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOE	rs141622900	GA	chip	A (–)	1	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A decreases the trait

No copies of the effect allele (34)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOE	rs7412	CC	chip	T (–)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p<1e-300); T allele decreases the trait
PCSK9	rs11591147	GG	chip	T (–)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p<1e-300); T allele decreases the trait
CETP	rs708272	GG	chip	A (+)	0	GWAS: Total Cholesterol to Total Lipids percentile (Farashi S; p=1e-13); allele A increases the trait
SIDT2	rs651821	TT	chip	C (+)	0	in PGS002700: C allele increases the score
APOE	rs12721109	GG	imputation DR2=0.96	A (–)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A decreases the trait
APOE	rs62117160	GG	chip	A (–)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A decreases the trait
LDLR	rs72658867	GG	chip	A (–)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=9e-195); allele A decreases the trait
CETP	rs5882	AA	chip	G (–)	0	in PGS002700: G allele decreases the score
APOE	rs429358	TT	imputation DR2=0.99	C (+)	0	GWAS: Total cholesterol minus HDL-C levels (Zoodma M; p<1e-300); C allele increases the trait
PCSK7	rs662799	AA	chip	G (+)	0	GWAS: Total cholesterol levels (Jee YH; p=1e-55); G allele increases the trait
LIPG	rs77960347	AA	chip	G (+)	0	GWAS: Total cholesterol levels (Graham SE; p=2e-132); G allele increases the trait
—	rs12713559	GG	chip	A (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=1e-21); allele A increases the trait
SNX8	rs144787122	AA	chip	G (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=7e-19); G allele increases the trait
LIPG	rs117623631	CC	chip	T (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=3e-30); T allele increases the trait
APOC4	rs140526515	AA	imputation DR2=0.93	G (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=5e-27); G allele increases the trait
NECTIN2	rs138914864	CC	imputation DR2=0.97	T (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=6e-26); T allele increases the trait
NECTIN2	rs117310449	CC	chip	T (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=2e-98); T allele increases the trait
NECTIN2	rs144261139	CC	chip	A (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=4e-90); allele A increases the trait
NECTIN2	rs76366838	GG	chip	A (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=5e-183); allele A increases the trait
APOE	rs769449	GG	chip	A (+)	0	

						GWAS: Total cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A increases the trait
ABCG5	rs141828689	CC	chip	T (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=3e-10); T allele increases the trait
CLPTM1	rs12691088	GG	chip	A (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=7e-89); allele A increases the trait
NECTIN2	rs138607350	TT	chip	G (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=7e-37); G allele increases the trait
NECTIN2	rs34095326	GG	imputation DR2=0.96	A (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A increases the trait
TOMM40	rs394819	GG	chip	T (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=4e-10); T allele increases the trait
LIPG	rs80175721	AA	chip	G (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=5e-40); G allele increases the trait
SLC22A3	rs3918291	TT	imputation DR2=1.00	C (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=4e-30); C allele increases the trait
CLPTM1	rs79429216	GG	chip	A (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=5e-44); allele A increases the trait
MAFB	rs2207132	GG	chip	A (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=9e-97); allele A increases the trait
NECTIN2	rs41289512	CC	chip	G (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=8e-122); G allele increases the trait
LPA	rs3798220	TT	chip	C (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=2e-52); C allele increases the trait
APOC1	rs60049679	GG	imputation DR2=0.81	C (+)	0	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=4e-191); C allele increases the trait
ABCA6	rs77542162	AA	chip	G (+)	0	GWAS: Total cholesterol levels (Graham SE; p=2e-130); G allele increases the trait
MAFB	rs1883711	GG	chip	C (+)	0	GWAS: Total cholesterol levels (Graham SE; p=1e-147); allele C increases the trait

Undefined effect allele (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PCSK9	rs562556	AA	chip	—	—	
LPL	rs328	CC	chip	—	—	
PCSK7	rs5128	CC	chip	—	—	
MICB	rs361525	GA	chip	—	—	
IL6	rs1800795	GG	chip	—	—	
FABP2	rs1799883	TC	imputation DR2=1.00	—	—	

SNP not found in the data (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ABCA1	rs188308962	—	SNP not found in the data (Imputed DR2=0.04, below the 0.8 threshold)	G (—)	—	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=3e-17); G allele decreases the trait
—	rs151135411	—	SNP not found in the data	A (+)	—	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=3e-15); allele A increases the trait
—	rs150401285	—	SNP not found in the data (Imputed DR2=0.04, below the 0.8 threshold)	G (+)	—	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=2e-11); G allele increases the trait
PVR	rs139267469	—	SNP not found in the data (Imputed DR2=0.03, below the 0.8 threshold)	A (+)	—	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=2e-10); allele A increases the trait
BCL3	rs114036675	—	SNP not found in the data (Imputed DR2=0.02, below the 0.8 threshold)	A (+)	—	GWAS: Total cholesterol levels (Sinnott-Armstrong N; p=5e-09); allele A increases the trait

—	rs145030841	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	T (+)	—	GWAS: Total cholesterol levels (Sakaue S; p=9e-25); T allele increases the trait
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LDL Cholesterol · Cholesterol LDL

LOWER LEVELS lower levels

PGS002703 (PGS Catalog): 948 324 of 970 081 variants with genotype (cover 98%). the 24th percentile in the reference population (n=633); 0% of the reference population has the same score.

LDL (and ApoB) is causal in atherosclerosis: the lower and longer the risk. Genomic polygenic score; a high or low genetic score does not change the target.

Example report: TYPICAL LEVELS (p44) · Category: HEART & BLOOD VESSELS · Mechanisms: Vascular Aging; Arterial stiffness; Inflammaging

Alleles that pull to «HIGHER LEVELS» — in the opposite direction to the result (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOE	rs7254892	GG	chip	G (+)	2	in PGS002703: G allele increases the score
PCSK9	rs11591147	GG	chip	G (+)	2	GWAS: LDL cholesterol levels (Richardson TG; p<1e-300); G allele increases the trait
LDLR	rs6511720	GT	chip	G (+)	1	in PGS002703: G allele increases the score
SLCO1B1	rs4149056	TT	chip	T (+)	2	in PGS002703: T allele increases the score
NLRC5	rs1800775	CC	chip	C (+)	2	in PGS002703: C allele increases the score
APOE	rs4420638	AG	imputation DR2=1.00	G (+)	1	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); G allele increases the trait

Alleles that pull to «LOWER LEVELS» — in the direction of the result (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOE	rs141622900	GA	chip	A (—)	1	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A decreases the trait
APOB	rs693	GA	chip	G (—)	1	in PGS002703: G allele decreases the score
LDLR	rs688	CC	chip	C (—)	2	in PGS002703: C allele decreases the score
SIDT2	rs964184	GC	chip	C (—)	1	GWAS: VLDL cholesterol levels (Zoddsma M; p<1e-300); C allele decreases the trait
LDLR	rs73015030	GA	chip	A (—)	1	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=7e-124); allele A decreases the trait

No copies of the effect allele (28)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOE	rs7412	CC	chip	T (—)	0	GWAS: LDL cholesterol (Sakaue S; p<1e-300); T allele decreases the trait
APOE	rs12721109	GG	imputation DR2=0.96	A (—)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A decreases the trait
APOE	rs62117160	GG	chip	A (—)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A decreases the trait
BCAM	rs28399654	GG	chip	A (—)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A decreases the trait
CETP	rs5882	AA	chip	G (—)	0	in PGS002703: G allele decreases the score
NECTIN2	rs365653	AA	imputation DR2=0.93	G (—)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); G allele decreases the trait
USP24	rs72660594	TT	chip	C (—)	0	

						GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=7e-87); C allele decreases the trait
NECTIN2	rs11668327	GG	imputation DR2=0.93	C (–)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); C allele decreases the trait
LDLR	rs72658867	GG	chip	A (–)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=7e-239); allele A decreases the trait
—	rs12713559	GG	chip	A (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=1e-26); allele A increases the trait
APOC4	rs140526515	AA	imputation DR2=0.93	G (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=9e-35); G allele increases the trait
NECTIN2	rs138914864	CC	imputation DR2=0.97	T (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=3e-33); T allele increases the trait
NECTIN2	rs117310449	CC	chip	T (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=2e-112); T allele increases the trait
APOE	rs769449	GG	chip	A (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A increases the trait
NECTIN2	rs144261139	CC	chip	A (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=4e-102); allele A increases the trait
NECTIN2	rs76366838	GG	chip	A (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=1e-207); allele A increases the trait
ABCG5	rs141828689	CC	chip	T (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=3e-11); T allele increases the trait
CLPTM1	rs12691088	GG	chip	A (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=3e-137); allele A increases the trait
APOE	rs429358	TT	imputation DR2=0.99	C (+)	0	GWAS: Clinical LDL cholesterol levels (Zoodma M; p<1e-300); C allele increases the trait
NECTIN2	rs34095326	GG	imputation DR2=0.96	A (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); allele A increases the trait
TOMM40	rs394819	GG	chip	T (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=1e-13); T allele increases the trait
NECTIN2	rs41289512	CC	chip	G (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=2e-188); G allele increases the trait
NECTIN2	rs138607350	TT	chip	G (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=8e-36); G allele increases the trait
SNX8	rs144787122	AA	chip	G (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=1e-15); G allele increases the trait
SLC22A3	rs3918291	TT	imputation DR2=1.00	C (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=6e-34); C allele increases the trait
LPA	rs3798220	TT	chip	C (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=3e-68); C allele increases the trait
MAFB	rs2207132	GG	chip	A (+)	0	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=3e-108); allele A increases the trait
APOC1	rs60049679	GG		C (+)	0	

imputation
DR2=0.81

GWAS: Low density lipoprotein cholesterol levels
(Sinnott-Armstrong N; p=3e-217); C allele increases the
trait

Undefined effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOB	rs5742904	CC	chip	—	—	
OLR1	rs12316150	AA	imputation DR2=0.97	—	—	

SNP not found in the data (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CEACAM20	rs200628672	—	SNP not found in the data	A (—)	—	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=1e-11); allele A decreases the trait
—	rs151135411	—	SNP not found in the data	A (+)	—	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=1e-17); allele A increases the trait
—	rs150401285	—	SNP not found in the data (Imputed DR2=0.04, below the 0.8 threshold)	G (+)	—	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=3e-11); G allele increases the trait
CLPTM1	rs490243	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	T (+)	—	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=1e-10); T allele increases the trait
APOC1	rs389261	—	SNP not found in the data (Imputed DR2=0.74, below the 0.8 threshold)	A (+)	—	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=1e-11); allele A increases the trait
PVR	rs139267469	—	SNP not found in the data (Imputed DR2=0.03, below the 0.8 threshold)	A (+)	—	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=3e-10); allele A increases the trait
BCL3	rs114036675	—	SNP not found in the data (Imputed DR2=0.02, below the 0.8 threshold)	A (+)	—	GWAS: Low density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=3e-10); allele A increases the trait

HDL Cholesterol · Cholesterol HDL

LOWER LEVELS lower levels

PGS002695 (PGS Catalog): 949 906 of 971 682 variants with genotype (cover 98%). the 18th percentile in the reference population (n=633); 0% of the reference population has the same score.

HDL participates in reverse cholesterol transport. High genetic HDL is associated with less cardiovascular risk in observational studies, but not in Mendelian randomization; the practical target remains LDL/ApoB.

Example report: TYPICAL LEVELS · Category: HEART & BLOOD VESSELS · Mechanisms: Vascular Aging

Alleles that pull to «HIGHER LEVELS» — in the opposite direction to the result (17)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
WDR11	rs10886863	TC	imputation DR2=1.00	T (+)	1	in PGS002695: T allele increases the score
PLTP	rs6073958	TC	chip	C (+)	1	GWAS: HDL cholesterol levels (Zoodma M; p=2e-205); C allele increases the trait
ABCA8	rs112001035	GG	imputation DR2=0.98	G (+)	2	GWAS: high density lipoprotein cholesterol (HDLc, mean, inv-norm transformed) (Verma A; p=2e-45); G allele increases the trait
UBE2L3	rs12158299	CC	imputation DR2=1.00	C (+)	2	in PGS002695: C allele increases the score
VEGFA	rs998584	CA	chip	C (+)	1	in PGS002695: C allele increases the score
MCUB	rs189866430	TT	imputation DR2=1.00	T (+)	2	GWAS: high density lipoprotein cholesterol (HDLc, minimum, inv-norm transformed) (Verma A; p=5e-37); T allele increases the trait
HNF4A	rs1800961	CC	chip	C (+)	2	in PGS002695: C allele increases the score

ADAL	rs150844304	AA	chip	A (+)	2	GWAS: HDL cholesterol levels (Richardson TG; p=2e-51); allele A increases the trait
ARID1A	rs193084249	AA	imputation DR2=0.92	A (+)	2	GWAS: HDL cholesterol levels (Richardson TG; p=4e-38); allele A increases the trait
SLC39A8	rs13107325	CC	chip	C (+)	2	GWAS: high density lipoprotein cholesterol (HDLc, mean, inv-norm transformed) (Verma A; p=1e-193); allele C increases the trait
APOE	rs429358	TT	imputation DR2=0.99	T (+)	2	GWAS: high density lipoprotein cholesterol (HDLc, mean, inv-norm transformed) (Verma A; p=1e-323); T allele increases the trait
FLT3	rs76428106	TT	chip	T (+)	2	GWAS: high density lipoprotein cholesterol (HDLc, mean, inv-norm transformed) (Verma A; p=1e-12); T allele increases the trait
PLG	rs571848809	GG	imputation DR2=1.00	G (+)	2	GWAS: HDL cholesterol levels (Richardson TG; p=7e-75); G allele increases the trait
NELFCD	rs76602912	TT	chip	T (+)	2	GWAS: HDL cholesterol levels (Richardson TG; p=4e-12); T allele increases the trait
PPARG	rs12485478	AA	imputation DR2=1.00	A (+)	2	in PGS002695: allele A increases the score
CNIH4	rs56105022	GG	imputation DR2=0.99	G (+)	2	GWAS: HDL cholesterol levels (Richardson TG; p=4e-10); G allele increases the trait
DGAT2	rs1219550	TT	imputation DR2=0.99	T (+)	2	in PGS002695: T allele increases the score

Alleles that pull to «LOWER LEVELS» — in the direction of the result (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOB	rs2678379	AG	chip	G (–)	1	GWAS: HDL cholesterol (Sakaue S; p=1e-102); G allele decreases the trait
IRS1	rs2138161	TC	imputation DR2=1.00	C (–)	1	GWAS: HDL cholesterol (Sakaue S; p=1e-74); allele C decreases the trait
ABCA1	rs2740488	AC	imputation DR2=1.00	C (–)	1	GWAS: High density lipoprotein cholesterol levels (Graham SE; p=2e-246); C allele decreases the trait

No copies of the effect allele (15)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SCARB1	rs921919	GG	imputation DR2=0.93	A (–)	0	GWAS: HDL cholesterol levels (Zoodma M; p=3e-123); allele A decreases the trait
GPAM	rs2792751	CC	imputation DR2=0.99	T (+)	0	in PGS002695: T allele increases the score
ETV5	rs57912727	AA	imputation DR2=1.00	C (–)	0	GWAS: HDL cholesterol (Sakaue S; p=4e-21); allele C decreases the trait
CD300LG	rs72836561	CC	chip	T (–)	0	GWAS: High density lipoprotein cholesterol levels (Graham SE; p=3e-275); T allele decreases the trait
LPL	rs328	CC	chip	G (+)	0	GWAS: HDL cholesterol levels (Farashi S; p=3e-120); G allele increases the trait
ADAM10	rs2070895	GG	chip	A (+)	0	GWAS: HDL cholesterol (Sakaue S; p<1e-300); allele A increases the trait
ACAD10	rs11066015	GG	chip	A (–)	0	GWAS: HDL cholesterol (Sakaue S; p=2e-109); allele A decreases the trait
ALPK2	rs41292412	CC	chip	T (–)	0	GWAS: HDL cholesterol levels (Zoodma M; p=3e-14); T allele decreases the trait
FADS1	rs174567	AA	imputation DR2=1.00	G (+)	0	GWAS: HDL cholesterol levels (Zoodma M; p=3e-302); G allele increases the trait
ZNF653	rs737338	CC	imputation DR2=1.00	T (–)	0	GWAS: HDL cholesterol levels (Zoodma M; p=3e-96); T allele decreases the trait
RSPO3	rs72959041	GG	chip	A (–)	0	GWAS: HDL cholesterol levels (Zoodma M; p=1e-37); allele A decreases the trait
PABPC4	rs17513135	CC	chip	T (–)	0	

						GWAS: HDL cholesterol (Sakaue S; p=1e-63); T allele decreases the trait
TSC22D2	rs9844972	GG	imputation DR2=0.97	C (–)	0	GWAS: HDL cholesterol (Sakaue S; p=5e-16); allele C decreases the trait
KANSL1	rs117499775	TT	chip	C (–)	0	GWAS: HDL cholesterol (Sakaue S; p=4e-09); allele C decreases the trait
ST3GAL4	rs112771035	CC	imputation DR2=0.99	G (+)	0	GWAS: HDL cholesterol levels (Zoodsma M; p=1e-34); G allele increases the trait

SNP not found in the data (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
—	rs200748895	—	SNP not found in the data	—	—	
—	rs71926466	—	SNP not found in the data	—	—	
CYP27A1	rs17572799	—	SNP not found in the data (Imputed DR2=0.70, below the 0.8 threshold)	G (–)	—	GWAS: HDL cholesterol (Sakaue S; p=2e-12); G allele decreases the trait
BTNL8	rs188238483	—	SNP not found in the data (Imputed DR2=0.48, below the 0.8 threshold)	T (+)	—	GWAS: high density lipoprotein cholesterol (HDLC, mean, inv-norm transformed) (Verma A; p=2e-43); T allele increases the trait
TCF15	rs151235402	—	SNP not found in the data (Imputed DR2=0.65, below the 0.8 threshold)	C (+)	—	GWAS: high density lipoprotein cholesterol (HDLC, minimum, inv-norm transformed) (Verma A; p=3e-29); allele C increases the trait
—	rs56271783	—	SNP not found in the data	G (+)	—	GWAS: high density lipoprotein cholesterol (HDLC, mean, inv-norm transformed) (Verma A; p=2e-51); G allele increases the trait

Lipid Oxidation · Lipid Oxidation

HIGHER higher

4 score of 4 variants (signal effect alleles). the 77th percentile (mean position) in the reference population (n=633); 31% of the reference population has exactly the same score.

It includes Lp(a) (which carries oxidized phospholipids) and PON1 paraoxonase, which protects LDL from oxidation. LPA rs10455872 G and rs3798220 C are the main alleles of high Lp(a).

Example report: TYPICAL · Category: HEART & BLOOD VESSELS · Mechanisms: Oxidative stress; Vascular Aging

Alleles that pull to «HIGHER» — in the direction of the result (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PON1	rs854560	TT	imputation DR2=1.00	T (+)	2	PON1 L55M: minus PON1
PLG	rs11751605	TT	imputation DR2=1.00	T (+)	2	PLG (as in the example report)
LPA	rs9457933	CT	imputation DR2=0.97	C (+)	1	LPA (as in the example report)

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
LPA	rs10455872	AA	chip	G (+)	0	LPA: much higher Lp(a)

Undefined effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PLG	rs3124785	GG	imputation DR2=0.98	—	—	
SLC22A3	rs10755578	CC	chip	—	—	
PON1	rs662	TT	chip	—	—	

SNP not found in the data (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
LPA	rs9355816	—		—	—	

SNP not found in the data (Imputed DR2=0.02, below the 0.8 threshold)

Oxidative Stress · Oxidative stress

LOWER lower

15 score of 15 variants (signal effect alleles). the 5th percentile (mean position) in the reference population (n=633); 3% of the reference population has exactly the same score.

Equilibrium between reactive species production and antioxidant defenses (SOD2, catalase, GPX1, NQO1, GST, PGC-1 α , FOXO3). Variants in these enzymes slightly alter the antioxidant capacity.

Example report: TYPICAL · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: Oxidative stress; Redox imbalance; Mitochondrial dysfunction

Alleles that pull to «HIGHER» — in the opposite direction to the result (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
NOS3	rs2070744	CT	chip	C (+)	1	NOS3 -786C: less eNOS
GPX1	rs1050450	GA	chip	A (+)	1	GPX1 Pro198Leu: less activity
GSTP1	rs1695	AG	chip	G (+)	1	GSTP1 Ile105Val
PON1	rs854560	TT	imputation DR2=1.00	T (+)	2	PON1 L55M

Alleles that pull to «LOWER» — in the direction of the result (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FOXO3	rs2802292	GG	chip	G (—)	2	FOXO3: longevity allele
FOXO3	rs2802288	AA	chip	A (—)	2	GWAS: Gamma glutamyl transferase levels (Ghouse J; p=2e-08); favourable allele A
FOXO3	rs2253310	CC	chip	C (—)	2	GWAS: Gamma glutamyl transferase levels (Jee YH; p=5e-11); favourable C allele

No copies of the effect allele (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SOD2	rs4880	GG	chip	A (+)	0	SOD2 Val16: less mitochondrial import
CAT	rs1001179	CC	chip	T (+)	0	CAT -262T
PPARGC1A	rs8192678	CC	chip	T (+)	0	PPARGC1A Gly482Ser
NQO1	rs1800566	GG	chip	A (+)	0	NQO1 P187S: loss of function
OGG1	rs1052133	CC	chip	G (+)	0	OGG1 S326C: less repair
ARMC2	rs768023	GG	chip	A (+)	0	GWAS: Gamma glutamyl transpeptidase (Sakaue S; p=1e-08); unfavourable allele A
ALDH2	rs671	GG	chip	A (+)	0	ALDH2*2
APOE	rs429358	TT	imputation DR2=0.99	C (+)	0	APOE ϵ 4

Undefined effect allele (33)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SOD3	rs2536512	AA	chip	—	—	
CAT	rs769217	CT	chip	—	—	
FOXO3	rs12212067	TG	chip	—	—	
FOXO3	rs4946936	TT	chip	—	—	
PON1	rs662	TT	chip	—	—	
FOXO3	rs12202234	CG	chip	—	—	
FOXO3	rs17069665	AG	chip	—	—	
FOXO3	rs9398171	CC	chip	—	—	
FOXO3	rs3800230	TG	chip	—	—	

FOXO3	rs9400239	TT	chip	—	—
FOXO3	rs479744	GT	imputation DR2=1.00	—	—
SIRT1	rs7895833	AA	chip	—	—
CAT	rs7943316	AT	imputation DR2=0.99	—	—
GPX4	rs713041	TC	chip	—	—
APEX1	rs1130409	TG	chip	—	—
GCLC	rs1555903	TT	imputation DR2=1.00	—	—
UGT1A6	rs1105879	AC	chip	—	—
NOS1	rs1879417	CT	chip	—	—
SIRT1	rs12778366	TT	chip	—	—
GSTO2	rs156697	AA	imputation DR2=1.00	—	—
UCP2	rs659366	CT	chip	—	—
TFAM	rs1937	GG	chip	—	—
CDKN2A	rs10811661	TC	chip	—	—
UCP1	rs1800592	TC	chip	—	—
HNRNPA3	rs13001694	AA	chip	—	—
SOD1	rs2234694	AA	imputation DR2=0.97	—	—
ARMC2	rs6911407	AA	chip	—	—
MRPS31	rs4581585	CC	chip	—	—
GCLM	rs41303970	GA	chip	—	—
TOM1	rs2071746	AA	imputation DR2=0.94	—	—
SLC23A1	rs33972313	CC	chip	—	—
NOS2	rs2297518	GG	chip	—	—
FOXO3	rs1935952	GG	chip	—	—

SNP not found in the data (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MVD	rs9932581	—	SNP not found in the data (Imputed DR2=0.74, below the 0.8 threshold)	—	—	
MPO	rs2333227	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—	

Detox · Detoxification

HIGHER ABILITY higher capacity

13 score of 13 variants (signal effect alleles). the 21st percentile (mean position) in the reference population (n=633); 12% of the reference population has exactly the same score.

Overall phase I capacity (CYP, EPHX1, ALDH2) and phase II (GST, NAT2, UGT, NQO1, COMT) and repair of DNA (XPC, XRCC1, OGG1).

Example report: TYPICAL ABILITY · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: DNA Damage and DNA Repair Deficit; Oxidative stress

Alleles that pull to «LOWER ABILITY» — in the opposite direction to the result (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTHFR	rs1801133	GA	chip	A (+)	1	MTHFR 677T
PON1	rs854560	TT	imputation DR2=1.00	T (+)	2	PON1 L55M
GSTP1	rs1695	AG	chip	G (+)	1	GSTP1 Ile105Val
GPX1	rs1050450	GA	chip	A (+)	1	GPX1 Pro198Leu

No copies of the effect allele (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
XPC	rs2228001	TT	chip	G (+)	0	XPC K939Q
COMT	rs4680	GG	chip	A (+)	0	COMT Met
NAT2	rs1799930	GG	chip	A (+)	0	NAT2*6 (slow)
ALDH2	rs671	GG	chip	A (+)	0	ALDH2*2
GSTP1	rs1138272	CC	chip	T (+)	0	GSTP1 Ala114Val
NFE2L2	rs6721961	GG	imputation DR2=1.00	T (+)	0	NFE2L2 -617A: minus NRF2
CYP2D6	rs3892097	CC	chip	T (+)	0	CYP2D6*4
NQO1	rs1800566	GG	chip	A (+)	0	NQO1*2
EPHX1	rs1051740	TT	chip	C (+)	0	EPHX1 Tyr113His (slow)

Undefined effect allele (36)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PON1	rs662	TT	chip	—	—	
SOD2	rs4880	GG	chip	—	—	
UGT2A1	rs10518065	AG	chip	—	—	
ADH1B	rs1229984	CC	imputation DR2=0.88	—	—	
NFE2L2	rs35652124	TC	chip	—	—	
CYP1B1	rs1056836	GG	chip	—	—	
NAT2	rs1495741	AA	chip	—	—	
GSTA1	rs3957357	AA	chip	—	—	
CYP1B1	rs1800440	TT	chip	—	—	
GSTO2	rs156697	AA	imputation DR2=1.00	—	—	
—	rs72547513	CC	chip	—	—	
BORCS7	rs743572	AG	chip	—	—	
NAT2	rs1041983	CC	chip	—	—	
ASAH1	rs4271002	GG	chip	—	—	
—	rs366631	AA	chip	—	—	
XRCC1	rs1799782	GG	chip	—	—	
PTGS2	rs5277	CG	imputation DR2=0.97	—	—	
MLLT3	rs76878079	GG	imputation DR2=0.83	—	—	
MTG1	rs2031920	CC	chip	—	—	
EGLN2	rs28399433	AA	chip	—	—	
CYP2D6	rs16947	AG	chip	—	—	
—	rs2279343	AG	chip	—	—	
CYP2E1	rs2070673	TT	chip	—	—	
GSDMB	rs7216389	CT	chip	—	—	
CSK	rs2606345	CA	chip	—	—	
UGT2B7	rs7439366	CC	chip	—	—	
CYP1B1	rs1056827	CA	imputation DR2=1.00	—	—	
ITCH	rs819147	CT	chip	—	—	
GSTM1	rs1056806	CC	chip	—	—	
CTH	rs1021737	GG	chip	—	—	
CYP1A1	rs1048943	TT	chip	—	—	
COX15	rs717620	CC	chip	—	—	
NAT1	rs4986782	GG	chip	—	—	
			chip			

TRIM4	rs2740574	TT		—	—
NQO1	rs1131341	GG	chip	—	—
CYP1A2	rs762551	CA	chip	—	—

SNP not found in the data (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
—	rs12228069	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)		—	—

Phase I Detox · Phase I Detoxification

TYPICAL ABILITY typical capacity

10 score of 10 variants (signal effect alleles). the 49th percentile (mean position) in the reference population (n=633); 24% of the reference population has exactly the same score.

CYP (1A1, 1A2, 1B1, 2C9, 2C19, 2D6, 2E1), EPHX1 and ALDH2: oxidize drugs, hormones and environmental toxics. Variants of function loss make phase I slower.

Example report: TYPICAL ABILITY · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: DNA Damage and DNA Repair Deficit

Alleles that pull to «LOWER ABILITY» (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CYP1A2	rs762551	CA	chip	C (+)	1	CYP1A2 (less inducible)
CYP2C9	rs1057910	AC	chip	C (+)	1	CYP2C9*3

No copies of the effect allele (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ADH1B	rs1229984	CC	imputation DR2=0.88	T (—)	0	ADH1B*2 (fast)
ALDH2	rs671	GG	chip	A (+)	0	ALDH2*2
CYP2D6	rs3892097	CC	chip	T (+)	0	CYP2D6*4
EPHX1	rs1051740	TT	chip	C (+)	0	EPHX1 Tyr113His
CYP2C19	rs4244285	GG	chip	A (+)	0	CYP2C19*2
CYP2D6	rs1065852	GG	chip	A (+)	0	CYP2D6*10
CYP2C9	rs1799853	CC	chip	T (+)	0	CYP2C9*2
NOC3L	rs12248560	CC	chip	T (—)	0	CYP2C19*17 (fast)

Undefined effect allele (25)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
UGT2A1	rs10518065	AG	chip	—	—	
CYP1B1	rs1056836	GG	chip	—	—	
—	rs72547513	CC	chip	—	—	
BORCS7	rs743572	AG	chip	—	—	
CYP1B1	rs1800440	TT	chip	—	—	
CYP1B1	rs10012	GC	chip	—	—	
CYP2D6	rs16947	AG	chip	—	—	
—	rs2279343	AG	chip	—	—	
CYP2E1	rs2070673	TT	chip	—	—	
CSK	rs2606345	CA	chip	—	—	
CYP1B1	rs1056827	CA	imputation DR2=1.00	—	—	
CSK	rs2472304	GA	chip	—	—	
CYP2E1	rs2480256	GG	imputation DR2=0.99	—	—	
CYP2E1	rs1329149	CC	imputation DR2=1.00	—	—	

CYP2E1	rs2070676	CC	chip	—	—
CYP1A1	rs1048943	TT	chip	—	—
NAT1	rs4986782	GG	chip	—	—
TRIM4	rs2740574	TT	chip	—	—
CYP1A1	rs4646903	AA	imputation DR2=0.99	—	—
CYP2A6	rs1801272	AA	chip	—	—
CYP1A1	rs1799814	GG	chip	—	—
SYCE1	rs8192772	TT	imputation DR2=0.99	—	—
CYP2E1	rs6413419	GG	chip	—	—
EGLN2	rs28399433	AA	chip	—	—
XRCC4	rs28383151	GG	chip	—	—

Phase II Detox · II phase detoxification

TYPICAL ABILITY typical capacity

14 score of 14 variants (signal effect alleles). the 56th percentile (mean position) in the reference population (n=633); 19% of the reference population has exactly the same score.

Conjugation with glutathione (GSTP1/GSTM1/GSTA1), acetylation (NAT1/NAT2), glucuronidation (UGT1A1/UGT2B7), methylation (COMT) and reduction of quinones (NQO1).

Example report: TYPICAL ABILITY · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: DNA Damage and DNA Repair Deficit; Redox imbalance

Alleles that pull to «LOWER ABILITY» (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
GSTP1	rs1695	AG	chip	G (+)	1	GSTP1 Ile105Val
GPX1	rs1050450	GA	chip	A (+)	1	GPX1 Pro198Leu
PON1	rs854560	TT	imputation DR2=1.00	T (+)	2	PON1 L55M
UGT1A6	rs6742078	GT	chip	T (+)	1	UGT1A1 (tag *28)
NAT2	rs1801280	CC	chip	C (+)	2	NAT2*5

No copies of the effect allele (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
GSTP1	rs1138272	CC	chip	T (+)	0	GSTP1 Ala114Val
XPC	rs2228001	TT	chip	G (+)	0	XPC K939Q
NFE2L2	rs6721961	GG	imputation DR2=1.00	T (+)	0	NFE2L2 -617A
NQO1	rs1800566	GG	chip	A (+)	0	NQO1*2
UGT1A1	rs4148323	GG	chip	A (+)	0	UGT1A1*6
COMT	rs4680	GG	chip	A (+)	0	COMT Met
NAT2	rs1799930	GG	chip	A (+)	0	NAT2*6
NAT2	rs1801279	GG	chip	A (+)	0	NAT2*14
NAT2	rs1799931	GG	chip	A (+)	0	NAT2*7

Undefined effect allele (21)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
NQO1	rs1131341	GG	chip	—	—	
NAT2	rs1495741	AA	chip	—	—	
CYP1B1	rs1800440	TT	chip	—	—	
GSTO2	rs156697	AA	imputation DR2=1.00	—	—	
NAT2	rs1041983	CC	chip	—	—	
ASAH1	rs4271002	GG	chip	—	—	

—	rs366631	AA	chip	—	—
ERCC5	rs17655	GG	chip	—	—
THSD7A	rs116864947	TT	imputation DR2=1.00	—	—
UGT2B7	rs7439366	CC	chip	—	—
GSTM1	rs1056806	CC	chip	—	—
COX15	rs717620	CC	chip	—	—
NAT1	rs4986782	GG	chip	—	—
NFE2L2	rs35652124	TC	chip	—	—
MTHFR	rs1801133	GA	chip	—	—
GSTA1	rs3957357	AA	chip	—	—
NAT2	rs1799929	TT	chip	—	—
NAT2	rs1208	GG	chip	—	—
GSTM3	rs7483	CT	chip	—	—
MTG1	rs2031920	CC	chip	—	—
SULT1A1	rs1042028	CC	imputation DR2=0.98	—	—

Inflammation (CRP) · Inflammation (CRP)

TYPICAL LEVELS typical levels

PGS002873 (PGS Catalog): 1 082 460 of 1 113 831 variants with genotype (97% coverage). the 26th percentile in the reference population (n=633); 0% of the reference population has the same score.

C-reactive protein is a marker of low-grade inflammation. About 25–40% of variation is genetic (CRP, HNF1A, IL6R, APOE, LEPR).

Example report: TYPICAL LEVELS (p36) · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: Chronic Inflammation; Inflammaging

Alleles that pull to «HIGHER LEVELS» (14)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IL6R	rs2228145	AC	chip	A (+)	1	GWAS: C-reactive protein levels (Pietzner M; p=2e-15); allele A increases the trait
FUT2	rs601338	GA	chip	A (+)	1	in PGS002873: allele A increases the score
IL1B	rs16944	AG	chip	G (+)	1	in PGS002873: G allele increases the score
IL19	rs1800896	TC	chip	C (+)	1	in PGS002873: C allele increases the score
IL21	rs6822844	GT	chip	T (+)	1	in PGS002873: T allele increases the score
LRP6	rs2160525	GG	chip	G (+)	2	in PGS002873: G allele increases the score
LRP6	rs2302685	TT	imputation DR2=0.99	T (+)	2	in PGS002873: T allele increases the score
FUT2	rs492602	AG	chip	G (+)	1	in PGS002873: G allele increases the score
FUT2	rs602662	GA	chip	A (+)	1	in PGS002873: allele A increases the score
LEPR	rs4394621	AA	imputation DR2=1.00	A (+)	2	GWAS: C-reactive protein levels (Sinnott-Armstrong N; p=9e-276); allele A increases the trait
APOE	rs429358	TT	imputation DR2=0.99	T (+)	2	GWAS: C-reactive protein levels (Ligthart S; p<1e-300); T allele increases the trait
SLC20A1	rs1800587	GA	chip	A (+)	1	in PGS002873: allele A increases the score
IL1A	rs17561	CA	chip	A (+)	1	in PGS002873: allele A increases the score
SOD2	rs4880	GG	chip	G (+)	2	in PGS002873: G allele increases the score

Alleles that pull to «LOWER LEVELS» (15)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CRP	rs1205	CT	chip	T (–)	1	in PGS002873: T allele decreases the score
IL19	rs1800872	GG	chip	G (–)	2	in PGS002873: G allele decreases the score

IL19	rs3024505	GA	chip	A (–)	1	in PGS002873: allele A decreases the score
IL4R	rs1805011	AC	chip	C (–)	1	in PGS002873: C allele decreases the score
IL1RN	rs419598	TC	chip	C (–)	1	in PGS002873: C allele decreases the score
ATG16L1	rs10210302	CT	chip	T (–)	1	in PGS002873: T allele decreases the score
ATG16L1	rs2241880	AG	chip	G (–)	1	in PGS002873: G allele decreases the score
STEAP1B	rs1554606	GG	imputation DR2=1.00	G (–)	2	in PGS002873: G allele decreases the score
IL13	rs20541	GG	chip	G (–)	2	in PGS002873: G allele decreases the score
ADRB2	rs1042713	GA	chip	A (–)	1	in PGS002873: allele A decreases the score
TIMP4	rs3755724	CT	chip	T (–)	1	in PGS002873: T allele decreases the score
IFIH1	rs1990760	TT	chip	T (–)	2	in PGS002873: T allele decreases the score
IL6	rs1524107	CT	chip	T (–)	1	in PGS002873: T allele decreases the score
IL6	rs2066992	GT	chip	T (–)	1	in PGS002873: T allele decreases the score
CFH	rs6677604	GA	chip	A (–)	1	in PGS002873: allele A decreases the score

No copies of the effect allele (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CTLA4	rs231775	AA	chip	G (–)	0	in PGS002873: G allele decreases the score
SIRT1	rs12778366	TT	chip	C (+)	0	in PGS002873: C allele increases the score
CRP	rs3093059	AA	chip	G (+)	0	in PGS002873: G allele increases the score
RAD50	rs2069812	AA	chip	G (–)	0	in PGS002873: G allele decreases the score
TNF	rs1800629	GG	chip	A (+)	0	in PGS002873: allele A increases the score
FADS2	rs174546	CC	chip	T (–)	0	in PGS002873: T allele decreases the score
IL1B	rs1143634	GG	chip	A (–)	0	in PGS002873: allele A decreases the score
CRP	rs10494326	CC	chip	T (+)	0	GWAS: C-reactive protein levels (Sinnott-Armstrong N; p=2e-65); T allele increases the trait
TLR4	rs4986790	AA	chip	G (–)	0	in PGS002873: G allele decreases the score

Undefined effect allele (12)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
KLC1	rs8702	CG	chip	—	—	
STAT4	rs10181656	CC	chip	—	—	
CYP1B1	rs1056836	GG	chip	—	—	
AOC1	rs1049793	CG	chip	—	—	
FUT2	rs281377	CT	chip	—	—	
MICB	rs361525	GA	chip	—	—	
SLC22A5	rs1800925	CC	chip	—	—	
HLA-DQA1	rs2187668	CC	chip	—	—	
SRA1	rs2569191	CT	chip	—	—	
IL6	rs1800795	GG	chip	—	—	
LGALS9	rs2248814	AA	chip	—	—	
TOM1	rs2071746	AA	imputation DR2=0.94	—	—	

Bone Health (Functional) · Bone health (functional)

TYPICAL typical

5 score of 5 variants (signal effect alleles). the 30th percentile (mean position) in the reference population (n=633); 25% of the reference population has exactly the same score.

Variants in VDR, COL1A1, GDF5, TNFRSF11B (osteoprotegerin), DIO2 and CYP2R1 associated with bone mineral density and collagen quality.

Alleles that pull to «WORSE» (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
VDR	rs2228570	AG	chip	A (+)	1	VDR FokI f
TNFRSF11B	rs2073618	GC	chip	C (+)	1	TNFRSF11B (OPG) N3K

No copies of the effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CPNE1	rs143383	GG	chip	A (+)	0	GDF5 +104T
COL1A1	rs1800012	CC	chip	A (–)	0	COL1A1 Sp1 s
ALDH7A1	rs13182402	AA	chip	G (+)	0	GWAS: Osteoporosis (Guo Y; p=2e-09); unfavourable G allele

Undefined effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
COPB1	rs10741657	AG	chip	—	—	
VDR	rs1544410	CT	chip	—	—	
VDR	rs731236	AG	chip	—	—	
DIO2	rs225014	TT	chip	—	—	

Type 2 Diabetes · Diabetes type 2

TYPICAL LIKELIHOOD typical probability

PGS002720 (PGS Catalog): 891 114 of 911 809 variants with genotype (cover 98%). the 59th percentile in the reference population (n=633); 0% of the reference population has the same score.

Polygenic score of 2 type diabetes. The lifestyle (fiber, muscle, sleep) weighs more than the score.

Example report: TYPICAL LIKELIHOOD (p49) · Category: BLOOD SUGAR CONTROL · Mechanisms: Insulin resistance; Hyperglycemia and Intolerance to Glucose; Accumulation of Advanced Glycation Final Products (AGEs)

Alleles that pull to «MORE LIKELY» (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SLC30A8	rs13266634	CC	chip	C (+)	2	in PGS002720: C allele increases the score
CDKN2A	rs10811661	TC	chip	T (+)	1	in PGS002720: T allele increases the score
ADCY5	rs11708067	AG	chip	A (+)	1	in PGS002720: allele A increases the score
BCL2	rs12454712	TC	chip	T (+)	1	in PGS002720: T allele increases the score
TCF7L2	rs4506565	AT	chip	T (+)	1	GWAS: Type 2 diabetes (Sakaue S; p=2e-47); T allele increases the trait
IGF2BP2	rs11927381	TC	chip	C (+)	1	GWAS: Type 2 diabetes (Cook JP; p=3e-14); allele C increases the trait
KCNJ11	rs5215	CC	chip	C (+)	2	in PGS002720: C allele increases the score
FTO	rs9939609	TA	chip	A (+)	1	GWAS: Type 2 diabetes (Perry JR; p=1e-20); allele A increases the trait

Alleles that pull to «LESS LIKELY» (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TCF7L2	rs7903146	CT	chip	C (–)	1	in PGS002720: C allele decreases the score
GCKR	rs780093	TC	chip	T (–)	1	in PGS002720: T allele decreases the score
HMGA2	rs2261181	CC	chip	C (–)	2	in PGS002720: C allele decreases the score
DGKB	rs2191349	GG	chip	G (–)	2	GWAS: Type 2 diabetes (Vujkovic M; p=6e-48); allele G decreases the trait
IGF2BP2	rs1470579	AC	chip	A (–)	1	in PGS002720: allele A decreases the score
IGF2BP2	rs11705701	GA	chip	G (–)	1	in PGS002720: G allele decreases the score

MC4R	rs12970134	GG	chip	G (–)	2	in PGS002720: G allele decreases the score
CDKAL1	rs7756992	AG	chip	A (–)	1	in PGS002720: allele A decreases the score
CDKAL1	rs9465871	TC	chip	T (–)	1	in PGS002720: T allele decreases the score

No copies of the effect allele (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IMPDH1	rs791595	GG	chip	A (–)	0	in PGS002720: allele A decreases the score
TNF	rs2857605	TT	chip	C (–)	0	GWAS: Type 2 diabetes (Xue A; p=6e-14); allele C decreases the trait
SLC38A11	rs10195252	CC	chip	T (+)	0	in PGS002720: T allele increases the score
TSPAN3	rs7177055	AA	chip	G (–)	0	in PGS002720: G allele decreases the score
CCND2	rs76895963	TT	chip	G (–)	0	GWAS: Type 2 diabetes (Huerta-Chagoya A; p=1e-126); allele G decreases the trait
WSB2	rs7973260	GG	imputation DR2=0.81	A (–)	0	in PGS002720: allele A decreases the score
JAZF1	rs1635852	CC	chip	T (+)	0	in PGS002720: T allele increases the score
TAP2	rs2071479	CC	chip	T (+)	0	GWAS: Type 2 diabetes (Xue A; p=7e-11); T allele increases the trait
—	rs184660829	TT	imputation DR2=0.82	C (+)	0	GWAS: Type 2 diabetes (Mahajan A; p=3e-08); allele C increases the trait

Undefined effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PEMT	rs12325817	CC	chip	—	—	
GIPR	rs10423928	TT	chip	—	—	

SNP not found in the data (15)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
—	rs569511541	—	SNP not found in the data	G (+)	—	GWAS: Type 2 diabetes (Mahajan A; p=2e-08); allele G increases the trait
—	rs562386202	—	SNP not found in the data	G (+)	—	GWAS: Type 2 diabetes (Mahajan A; p=4e-08); allele G increases the trait
—	rs543786825	—	SNP not found in the data	—	—	
—	rs759111467	—	SNP not found in the data	—	—	
CCDC68	rs76197067	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—	
RNASEH2A	rs755734872	—	SNP not found in the data	—	—	
NEDD1	rs557027608	—	SNP not found in the data	—	—	
—	rs533172266	—	SNP not found in the data	—	—	
QSER1	rs528122639	—	SNP not found in the data	—	—	
BPTF	rs558308082	—	SNP not found in the data (Imputed DR2=0.12, below the 0.8 threshold)	—	—	
ZC3H11B	rs553014999	—	SNP not found in the data	—	—	
ABCC8	rs67254669	—	SNP not found in the data (Imputed DR2=0.02, below the 0.8 threshold)	G (+)	—	GWAS: Type 2 diabetes (Mahajan A; p=1e-08); allele G increases the trait
INS	rs571342427	—	SNP not found in the data	—	—	
FAM13A	rs576406049	—	SNP not found in the data	—	—	
—	rs745903616	—	SNP not found in the data	—	—	

APOC3 (Blood Lipids/ Longevity) · APOC3 (lipids/longevity)

LOWER ACTIVITY lower activity

2 score of 2 variants (signal effect alleles). the 18th percentile (mean position) in the reference population (n=633); 37% of the reference population has exactly the same score.

ApoC-III inhibits lipoprotein lipase and increases triglycerides. Less ApoC-III (e.g. -641C, rs2542052) is associated with low triglycerides and longevity; the S2 allele (rs5128 G) is associated with higher triglycerides.

Example report: TYPICAL ACTIVITY · Category: HEART & BLOOD VESSELS · Mechanisms: Vascular Aging; Metabolic Homeostasis Decline

Alleles that pull to «LOWER ACTIVITY» — in the direction of the result (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SIDT2	rs2542052	CC	chip	C (–)	2	APOC3 -641C: less ApoC-III, longevity

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PCSK7	rs5128	CC	chip	G (+)	0	APOC3 SstI S2: higher triglycerides

IL6 Gene (Inflammation) · Gene IL6 (inflammation)

TYPICAL ACTIVITY typical activity

Genotype rule (as in the example report): rs1800795 (-174) GG.

IL-6 is a central cytokine of inflammation. The -174 G>C polymorphism (rs1800795) alters the transcription; the G allele is associated with more IL-6 in some studies (mixed results).

Example report: TYPICAL ACTIVITY · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: Inflammaging; Chronic Inflammation

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IL6	rs1800795	GG	chip	C (+)	0	-174C: lower IL-6 transcription (mixed results)

TNF Gene (Inflammation) · Gene TNF (inflammation)

LOWER ACTIVITY lower activity

15 score of 15 variants (signal effect alleles). the 20th percentile (mean position) in the reference population (n=633); 0% of the reference population has exactly the same score.

TNF-α is a pro-inflammatory cytokine. The allele A of -308 (rs1800629) is associated with higher production of TNF-α. The remaining SNPs come from a GWAS of TNF-α levels.

Example report: TYPICAL ACTIVITY · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: Inflammaging; Chronic Inflammation

Alleles that pull to «HIGHER ACTIVITY» — in the opposite direction to the result (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
YTHDF2	rs185682149	CC	imputation DR2=0.85	C (+)	2	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=3e-06); unfavourable C allele
TMEM71	rs72725197	CC	imputation DR2=0.92	C (+)	2	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=5e-06); unfavourable C allele
TRIB2	rs116434579	GG	imputation DR2=0.96	G (+)	2	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=8e-06); unfavourable G allele
—	rs7256693	TC	imputation DR2=0.93	C (+)	1	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=5e-06); unfavourable C allele
ATP10B	rs149126334	AG	imputation DR2=0.99	G (+)	1	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=5e-06); unfavourable G allele
—	rs72841564	CC	imputation DR2=0.95	C (+)	2	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=8e-06); unfavourable C allele
GBA2	rs10814274	CT	imputation DR2=1.00	C (+)	1	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=8e-06); unfavourable C allele

Alleles that pull to «LOWER ACTIVITY» — in the direction of the result (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SPRY1	rs115669577	GG	imputation DR2=0.97	G (—)	2	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=8e-07); favourable G allele
ERICH1	rs138808635	CC	imputation DR2=0.96	C (—)	2	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=9e-06); favourable C allele
GTF2E1	rs10511404	GA	imputation DR2=0.95	G (—)	1	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=6e-06); favourable G allele
—	rs8121916	CC	chip	C (—)	2	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=3e-06); favourable C allele

No copies of the effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ROBO1	rs149420276	AA	imputation DR2=0.91	G (+)	0	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=2e-06); unfavourable G allele
GPX3	rs111332265	AA	imputation DR2=0.97	G (+)	0	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=7e-07); unfavourable G allele
ANO3	rs10834997	AA	imputation DR2=0.97	G (+)	0	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=1e-06); unfavourable G allele
TNF	rs1800629	GG	chip	A (+)	0	TNF -308A: plus TNF-α

Undefined effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TNF	rs1799724	CC	chip	—	—	
NCR3	rs1800610	GG	chip	—	—	
MICB	rs361525	GA	chip	—	—	
TNF	rs1799964	TC	chip	—	—	

SNP not found in the data (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PHACTR4	rs188618085	—	SNP not found in the data (Imputed DR2=0.69, below the 0.8 threshold)	C (+)	—	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=7e-07); unfavourable C allele
CRISPLD2	rs118135095	—	SNP not found in the data (Imputed DR2=0.19, below the 0.8 threshold)	G (+)	—	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=1e-06); unfavourable G allele
PSD3	rs79105320	—	SNP not found in the data (Imputed DR2=0.71, below the 0.8 threshold)	G (—)	—	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=4e-06); favourable G allele
RBM11	rs72490194	—	SNP not found in the data (Imputed DR2=0.51, below the 0.8 threshold)	C (—)	—	GWAS: Necrosis factor alpha levels (Ahola-Olli AV; p=7e-06); favourable C allele

CETP (Cholesterol/ Longevity) · CETP (cholesterol/longevity)

HIGHER ACTIVITY higher activity

2 score of 2 variants (signal effect alleles). the 11th percentile (mean position) in the reference population (n=633); 22% of the reference population has exactly the same score.

CETP transfers cholesterol esters from HDL to ApoB lipoproteins. Lower genetic activity (TaqIB B2 = rs708272 A; I405V = rs5882 G) is associated with higher HDL and, by Mendelian randomization, with less coronary disease, proportional to ApoB decrease.

Example report: LOWER ACTIVITY · Category: HEART & BLOOD VESSELS · Mechanisms: Vascular Aging

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
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CETP	rs5882	AA	chip	G (+)	0	I405V (V): minus CETP
CETP	rs708272	GG	chip	A (+)	0	TaqIB B2: minus CETP

IL1B (Inflammation/ Fatigue) · IL1B (flammation/fatigue)

TYPICAL ACTIVITY typical activity

3 score of 3 variants (signal effect alleles). the 55th percentile (mean position) in the reference population (n=633); 31% of the reference population has exactly the same score.

IL-1β is a cytokine of the NLRP3 inflammasome. The -511T (rs16944 A), -31C (rs1143627 G) and +3954T (rs1143634 A) alleles are associated with higher production.

Example report: LOWER ACTIVITY · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: Inflammaging; Neuroinflammation

Alleles that pull to «HIGHER ACTIVITY» (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IL1B	rs16944	AG	chip	A (+)	1	IL1B -511T
IL1A	rs1143627	GA	chip	G (+)	1	IL1B -31C

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IL1B	rs1143634	GG	chip	A (+)	0	IL1B +3954T

Undefined effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IL1A	rs1143633	CT	chip	—	—	
IL1A	rs1143623	CG	chip	—	—	
IL1B	rs1143643	CT	chip	—	—	

Results in detail: Nutrition

MTHFR · MTHFR

LOWER ACTIVITY lower activity

Genotype rule (as in the example report): C677T rs1801133 GA, A1298C rs1801131 TG — compound heterozygous, ~50–60% of activity.

MTHFR produces 5-metiltetra-hidrofolato. C677T (rs1801133 A) reduces ~35% activity by allele; A1298C (rs1801131 G) reduces less. The combination 677CT + 1298AC (compound heterozygous) gives ~50–60% of activity.

Example report: LOWER ACTIVITY · Category: NUTRITION · Mechanisms: Epigenetic Changes; Vascular Aging

With the effect allele — in the direction of the result (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTHFR	rs1801133	GA	chip	A (+)	1	677T: –35% activity by allele
MTHFR	rs1801131	TG	chip	G (+)	1	1298C: slightly reduced activity

MUFA Response · Response to monounsaturated fat

TYPICAL typical

9 score of 9 variants (signal effect alleles). the 62nd percentile (mean position) in the reference population (n=633); 0% of the reference population has exactly the same score.

Variants (PPARG, FABP2, TCF7L2, APOE, PNPLA3, IQCJ, ADRB3...) that, in gene-diet interaction studies, modify the metabolic benefit of olive oil and other monounsaturated fats. Approximation: unfavourable alleles for lipids/ blood glucose in GWAS Catalog.

Example report: WORSE · Category: HEART & BLOOD VESSELS · Mechanisms: Vascular Aging; Dysfunction of adipose tissue

Alleles that pull to «WORSE» (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
NECTIN2	rs1064725	TT	imputation DR2=0.81	T (+)	2	GWAS: Triglyceride to HDL cholesterol ratio (Oliveri A; p=6e-23); unfavourable T allele
TCF7L2	rs7903146	CT	chip	T (+)	1	TCF7L2
BDNF	rs6265	CT	chip	T (+)	1	GWAS: Type 2 diabetes (Elashi AA; p=3e-08); unfavourable T allele
FABP2	rs1799883	TC	imputation DR2=1.00	T (+)	1	FABP2 Thr54

Alleles that pull to «BETTER» (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
GCKR	rs1260326	TC	chip	C (–)	1	GWAS: Triglyceride levels (Graham SE; p<1e-300); favourable C allele
PNPLA3	rs738409	CG	chip	G (–)	1	GWAS: Total cholesterol levels (Graham SE; p=2e-39); favourable G allele

No copies of the effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PPARG	rs1801282	CC	chip	G (–)	0	PPARG Ala12
RFC4	rs17300539	GG	chip	A (+)	0	ADIPOQ -11391A (as in the example report)
APOE	rs429358	TT	imputation DR2=0.99	C (+)	0	APOE ε4

Undefined effect allele (14)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
UCP3	rs1800849	GG	chip	—	—	

FAAH	rs324420	CA	chip	—	—
ADRB3	rs4994	AA	chip	—	—
ABCG1	rs4148102	GG	imputation DR2=0.97	—	—
MED24	rs709592	CT	chip	—	—
ACTRT3	rs12696304	CG	chip	—	—
MGLL	rs9877819	GG	imputation DR2=0.95	—	—
ST6GAL1	rs3774261	AA	chip	—	—
RFC4	rs266729	CC	chip	—	—
IQCJ	rs17782879	GG	imputation DR2=0.99	—	—
PCSK5	rs1029035	TT	imputation DR2=1.00	—	—
ANKK1	rs1800497	GG	chip	—	—
ABTB1	rs782440	CC	chip	—	—
IQCJ	rs1868414	CT	imputation DR2=0.98	—	—

Protein Metabolism · Protein Metabolism

WORSE RESPONSE worst response

5 score of 5 variants (signal effect alleles). the 82nd percentile (mean position) in the reference population (n=633); 12% of the reference population has exactly the same score.

Variants (FTO, GLP1R, MTNR1B, TFAP2B, CNDP1...) that modify the response to diets richer in protein (society, body composition, glycemia).

Example report: WORSE RESPONSE · Category: BODY FAT & APPETITE · Mechanisms: Sarcopenia; Thin Mass Loss

Alleles that pull to «WORSE RESPONSE» — in the direction of the result (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FTO	rs9939609	TA	chip	A (+)	1	FTO
FTO	rs1558902	TA	chip	A (+)	1	FTO
GLP1R	rs6923761	AA	imputation DR2=1.00	A (+)	2	GLP1R Gly168Ser
MTNR1B	rs10830963	CG	chip	G (+)	1	MTNR1B

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TFAP2B	rs987237	AA	chip	G (+)	0	GWAS: adiposity (Huang J; p=3e-109); unfavourable G allele

Undefined effect allele (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CNDP2	rs4891558	TC	imputation DR2=1.00	—	—	
NADSYN1	rs12785878	TT	chip	—	—	
FUCA1	rs3123554	AG	imputation DR2=1.00	—	—	
NTN5	rs838147	AG	imputation DR2=1.00	—	—	
CLOCK	rs3749474	CT	chip	—	—	
ADRB3	rs4994	AA	chip	—	—	
ST6GAL1	rs1501299	GT	chip	—	—	
CNDP1	rs7244647	TC	imputation DR2=0.99	—	—	
TNF	rs1800629	GG	chip	—	—	

Riboflavin (Vitamin B2) · Riboflavin (vitamin B2)

TYPICAL NEED typical necessity

Genotype rule (as in the example report): C677T rs1801133 GA, A1298C rs1801131 TG; without 677TT.

Riboflavin (FAD) is a cofactor of MTHFR. In the genotype 677TT riboflavin stabilizes the enzyme and low to homocysteine; in 677CT the effect is small.

Example report: INCREASED NEED · Category: NUTRITION · Mechanisms: Epigenetic Changes; Oxidative Phosphorylation Decline

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTHFR	rs1801133	GA	chip	A (+)	1	677T: riboflavin stabilizes the enzyme mainly in TT

Undefined effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTRR	rs1801394	AA	chip	—	—	
MTHFR	rs1801131	TG	chip	—	—	

Vitamin B6 (Pyridoxine) · Vitamin B6 (pyridoxine)

TYPICAL NEED typical necessity

1 score of 1 variants (signal effect alleles). the 47th percentile (mean position) in the reference population (n=633); 50% of the reference population has exactly the same score.

ALPL (alkaline phosphatase) and NBPf3 influence plasma levels of PLP (active form of B6). PDXK activates pyridoxine.

Example report: INCREASED NEED · Category: NUTRITION · Mechanisms: Neuroinflammation

Alleles that pull to «INCREASED NEED» (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
NBPf3	rs4654748	CT	chip	C (+)	1	NBPf3/ALPL: lower PLP

Undefined effect allele (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ALPL	rs1256341	TC	imputation DR2=0.99	—	—	
NBPf3	rs1697421	CT	chip	—	—	
CBS	rs234706	GA	chip	—	—	
MTHFR	rs1801133	GA	chip	—	—	
PDXK	rs2010795	GG	chip	—	—	
ALPL	rs1772719	AC	chip	—	—	
ALPL	rs1256335	GA	chip	—	—	

SNP not found in the data (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PDXK	rs147242481	—	SNP not found in the data (Imputed DR2=0.03, below the 0.8 threshold)		—	—

Folate · Folate (functional)

DECREASED NEED decreased need

5 score of 5 variants (signal effect alleles). the 10th percentile (mean position) in the reference population (n=633); 11% of the reference population has exactly the same score.

Folate pathway variants (MTHFR, MTHFD1, SHMT1, MTRR, MTR, Packages1, MTHFS) that increase the need for food folate or active folate.

Example report: INCREASED NEED · Category: NUTRITION · Mechanisms: Epigenetic Changes; DNA Damage and DNA Repair Deficit

Alleles that pull to «INCREASED NEED» — in the opposite direction to the result (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTHFR	rs1801133	GA	chip	A (+)	1	MTHFR 677T

MTHFR	rs1801131	TG	chip	G (+)	1	MTHFR 1298C
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No copies of the effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SHMT1	rs1979277	GG	chip	A (+)	0	SHMT1 1420T
MTHFD1	rs2236225	GG	chip	A (+)	0	MTHFD1 1958A
MTRR	rs1801394	AA	chip	G (+)	0	MTRR 66G

Undefined effect allele (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTHFS	rs6495446	CC	chip	—	—	
FOLH1	rs61886492	GA	imputation DR2=1.00	—	—	
MSH3	rs408626	CC	imputation DR2=1.00	—	—	
MSH3	rs1650697	GG	chip	—	—	
MTHFD1L	rs11754661	GG	chip	—	—	
FOLH1	rs202676	AA	chip	—	—	
MTR	rs1805087	AA	chip	—	—	

Alcohol Sensitivity · Alcohol sensitivity

TYPICAL typical

Genotype rule (as in the example report): ADH1B rs1229984 CC, ALDH2 rs671 GG.

ADH1B*2 (rs1229984 T) accelerates the formation of acetaldehyde and ALDH2*2 (rs671 A) delay its elimination; both increase the sensitivity to alcohol.

Example report: HIGHER · Category: NUTRITION · Mechanisms: DNA Damage and DNA Repair Deficit

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ADH1B	rs1229984	CC	imputation DR2=0.88	T (+)	0	ADH1B*2: faster acetaldehyde
ALDH2	rs671	GG	chip	A (+)	0	ALDH2*2: accumulation of acetaldehyde

Calcium · Calcium

INCREASED NEED increased need

PGS001893 (PGS Catalog): 46 656 of 48 395 variants with genotype (cover 96%). the 22nd percentile in the reference population (n=633); 0% of the reference population has the same score.

Polygenic serum calcium score. A low score (lower genetic calcium) is read as an increased need for food calcium.

Example report: INCREASED NEED · Category: NUTRITION · Mechanisms: Osteoporosis

Undefined effect allele (34)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ALDH7A1	rs13182402	AA	chip	—	—	
FAM216B	rs9525667	TT	imputation DR2=0.99	—	—	
GAL	rs880610	GG	imputation DR2=1.00	—	—	
CDC42SE1	rs2864700	TT	imputation DR2=0.94	—	—	
CTNNB1	rs389264	CT	imputation DR2=0.98	—	—	
RSPO3	rs9482772	TT	imputation DR2=1.00	—	—	
GALNT3	rs10204976	CG	imputation DR2=1.00	—	—	
CPED1	rs10242100	AG	chip	—	—	
MRPL20	rs12408050	GG	imputation DR2=0.98	—	—	
ARL4C	rs12151790	GG	imputation DR2=0.88	—	—	

DOK6	rs17184557	TT	chip	—	—
FKBP11	rs3741619	GA	imputation DR2=0.97	—	—
ETS2	rs11088458	GG	imputation DR2=0.97	—	—
HBZ	rs10794639	AA	imputation DR2=1.00	—	—
SMOC1	rs3742909	GG	imputation DR2=1.00	—	—
HOXC6	rs7308105	TC	imputation DR2=1.00	—	—
TNFRSF11B	rs2062375	GC	chip	—	—
MARCO	rs115242848	CC	chip	—	—
PRSS3	rs10814041	GG	imputation DR2=0.98	—	—
MN1	rs139959245	CC	imputation DR2=0.91	—	—
NUDT2	rs307646	AA	imputation DR2=1.00	—	—
CDH6	rs2173682	AG	imputation DR2=1.00	—	—
IDH3A	rs2028548	CT	imputation DR2=0.99	—	—
HLA-DQA1	rs2071805	CT	imputation DR2=1.00	—	—
MECOM	rs784288	AA	chip	—	—
VAR52	rs9262558	CC	chip	—	—
SP7	rs144680237	CC	imputation DR2=0.99	—	—
CPED1	rs3779381	AG	chip	—	—
ZBTB40	rs34414754	AC	imputation DR2=1.00	—	—
GAL	rs56154705	CC	imputation DR2=0.99	—	—
WLS	rs2566755	TT	chip	—	—
SEM1	rs4448201	CC	imputation DR2=0.98	—	—
CCDC170	rs4869744	TT	imputation DR2=0.99	—	—
—	rs12808199	GG	chip	—	—

SNP not found in the data (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
AMT	rs34240317	—	SNP not found in the data	—	—	
STK39	rs578031265	—	SNP not found in the data (Imputed DR2=0.02, below the 0.8 threshold)	—	—	
ARRDC3	rs7733007	—	SNP not found in the data (Imputed DR2=0.67, below the 0.8 threshold)	—	—	

FUT2 (Gut/vitamin B12) · FUT2 (dentestatin/vitamin B12)

SECRETOR secretor

Genotype rule (as in the example report): rs601338 GA: secretor.

FUT2 places antigens of the blood group in the mucous membranes. Non-secretors (rs601338 AA) have higher B12, different microbiota (less bifidobacteria) and protection against norovirus.

Example report: SECRETOR · Category: NUTRITION · Mechanisms: Dysbiosis; Intestinal Barrier dysfunction

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FUT2	rs601338	GA	chip	A (+)	1	W154X: non-secretor in AA

Undefined effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FUT2	rs1047781	AA	chip	—	—	

Carbohydrate Metabolism · Metabolism of carbohydrates

TYPICAL typical

8 score of 8 variants (signal effect alleles). the 61st percentile (mean position) in the reference population (n=633); 16% of the reference population has exactly the same score.

Variants (TCF7L2, IRS1, PPARG, FTO, SLC2A2, AMY...) linked to the glycemic response to carbohydrates.

Example report: TYPICAL · Category: BLOOD SUGAR CONTROL · Mechanisms: Hyperglycemia and Intolerance to Glucose; Insulin resistance

Alleles that pull to «WORSE» (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TCF7L2	rs7903146	CT	chip	T (+)	1	TCF7L2
IRS1	rs2943641	TC	chip	C (+)	1	IRS1 (insulin resistance)
FTO	rs1121980	GA	chip	A (+)	1	GWAS: adiposity (Pulit SL; p=1e-206); unfavourable A allele
FTO	rs9939609	TA	chip	A (+)	1	FTO
FABP2	rs1799883	TC	imputation DR2=1.00	T (+)	1	FABP2 Thr54

No copies of the effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SLC2A2	rs5400	GG	chip	A (+)	0	SLC2A2 Ile110
PPARG	rs1801282	CC	chip	G (–)	0	PPARG Ala12
PPARG	rs3856806	CC	chip	T (+)	0	GWAS: Waist-hip index (Christakoudi S; p=2e-08); unfavourable T allele

Undefined effect allele (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
AMY2B	rs4244372	TT	chip	—	—	
TAS1R2	rs35874116	CC	chip	—	—	
ADRB2	rs1042714	GC	imputation DR2=1.00	—	—	
ANKK1	rs1800497	GG	chip	—	—	
LEPR	rs1137101	AG	chip	—	—	
CETP	rs5883	CC	chip	—	—	
NLRC5	rs3764261	CC	chip	—	—	
ADRB3	rs4994	AA	chip	—	—	
RFC4	rs17300539	GG	chip	—	—	

Saturated Fat · saturated fat

TYPICAL RESPONSE typical response

21 score of 21 variants (signal effect alleles). the 46th percentile (mean position) in the reference population (n=633); 9% of the reference population has exactly the same score.

Variants (APOE, APOA2, FTO, PPARG, CETP, LPL, TCF7L2...) that modify the effect of saturated fat on LDL and adiposity. Approach: unfavourable alleles for lipids and adiposity.

Example report: TYPICAL RESPONSE · Category: HEART & BLOOD VESSELS · Mechanisms: Vascular Aging; Dysfunction of adipose tissue

Alleles that pull to «WORSE RESPONSE» (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TCF7L2	rs7903146	CT	chip	C (+)	1	GWAS: adiposity (Huang J; p=2e-66); unfavourable C allele
FTO	rs1121980	GA	chip	A (+)	1	GWAS: adiposity (Pulit SL; p=1e-206); unfavourable A allele

CETP	rs5882	AA	chip	A (+)	2	GWAS: High density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); unfavourable A allele
APOB	rs693	GA	chip	A (+)	1	APOB XbaI
FTO	rs1558902	TA	chip	A (+)	1	GWAS: adiposity (Akiyama M; p=2e-223); unfavourable A allele
FTO	rs17817449	TG	chip	G (+)	1	GWAS: High density lipoprotein cholesterol levels (Sinnott-Armstrong N; p=2e-09); unfavourable G allele
LPL	rs13702	TC	chip	T (+)	1	GWAS: Triglycerides (Hoffmann TJ; p=5e-177); unfavourable T allele
FCER1G	rs5082	GA	chip	G (+)	1	APOA2 -265C (CC: more adiposity with saturated fat)
FTO	rs9939609	TA	chip	A (+)	1	FTO

Alleles that pull to «BETTER RESPONSE» (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SIDT2	rs2854117	CC	imputation DR2=0.98	C (–)	2	GWAS: Triglyceride levels (UKB data field 30870) (Loya H; p=8e-99); favourable C allele
FTO	rs1421085	TC	chip	T (–)	1	GWAS: adiposity (Sidorenko J; p<1e-300); favourable T allele
PCSK7	rs662799	AA	chip	A (–)	2	GWAS: Triglyceride levels (Graham SE; p<1e-300); favourable allele A
SIDT2	rs964184	GC	chip	C (–)	1	GWAS: VLDL cholesterol levels (Zoodma M; p<1e-300); favourable C allele

No copies of the effect allele (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
APOE	rs429358	TT	imputation DR2=0.99	C (+)	0	APOE ε4
ADAM10	rs2070895	GG	chip	A (+)	0	GWAS: Total cholesterol levels (Jee YH; p<1e-300); unfavourable A allele
PPARG	rs1801282	CC	chip	G (–)	0	PPARG Ala12
ADAM10	rs1800588	CC	chip	T (+)	0	GWAS: Triglyceride levels in HDL (Karjalainen MK; p<1e-300); unfavourable T allele
APOE	rs7412	CC	chip	T (–)	0	APOE ε2
PPARG	rs3856806	CC	chip	T (+)	0	GWAS: Waist-hip index (Christakoudi S; p=2e-08); unfavourable T allele
LPL	rs328	CC	chip	G (–)	0	GWAS: Triglyceride levels (Graham SE; p<1e-300); favourable G allele
MC4R	rs12970134	GG	chip	A (+)	0	GWAS: Triglycerides to Total Lipids in Medium HDL percentile (Zoodma M; p=1e-15); unfavourable A allele

Undefined effect allele (23)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TLR4	rs5030728	GG	chip	—	—	
PPARA	rs135549	CC	chip	—	—	
APOA1	rs670	CC	chip	—	—	
ABCA1	rs2230806	TT	chip	—	—	
SIDT2	rs5070	AG	imputation DR2=0.99	—	—	
STAT6	rs1799986	CC	chip	—	—	
STAT3	rs2293152	GC	chip	—	—	
AHSG	rs4917	CC	chip	—	—	
CD36	rs1984112	GG	chip	—	—	
CLOCK	rs1801260	AG	chip	—	—	
CLOCK	rs4580704	CC	imputation DR2=1.00	—	—	
PKDREJ	rs4253778	GC	chip	—	—	

PEX11A	rs894160	TT	chip	—	—
AGT	rs699	AG	chip	—	—
ACE	rs4343	GA	chip	—	—
APOC1	rs405509	GG	chip	—	—
PPARA	rs1800206	CC	chip	—	—
MED24	rs1568400	TT	imputation DR2=0.99	—	—
PPARG	rs10865710	CC	chip	—	—
STAT3	rs8069645	AG	chip	—	—
STAT3	rs744166	GG	chip	—	—
LPL	rs1121923	GG	chip	—	—
STAT3	rs1053005	TC	chip	—	—

Unsaturated Fat · Unsaturated Fat

TYPICAL RESPONSE typical response

12 score of 12 variants (signal effect alleles). the 68th percentile (mean position) in the reference population (n=633); 0% of the reference population has exactly the same score.

Variants (FADS1/2, PPARG, APOA1, PNPLA3, IQCJ...) that modify the benefit of unsaturated fats. Approximation: unfavourable alleles for lipids/blood glucose.

Example report: TYPICAL RESPONSE · Category: HEART & BLOOD VESSELS · Mechanisms: Vascular Aging; Inflammaging

Alleles that pull to «WORSE RESPONSE» (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FTO	rs9939609	TA	chip	A (+)	1	GWAS: Type 2 diabetes (Perry JR; p=1e-20); unfavourable A allele
FADS2	rs1535	AA	chip	A (+)	2	GWAS: LDL cholesterol levels (Spracklen CN; p=5e-45); unfavourable A allele
TCF7L2	rs7903146	CT	chip	T (+)	1	TCF7L2
BDNF	rs6265	CT	chip	T (+)	1	GWAS: Type 2 diabetes (Elashi AA; p=3e-08); unfavourable T allele
PNPLA3	rs738409	CG	chip	G (+)	1	PNPLA3 I148M
NECTIN2	rs1064725	TT	imputation DR2=0.81	T (+)	2	GWAS: Triglyceride to HDL cholesterol ratio (Oliveri A; p=6e-23); unfavourable T allele

Alleles that pull to «BETTER RESPONSE» (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FADS2	rs174550	TT	chip	T (—)	2	GWAS: Triglycerides to total lipids ratio in very large HDL (Karjalainen MK; p=3e-67); favourable T allele
GCKR	rs1260326	TC	chip	C (—)	1	GWAS: Triglyceride levels (Graham SE; p<1e-300); favourable C allele

No copies of the effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PPARG	rs1801282	CC	chip	G (—)	0	PPARG Ala12
MAU2	rs58542926	CC	chip	T (—)	0	GWAS: Total cholesterol levels (Graham SE; p<1e-300); favourable T allele
APOE	rs429358	TT	imputation DR2=0.99	C (+)	0	APOE ε4
FADS2	rs174546	CC	chip	T (+)	0	GWAS: Triglycerides to total lipids ratio in very small VLDL (Karjalainen MK; p=1e-204); unfavourable T allele

Undefined effect allele (30)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
RFC4	rs17300539	GG	chip	—	—	
LEPR	rs1805094	GG	chip	—	—	
FAAH	rs324420	CA	chip	—	—	
APOA1	rs670	CC	chip	—	—	
PON1	rs662	TT	chip	—	—	
ABCG1	rs4148102	GG	imputation DR2=0.97	—	—	
MTHFR	rs1801131	TG	chip	—	—	
TNF	rs1800629	GG	chip	—	—	
PPARA	rs1800206	CC	chip	—	—	
FABP2	rs1799883	TC	imputation DR2=1.00	—	—	
MGLL	rs9877819	GG	imputation DR2=0.95	—	—	
KCNMB3	rs7645550	CT	chip	—	—	
ABTB1	rs6776142	CC	imputation DR2=1.00	—	—	
ABTB1	rs549662	AA	imputation DR2=0.99	—	—	
ST6GAL1	rs3774261	AA	chip	—	—	
SIDT2	rs2854116	TT	chip	—	—	
RFC4	rs266729	CC	chip	—	—	
IQCJ	rs2044704	GG	imputation DR2=0.99	—	—	
IQCJ	rs1962071	AA	imputation DR2=0.97	—	—	
ANKK1	rs1800497	GG	chip	—	—	
ABTB1	rs13076593	CC	imputation DR2=0.90	—	—	
PCSK5	rs1029035	TT	imputation DR2=1.00	—	—	
UCP3	rs1800849	GG	chip	—	—	
ABTB1	rs782444	TC	imputation DR2=0.97	—	—	
TPRA1	rs6787155	CA	chip	—	—	
ABTB1	rs555183	AG	imputation DR2=1.00	—	—	
ADRB3	rs4994	AA	chip	—	—	
PODXL2	rs3773155	AA	imputation DR2=0.98	—	—	
ABTB1	rs1466571	GG	chip	—	—	
ADAM17	rs10495563	GA	chip	—	—	

Vitamin A · Vitamin A

TYPICAL NEED typical necessity

4 score of 4 variants (signal effect alleles). the 46th percentile (mean position) in the reference population (n=633); 26% of the reference population has exactly the same score.

BCO1 converts beta-carotene into retinol. R267S (rs12934922 T) and A379V (rs7501331 T) reduce conversion (up to ~50–70% in carriers of both).

Example report: TYPICAL NEED · Category: NUTRITION · Mechanisms: Immunosenescence

Alleles that pull to «INCREASED NEED» (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FFAR4	rs10882272	TC	chip	C (+)	1	GWAS: Retinol levels (Mondul AM; p=7e-15); unfavourable C allele
BCO1	rs12934922	AT	chip	T (+)	1	BCO1 R267S
C16ORF46	rs6564851	TG	chip	T (+)	1	BCO1 (less conversion)

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
BCO1	rs7501331	CC	chip	T (+)	0	BCO1 A379V

Undefined effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
C16ORF46	rs11645428	GA	chip	—	—	
BCO1	rs6420424	GA	chip	—	—	
BCO1	rs4889294	TC	chip	—	—	

Vitamin B12 · Vitamin B12

TYPICAL NEED typical necessity

PGS003158 (PGS Catalog): 1 082 449 of 1 113 819 variants with genotype (97% coverage). the 34th percentile in the reference population (n=633); 0% of the reference population has the same score.

Polygenic score of B12 levels (FUT2, TCN1, CUBN, MMAA, CLYBL...). Low score = lower genetic B12 : increased need.

Example report: TYPICAL NEED · Category: NUTRITION · Mechanisms: Epigenetic Changes; Mitochondrial dysfunction

Alleles that pull to «DECREASED NEED» (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FUT2	rs602662	GA	chip	A (+)	1	in PGS003158: allele A increases the score
CUBN	rs11254363	AG	chip	G (+)	1	in PGS003158: G allele increases the score
TCN1	rs526934	GA	chip	A (+)	1	in PGS003158: allele A increases the score
FUT3	rs708686	CT	chip	T (+)	1	in PGS003158: T allele increases the score
CUBN	rs1801222	GG	chip	G (+)	2	in PGS003158: G allele increases the score
RGS7	rs7544372	TT	chip	T (+)	2	in PGS003158: T allele increases the score
—	rs1990193	CA	chip	A (+)	1	in PGS003158: allele A increases the score
TCN2	rs9606756	AG	chip	G (+)	1	in PGS003158: G allele increases the score

Alleles that pull to «INCREASED NEED» (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FUT2	rs601338	GA	chip	A (−)	1	in PGS003158: allele A decreases the score
FUT2	rs516246	CT	chip	T (−)	1	in PGS003158: T allele decreases the score
CBS	rs234706	GA	chip	A (−)	1	in PGS003158: allele A decreases the score
MMAB	rs7134594	CT	chip	T (−)	1	in PGS003158: T allele decreases the score
TCN1	rs34324219	CC	chip	C (−)	2	GWAS: Other vitamin B12 deficiency anemia (PheCode 281.12) (Verma A; p=6e-62); C allele decreases the trait

No copies of the effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FUT2	rs1047781	AA	chip	T (+)	0	GWAS: Vitamin B12 in pregnancy (Liu S; p=1e-43); allele T increases the trait
ABCD4	rs4148077	CC	imputation DR2=1.00	T (−)	0	in PGS003158: T allele decreases the score
FUT5	rs3760775	GG	chip	T (+)	0	in PGS003158: T allele increases the score

Undefined effect allele (25)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TCN2	rs1801198	GC	chip	—	—	
MTRR	rs1801394	AA	chip	—	—	

FAM240C	rs12478296	CC	imputation DR2=0.99	—	—
SLC25A2	rs3749779	AA	chip	—	—
FOXK1	rs314590	AA	imputation DR2=0.94	—	—
CFAP299	rs1385890	GA	imputation DR2=0.99	—	—
LAMA4	rs76190642	GG	imputation DR2=0.96	—	—
ARAP2	rs142554771	TT	imputation DR2=0.94	—	—
LAMA4	rs144505878	GG	imputation DR2=0.95	—	—
RGS18	rs114973754	CC	imputation DR2=1.00	—	—
ADGRL3	rs545255284	TT	imputation DR2=0.98	—	—
C16ORF82	rs139645308	CC	imputation DR2=0.85	—	—
POU3F3	rs188141458	GG	imputation DR2=0.88	—	—
KCNK2	rs72761546	TT	imputation DR2=0.80	—	—
KCNK2	rs189754522	AA	imputation DR2=0.81	—	—
PCSK2	rs141477158	GG	imputation DR2=1.00	—	—
LRRC6	rs117429467	AA	imputation DR2=0.92	—	—
SPATA18	rs142766122	CC	imputation DR2=0.95	—	—
SRRM4	rs73215576	CC	chip	—	—
MICA	rs556990455	GG	imputation DR2=1.00	—	—
CADM2	rs188586547	AA	imputation DR2=1.00	—	—
CENPF	rs72759663	GG	imputation DR2=0.80	—	—
SMYD3	rs148487271	TT	imputation DR2=0.91	—	—
DACT1	rs118119041	GG	imputation DR2=0.90	—	—
MMUT	rs9473555	GG	chip	—	—

SNP not found in the data (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ADGRE1	rs62123070	—	SNP not found in the data (Imputed DR2=0.43, below the 0.8 threshold)	—	—	
—	rs1513859	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—	
CHODL	rs34988353	—	SNP not found in the data (Imputed DR2=0.74, below the 0.8 threshold)	—	—	
C1QL3	rs79770840	—	SNP not found in the data (Imputed DR2=0.60, below the 0.8 threshold)	—	—	
TMEM179	rs79885401	—	SNP not found in the data (Imputed DR2=0.76, below the 0.8 threshold)	—	—	
STT3B	rs188968123	—	SNP not found in the data (Imputed DR2=0.17, below the 0.8 threshold)	—	—	
HSPB7	rs144839376	—	SNP not found in the data (Imputed DR2=0.76, below the 0.8 threshold)	—	—	
AKAIN1	rs7239302	—	SNP not found in the data (Imputed DR2=0.09, below the 0.8 threshold)	—	—	
ST8SIA6	rs188363440	—	SNP not found in the data (Imputed DR2=0.74, below the 0.8 threshold)	—	—	

Vitamin C · Vitamin C

TYPICAL NEED typical necessity

10 score of 11 variants (signal effect alleles). the 69th percentile (mean position) in the reference population (n=633); 0% of the reference population has exactly the same score.

The SLC23A1 transporter (SVCT1) and other GWAS loci influence plasma ascorbate.

Alleles that pull to «DECREASED NEED» (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
GLB1L	rs13028225	TC	imputation DR2=1.00	T (–)	1	GWAS: Vitamin C levels (Zheng JS; p=2e-30); favourable T allele
AKT1	rs10136000	GA	imputation DR2=0.97	A (–)	1	GWAS: Vitamin C levels (Zheng JS; p=1e-08); favourable allele A
SYCP3	rs2559850	GA	imputation DR2=0.99	A (–)	1	GWAS: Vitamin C levels (Zheng JS; p=6e-20); favourable allele A
MAF	rs56738967	CC	imputation DR2=0.99	C (–)	2	GWAS: Vitamin C levels (Zheng JS; p=8e-10); favourable C allele
TBX2	rs9895661	TT	chip	T (–)	2	GWAS: Vitamin C levels (Zheng JS; p=1e-14); favourable T allele
GSTA1	rs7740812	GA	imputation DR2=1.00	G (–)	1	GWAS: Vitamin C levels (Zheng JS; p=2e-09); favourable G allele

No copies of the effect allele (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
LTA4H	rs117885456	GG	imputation DR2=0.88	A (–)	0	GWAS: Vitamin C levels (Zheng JS; p=2e-11); favourable allele A
FADS2	rs174547	TT	chip	C (–)	0	GWAS: Vitamin C levels (Zheng JS; p=4e-08); favourable C allele
SLC23A1	rs33972313	CC	chip	T (+)	0	SLC23A1 V264M
RER1	rs6693447	GG	chip	T (–)	0	GWAS: Vitamin C levels (Zheng JS; p=6e-10); favourable T allele
RGS14	rs10051765	TT	imputation DR2=1.00	C (–)	0	GWAS: Vitamin C levels (Zheng JS; p=4e-09); favourable C allele

Undefined effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SLC23A1	rs4257763	GG	chip	—	—	
GSTO2	rs156697	AA	imputation DR2=1.00	—	—	

Vitamin D · Vitamin D

DECREASED NEED decreased need

PGS002721 (PGS Catalog): 960 019 of 981 998 variants with genotype (cover 98%). the 84th percentile in the reference population (n=633); 0% of the reference population has the same score.

Polygenic score of 25(OH)D (GC, DHCR7, CYP2R1, CYP24A1...). Low score = lower genetic vitamin D : increased need.

Example report:
TYPICAL NEED · Category: NUTRITION · Mechanisms: Osteoporosis; Immunosenescence

Alleles that pull to «DECREASED NEED» — in the direction of the result (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
COPB1	rs12794714	GG	chip	G (+)	2	in PGS002721: G allele increases the score
GC	rs2282679	TT	chip	T (+)	2	in PGS002721: T allele increases the score
COPB1	rs10741657	AG	chip	A (+)	1	GWAS: Vitamin D levels (Jiang X; p=2e-46); allele A increases the trait
CYP1B1	rs1800440	TT	chip	T (+)	2	in PGS002721: T allele increases the score
VDR	rs1544410	CT	chip	C (+)	1	in PGS002721: C allele increases the score
GC	rs4588	GG	chip	G (+)	2	in PGS002721: G allele increases the score

Alleles that pull to «INCREASED NEED» — in the opposite direction to the result (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
COPB1	rs2060793	AG	chip	A (—)	1	in PGS002721: allele A decreases the score
COPB1	rs117913124	GA	chip	A (—)	1	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p<1e-300); allele A decreases the trait
VDR	rs731236	AG	chip	A (—)	1	in PGS002721: allele A decreases the score
ADH1B	rs1229984	CC	imputation DR2=0.88	C (—)	2	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=5e-13); allele C decreases the trait
CYP2R1	rs117576073	GG	chip	G (—)	2	GWAS: Serum 25-Hydroxyvitamin D levels (conditioned to adiposity) (Revez JA; p=3e-64); G allele decreases the trait
PDE3B	rs188480917	CG	imputation DR2=0.99	G (—)	1	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=5e-275); allele G decreases the trait

No copies of the effect allele (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
COPB1	rs10832289	AA	imputation DR2=0.98	T (—)	0	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=2e-266); T allele decreases the trait
GC	rs7041	CC	chip	A (—)	0	in PGS002721: allele A decreases the score
GC	rs11723621	AA	imputation DR2=1.00	G (—)	0	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p<1e-300); allele G decreases the trait
PDE3B	rs201501563	CC	imputation DR2=1.00	T (—)	0	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=9e-67); T allele decreases the trait
PSMA1	rs577185477	TT	imputation DR2=1.00	C (—)	0	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p<1e-300); allele C decreases the trait
NADSYN1	rs12785878	TT	chip	G (—)	0	in PGS002721: G allele decreases the score
GC	rs3775150	TT	imputation DR2=0.98	C (—)	0	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=4e-295); allele C decreases the trait
NPFFR2	rs143106299	AA	imputation DR2=0.99	T (—)	0	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=2e-28); T allele decreases the trait

Undefined effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
VDR	rs2228570	AG	chip	—	—	

SNP not found in the data (17)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
—	rs189918701	—	SNP not found in the data (Imputed DR2=0.07, below the 0.8 threshold)	G (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=2e-16); allele G decreases the trait
—	rs558560635	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	G (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=6e-16); allele G decreases the trait
—	rs375984409	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	G (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=3e-38); allele G decreases the trait
PDE3B	rs571484036	—	SNP not found in the data	A (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=4e-16); allele A decreases the trait
GC	rs222026	—	SNP not found in the data	T (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=7e-68); T allele decreases the trait
RRAS2	rs117206369	—	SNP not found in the data (Imputed DR2=0.73, below the 0.8 threshold)	T (+)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=1e-48); allele T increases the trait
—	rs201561609	—		T (—)	—	

			SNP not found in the data (Imputed DR2=0.14, below the 0.8 threshold)			GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=7e-28); T allele decreases the trait
GC	rs113938679	—	SNP not found in the data (Imputed DR2=0.12, below the 0.8 threshold)	A (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=6e-36); allele A decreases the trait
—	rs561089663	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	G (+)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=5e-43); allele G increases the trait
—	rs557657187	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	G (+)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=6e-16); allele G increases the trait
PSMA1	rs554808052	—	SNP not found in the data	C (+)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=5e-40); allele C increases the trait
GC	rs565277381	—	SNP not found in the data (Imputed DR2=0.02, below the 0.8 threshold)	T (+)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=7e-11); allele T increases the trait
—	rs567415847	—	SNP not found in the data (Imputed DR2=0.54, below the 0.8 threshold)	G (+)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=1e-14); allele G increases the trait
—	rs529640451	—	SNP not found in the data (Imputed DR2=0.07, below the 0.8 threshold)	C (+)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=2e-17); allele C increases the trait
NADSYN1	rs536006581	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	G (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=9e-35); allele G decreases the trait
COPB1	rs148514005	—	SNP not found in the data (Imputed DR2=0.18, below the 0.8 threshold)	T (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=1e-184); T allele decreases the trait
—	rs185433896	—	SNP not found in the data	A (—)	—	GWAS: Serum 25-Hydroxyvitamin D levels (Manousaki D; p=2e-38); allele A decreases the trait

Choline · Colina

DECREASED NEED decreased need

8 score of 8 variants (signal effect alleles). the 5th percentile (mean position) in the reference population (n=633); 5% of the reference population has exactly the same score.

PEMT (hepatic synthesis of phosphatidylcholine), CHDH, CHKA, MTHFD1 and MTRR modify choline requirement. Effect alleles as in the example report.

Example report: TYPICAL NEED · Category: NUTRITION · Mechanisms: Dysfunction of adipose tissue; Epigenetic Changes

Alleles that pull to «INCREASED NEED» — in the opposite direction to the result (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CHDH	rs9001	TT	chip	T (+)	2	CHDH
PEMT	rs7946	CT	chip	T (+)	1	PEMT V175M
ALDH3B1	rs10791957	AA	chip	A (+)	2	CHKA

No copies of the effect allele (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTRR	rs1532268	CC	chip	T (+)	0	MTRR S175L
MTHFD1	rs2236225	GG	chip	A (+)	0	MTHFD1 1958A
MTRR	rs1801394	AA	chip	G (+)	0	MTRR 66G
PEMT	rs12325817	CC	chip	G (+)	0	PEMT
CHDH	rs12676	CC	chip	A (+)	0	CHDH

Undefined effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
BHMT	rs3733890	GA	chip	—	—	
PEMT	rs4646343	GG	chip	—	—	
PEMT	rs3760188	CC	chip	—	—	
CHKA	rs7928739	AA	chip	—	—	

Iron Overload · Iron overload

TYPICAL LIKELIHOOD typical probability

Genotype rule (as in the example report): C282Y rs1800562 GG, H63D rs1799945 CG — H63D heterozygous; target ferritin <100.

Hereditary hemochromatosis is mainly due to HFE C282Y homozygous (rs1800562 AA). H63D (rs1799945 G) and S65C (rs1800730 T) have small effect, except in combination with C282Y.

Example report: TYPICAL LIKELIHOOD · Category: NUTRITION · Mechanisms: Oxidative stress; Autophagy / Deregulated Macroautophagy

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
HFE	rs1799945	CG	chip	G (+)	1	HFE H63D

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
HFE	rs1800562	GG	chip	A (+)	0	HFE C282Y
HFE	rs1800730	AA	chip	T (+)	0	HFE S65C

Undefined effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
BMP2	rs235756	GG	imputation DR2=0.86	—	—	
TF	rs1049296	CC	chip	—	—	

HLA-DQ (Gluten) · HLA-DQ (gluten)

TYPICAL GENETICS typical genetics

Genotype rule (as in the example report): rs2187668 CC (DQ2.5), rs7454108 TT (DQ8).

Almost all celiac patients have HLA-DQ2.5 (tag rs2187668 T) or DQ8 (tag rs7454108 C). Without these haplotypes, celiac disease is very unlikely.

Example report: TYPICAL GENETICS · Category: INFLAMMATION & AUTOIMMUNITY · Mechanisms: Intestinal Barrier dysfunction; Chronic Inflammation

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
HLA-DQA1	rs2187668	CC	chip	T (+)	0	HLA-DQ2.5 tag
HLA-DQA2	rs7454108	TT	chip	C (+)	0	HLA-DQ8 tag

Undefined effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
HLA-DQB1	rs2858331	AG	chip	—	—	

Coffee-Related Sleep Problems · Sleep problems related to caffeine

LESS LIKELY less probable

2 score of 2 variants (signal effect alleles). the 13th percentile (mean position) in the reference population (n=633); 26% of the reference population has exactly the same score.

Variants of a GWAS caffeine-induced insomnia and ADORA2A (receptor A2A adenosine).

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ADORA2A	rs5751876	CC	chip	T (+)	0	ADORA2A 1976T
MDFIC2	rs7628219	CC	chip	T (+)	0	GWAS: Insomnia (caffeine-induced) (Byrne EM; p=2e-06); unfavourable T allele

Undefined effect allele (17)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
NEBL	rs9665295	CC	chip	—	—	
ALDH7A1	rs13172305	GG	imputation DR2=0.97	—	—	
FCHSD2	rs1791933	CC	chip	—	—	
KCTD15	rs11878836	TT	imputation DR2=0.99	—	—	
GBP7	rs521704	CA	imputation DR2=1.00	—	—	
ADGRL2	rs12725617	CC	imputation DR2=0.99	—	—	
COMT	rs4680	GG	chip	—	—	
PRIMA1	rs6575353	AA	chip	—	—	
PRIMA1	rs12895096	GG	imputation DR2=0.96	—	—	
—	rs16905439	CC	chip	—	—	
TMEM51	rs2103117	GG	chip	—	—	
GBP3	rs12407812	CT	imputation DR2=0.98	—	—	
ADRA2A	rs2065779	GG	chip	—	—	
SLC7A1	rs2388082	CC	chip	—	—	
MTNR1B	rs10830964	CT	imputation DR2=0.95	—	—	
LRTM1	rs11706236	AA	chip	—	—	
NEDD4L	rs158856	CC	chip	—	—	

Salt Sensitivity
·
Salt Sensitivity

TYPICAL
typical

5 score of 5 variants (signal effect alleles). the 49th percentile (mean position) in the reference population (n=633); 29% of the reference population has exactly the same score.

Variants of the renin-angiotensin system (AGT, ACE), sodium channels (SCNN1A/G), SGK1, WNK1 and other sodium regulation loci.

Example report:
TYPICAL
·
Category:
HEART & BLOOD VESSELS
·
Mechanisms:
Vascular Aging; Arterial stiffness

Alleles that pull to «HIGHER» (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
POC1B	rs2681472	AA	chip	A (+)	2	GWAS: Systolic cardiovascular load (Takeuchi F; p=1e-20); unfavourable A allele
ADRB2	rs1042714	GC	imputation DR2=1.00	G (+)	1	ADRB2 Glu27
AGT	rs699	AG	chip	G (+)	1	AGT M235T: more angiotensinogen
ACE	rs4343	GA	chip	G (+)	1	ACE (D allele tag)

Alleles that pull to «LOWER» (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FGF5	rs16998073	AA	chip	A (—)	2	GWAS: Systolic cardiovascular load (Jee YH; p=3e-216); favourable allele A

Undefined effect allele (31)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
BCAT1	rs7961152	AC	chip	—	—	
SGK1	rs9389154	GG	imputation DR2=0.80	—	—	
NR2F2	rs2398162	GG	chip	—	—	
PRKG1	rs7905063	CC	imputation DR2=0.99	—	—	
PRKG1	rs7897633	CC	chip	—	—	
SLC4A5	rs7571842	AG	chip	—	—	
SCNN1G	rs4299163	CC	imputation DR2=0.98	—	—	
CLGN	rs2567241	CC	chip	—	—	
RAD52	rs2301880	CC	imputation DR2=0.99	—	—	
WNK1	rs12828016	GT	chip	—	—	
CSTF2T	rs12414562	GA	chip	—	—	
HYAL1	rs10510755	CT	imputation DR2=1.00	—	—	
SCNN1G	rs7404408	CT	imputation DR2=1.00	—	—	
SCNN1G	rs5735	CC	chip	—	—	
SCNN1G	rs4073930	TC	chip	—	—	
SCNN1G	rs4073291	AC	imputation DR2=1.00	—	—	
RENBP	rs78377269	G	chip	—	—	
ACE2	rs714205	G	imputation DR2=0.99	—	—	
SLC24A3	rs3790261	AA	imputation DR2=0.99	—	—	
ACE2	rs2285666	T	chip	—	—	
GSKIP	rs11847625	GG	imputation DR2=0.97	—	—	
SGK1	rs9376026	TT	imputation DR2=0.86	—	—	
RAD52	rs880054	CT	chip	—	—	
SLC8A1	rs434082	CC	imputation DR2=0.96	—	—	
SLC8A1	rs11893826	GA	imputation DR2=0.97	—	—	
CSTF2T	rs10997916	GG	chip	—	—	
SLC4A5	rs10177833	AC	chip	—	—	
SLC4A4	rs10022637	TT	imputation DR2=0.98	—	—	
KL	rs9536314	TT	chip	—	—	
SCNN1G	rs4499238	TT	imputation DR2=1.00	—	—	
SCNN1A	rs3741914	CC	imputation DR2=0.92	—	—	

SNP not found in the data (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ACE2	rs184874220	—	SNP not found in the data	—	—	
SCNN1A	rs4764586	—	SNP not found in the data (Imputed DR2=0.77, below the 0.8 threshold)	—	—	
GC	rs4254735	—	SNP not found in the data (Imputed DR2=0.11, below the 0.8 threshold)	—	—	
CPA3	rs75367686	—	SNP not found in the data (Imputed DR2=0.08, below the 0.8 threshold)	—	—	
TNFRSF1A	rs11614164	—	SNP not found in the data (Imputed DR2=0.70, below the 0.8 threshold)	—	—	

MTRR (Methylation) · MTRR (methylation)

TYPICAL ACTIVITY typical activity

Genotype rule (as in the example report): A66G rs1801394 AA.

MTRR regenerates methionine synthase (B12 dependent). A66G (rs1801394 G) slightly reduces affinity; the effect on homocysteine is small and depends on B12.

Example report: TYPICAL ACTIVITY · Category: NUTRITION · Mechanisms: Epigenetic Changes

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTRR	rs1801394	AA	chip	G (+)	0	MTRR 66G

Undefined effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MTRR	rs1532268	CC	chip	—	—	

VDR (Vitamin D) · VDR (vitamin D)

TYPICAL ACTIVITY typical activity

1 score of 1 variants (signal effect alleles). the 62nd percentile (mean position) in the reference population (n=633); 45% of the reference population has exactly the same score.

Vitamin D receptor. FokI (rs2228570 A = f allele, longer and less active protein) is the only one with clear functional effect; BsmI/TaqI/ApaI are markers.

Example report: TYPICAL ACTIVITY · Category: NUTRITION · Mechanisms: Osteoporosis; Immunosenescence

Alleles that pull to «LOWER ACTIVITY» (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
VDR	rs2228570	AG	chip	A (+)	1	FokI f (less active)

Undefined effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
VDR	rs1544410	CT	chip	—	—	
VDR	rs7975232	AA	chip	—	—	
VDR	rs731236	AG	chip	—	—	

PUFA Response · Response to polyunsaturated fat

TYPICAL typical

10 score of 10 variants (signal effect alleles). the 52nd percentile (mean position) in the reference population (n=633); 0% of the reference population has exactly the same score.

Variants (FADS1/2, IQCJ, NXPH1, MYB, PPARA...) that modify the lipid response to omega-3/omega-6. Approximation: unfavourable alleles for lipids/blood glucose.

Example report: TYPICAL · Category: HEART & BLOOD VESSELS · Mechanisms: Inflammaging; Vascular Aging

Alleles that pull to «WORSE» (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
NECTIN2	rs1064725	TT	imputation DR2=0.81	T (+)	2	GWAS: Triglyceride to HDL cholesterol ratio (Oliveri A; p=6e-23); unfavourable T allele
TCF7L2	rs7903146	CT	chip	T (+)	1	TCF7L2
FTO	rs9939609	TA	chip	A (+)	1	GWAS: Type 2 diabetes (Perry JR; p=1e-20); unfavourable A allele
PNPLA3	rs738409	CG	chip	G (+)	1	PNPLA3 I148M

Alleles that pull to «BETTER» (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
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FADS2	rs174550	TT	chip	T (–)	2	GWAS: Triglycerides to total lipids ratio in very large HDL (Karjalainen MK; p=3e-67); favourable T allele
GCKR	rs1260326	TC	chip	C (–)	1	GWAS: Triglyceride levels (Graham SE; p<1e-300); favourable C allele

No copies of the effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MAU2	rs58542926	CC	chip	T (–)	0	GWAS: Total cholesterol levels (Graham SE; p<1e-300); favourable T allele
PPARG	rs1801282	CC	chip	G (–)	0	PPARG Ala12
FADS2	rs174546	CC	chip	T (+)	0	FADS1: less conversion
LPL	rs328	CC	chip	G (–)	0	GWAS: Triglyceride levels (Graham SE; p<1e-300); favourable G allele

Undefined effect allele (38)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IQCJ	rs9827242	TC	imputation DR2=1.00	—	—	
ABTB1	rs782444	TC	imputation DR2=0.97	—	—	
TPRA1	rs6787155	CA	chip	—	—	
ABTB1	rs555183	AG	imputation DR2=1.00	—	—	
ADRB3	rs4994	AA	chip	—	—	
MYB	rs210936	AG	imputation DR2=0.96	—	—	
ADAM17	rs10495563	GA	chip	—	—	
ABCG1	rs4148102	GG	imputation DR2=0.97	—	—	
MTHFR	rs1801131	TG	chip	—	—	
TNF	rs1800629	GG	chip	—	—	
MED24	rs709592	CT	chip	—	—	
PSMD3	rs4065321	CT	chip	—	—	
ACTRT3	rs12696304	CG	chip	—	—	
MGLL	rs9877819	GG	imputation DR2=0.95	—	—	
NXPH1	rs7806226	CA	imputation DR2=0.94	—	—	
NXPH1	rs7805772	GA	chip	—	—	
KCNMB3	rs7645550	CT	chip	—	—	
IQCJ	rs61332355	CC	imputation DR2=0.97	—	—	
RFC4	rs266729	CC	chip	—	—	
IQCJ	rs2044704	GG	imputation DR2=0.99	—	—	
IQCJ	rs1962071	AA	imputation DR2=0.97	—	—	
IQCJ	rs17782879	GG	imputation DR2=0.99	—	—	
IQCJ	rs1449009	AA	imputation DR2=1.00	—	—	
ABTB1	rs13076593	CC	imputation DR2=0.90	—	—	
ALDH8A1	rs11154794	TT	imputation DR2=0.97	—	—	
PCSK5	rs1029035	TT	imputation DR2=1.00	—	—	
ABTB1	rs6776142	CC	imputation DR2=1.00	—	—	
ABTB1	rs549662	AA	imputation DR2=0.99	—	—	
SIDT2	rs2854116	TT	chip	—	—	
ANKK1	rs1800497	GG	chip	—	—	
RFC4	rs182052	GG	chip	—	—	
RFC4	rs17300539	GG	chip	—	—	
APOA1	rs670	CC	chip	—	—	

PON1	rs662	TT	chip	—	—
PODXL2	rs3773155	AA	imputation DR2=0.98	—	—
IQCJ	rs1868414	CT	imputation DR2=0.98	—	—
PPARA	rs1800206	CC	chip	—	—
FABP2	rs1799883	TC	imputation DR2=1.00	—	—

Fat Metabolism · Metabolism of fats

TYPICAL typical

22 score of 22 variants (signal effect alleles). the 63rd percentile (mean position) in the reference population (n=633); 15% of the reference population has exactly the same score.

Variants of lipases, transporters and regulators (LPL, APOA4, FABP1/2, MTP, CD36, PPARA/G, CETP...) that affect the use and storage of fat.

Example report: BETTER · Category: BODY FAT & APPETITE · Mechanisms: Dysfunction of adipose tissue; Metabolic Homeostasis Decline

Alleles that pull to «WORSE» (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TCF7L2	rs7903146	CT	chip	T (+)	1	TCF7L2
MC4R	rs2229616	CC	chip	C (+)	2	GWAS: adiposity (Huang J; p=2e-78); unfavourable C allele
CETP	rs5882	AA	chip	A (+)	2	GWAS: High density lipoprotein cholesterol levels (Sinnott-Armstrong N; p<1e-300); unfavourable A allele
APOB	rs693	GA	chip	A (+)	1	GWAS: Apolipoprotein B levels (Zoddsma M; p=3e-174); unfavourable A allele
LPL	rs13702	TC	chip	T (+)	1	GWAS: Triglycerides (Hoffmann TJ; p=5e-177); unfavourable T allele
FTO	rs9939609	TA	chip	A (+)	1	FTO
FABP2	rs1799883	TC	imputation DR2=1.00	T (+)	1	FABP2 Thr54

Alleles that pull to «BETTER» (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IRS1	rs2943641	TC	chip	T (—)	1	GWAS: High density lipoprotein cholesterol levels (Hoffmann TJ; p=3e-26); favourable T allele
FCER1G	rs5082	GA	chip	A (—)	1	GWAS: Triglycerides in very large HDL (Karjalainen MK; p=7e-27); favourable A allele
PCSK7	rs662799	AA	chip	A (—)	2	GWAS: Triglyceride levels (Graham SE; p<1e-300); favourable allele A
SIDT2	rs964184	GC	chip	C (—)	1	GWAS: VLDL cholesterol levels (Zoddsma M; p<1e-300); favourable C allele

No copies of the effect allele (11)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ADRB3	rs4994	AA	chip	G (+)	0	ADRB3 Arg64
GIPR	rs2287019	CC	chip	T (—)	0	GWAS: adiposity (p=4e-25); favourable T allele
LPL	rs328	CC	chip	G (—)	0	LPL S447X (win)
PPARG	rs1801282	CC	chip	G (—)	0	PPARG Ala12
APOE	rs7412	CC	chip	T (—)	0	GWAS: LDL cholesterol (Sakaue S; p<1e-300); favourable T allele
CETP	rs708272	GG	chip	A (—)	0	GWAS: Triglycerides in very large VLDL (Farashi S; p=2e-18); favourable A allele
APOE	rs429358	TT	imputation DR2=0.99	C (+)	0	APOE ε4

PPARG	rs3856806	CC	chip	T (+)	0	GWAS: Waist-hip index (Christakoudi S; p=2e-08); unfavourable T allele
NSMAF	rs3808607	GG	chip	T (+)	0	GWAS: Total cholesterol levels (MTAG) (Kamiza AB; p=1e-14); unfavourable T allele
ADAM10	rs2070895	GG	chip	A (+)	0	GWAS: Total cholesterol levels (Jee YH; p<1e-300); unfavourable A allele
ADAM10	rs1800588	CC	chip	T (+)	0	GWAS: Triglyceride levels in HDL (Karjalainen MK; p<1e-300); unfavourable T allele

Undefined effect allele (28)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PPARA	rs135549	CC	chip	—	—	
APOA1	rs670	CC	chip	—	—	
ABCA1	rs2230806	TT	chip	—	—	
ACSL5	rs2419621	CC	chip	—	—	
STAT3	rs2293152	GC	chip	—	—	
STAT6	rs1799986	CC	chip	—	—	
MTTP	rs1800591	GG	chip	—	—	
AHSG	rs4917	CC	chip	—	—	
CD36	rs1984112	GG	chip	—	—	
CLOCK	rs1801260	AG	chip	—	—	
PEX11A	rs894160	TT	chip	—	—	
CLOCK	rs3749474	CT	chip	—	—	
LPL	rs1121923	GG	chip	—	—	
TLR4	rs5030728	GG	chip	—	—	
CLOCK	rs4580704	CC	imputation DR2=1.00	—	—	
APOA4	rs5110	CC	imputation DR2=0.94	—	—	
SIDT2	rs5070	AG	imputation DR2=0.99	—	—	
FABP1	rs2241883	CC	chip	—	—	
UCP3	rs1800849	GG	chip	—	—	
TCF7L2	rs12255372	GT	chip	—	—	
RFC4	rs17300539	GG	chip	—	—	
APOC1	rs405509	GG	chip	—	—	
STAT3	rs8069645	AG	chip	—	—	
STAT3	rs744166	GG	chip	—	—	
AGT	rs699	AG	chip	—	—	
MICB	rs361525	GA	chip	—	—	
TNF	rs1800629	GG	chip	—	—	
PPARA	rs1800206	CC	chip	—	—	

Lactose Intolerance · Lactose intolerance

LIKELY TOLERANT probably tolerant

Genotype rule (as in the example report): -13910 rs4988235 GA.

The persistence of lactase is due to the -13910T allele (rs4988235 A in strand +). GG : probable non-persistence (intolerance in adulthood).

Example report: LIKELY TOLERANT · Category: NUTRITION · Mechanisms: Dysbiosis

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
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Caffeine Sensitivity · Caffeine sensitivity

LOWER lower

3 score of 3 variants (signal effect alleles). the 18th percentile (mean position) in the reference population (n=633); 22% of the reference population has exactly the same score.

CYP1A2 (rs762551 C = slower metabolism), AHR (rs4410790 T = less induction) and ADORA2A (rs5751876 T = greater sensitivity).

Example report: LOWER · Category: NUTRITION · Mechanisms: Sleep Disorder

Alleles that pull to «HIGHER» — in the opposite direction to the result (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CYP1A2	rs762551	CA	chip	C (+)	1	C allele (less inducible)

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
AHR	rs4410790	CC	chip	T (+)	0	AHR: less induction
ADORA2A	rs5751876	CC	chip	T (+)	0	ADORA2A

Undefined effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
NAT2	rs1495741	AA	chip	—	—	
COMT	rs4680	GG	chip	—	—	

Caffeine-Related Anxiety · Caffeine Anxiety

LESS LIKELY less probable

Genotype rule (as in the example report): ADORA2A rs5751876 CC.

ADORA2A rs5751876 TT is associated with more anxiety after caffeine in controlled trials.

Example report: LESS LIKELY · Category: NUTRITION · Mechanisms: Sleep Disorder

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ADORA2A	rs5751876	CC	chip	T (+)	0	ADORA2A 1976T: more anxiety with caffeine

CYP1A2 (Detox) · CYP1A2 (detoxification)

TYPICAL ACTIVITY typical activity

Genotype rule (as in the example report): rs762551 CA.

CYP1A2 metabolizes caffeine, melatonin and some drugs. C = slower; *1C (rs2069514 A) reduces induction.

Example report: HIGHER ACTIVITY · Category: NUTRITION · Mechanisms: Sleep Disorder

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CYP1A2	rs762551	CA	chip	A (+)	1	CYP1A2*1F: more inducible (fast metabolism)

Undefined effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
LMAN1L	rs2069514	GG	chip	—	—	

GC (Vitamin D) · GC (vitamin D)

TYPICAL ACTIVITY typical activity

2 score of 2 variants (signal effect alleles). the 27th percentile (mean position) in the reference population (n=633); 55% of the reference population has exactly the same score.

Vitamin D binding protein (GC/DBP) transports 25(OH)D. Alleles associated with lower 25(OH)D count as lower activity (GWAS Catalog).

Example report: HIGHER ACTIVITY · Category: NUTRITION · Mechanisms: Osteoporosis

Alleles that pull to «HIGHER ACTIVITY» (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
GC	rs2282679	TT	chip	T (–)	2	GWAS: Serum 25-Hydroxyvitamin D levels (Hong J; p=4e-63); favourable T allele

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
GC	rs4588	GG	chip	T (+)	0	GWAS: Vitamin D levels (Manousaki D; p=2e-263); unfavourable T allele

Undefined effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
GC	rs7041	CC	chip	—	—	

Results in detail: Body Composition

Bitter Taste Sensitivity · Sensitivity to bitter

TASTER **taster**

Genotype rule (as in the example report): TAS2R38 rs713598 CG, rs1726866 GA, rs10246939 TC — PAV/AVI.

TAS2R38 defines the perception of PTC/PROP. The haplotype PAV (rs713598 G, rs1726866 G, rs10246939 C) gives perception; AVI/AVI = non-tester.

Example report: TASTER · Category: BODY FAT & APPETITE · Mechanisms: Dysbiosis

With the effect allele — in the direction of the result (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TAS2R38	rs713598	CG	chip	G (+)	1	haplotype PAV (probe)
TAS2R38	rs1726866	GA	chip	G (+)	1	haplotype PAV (probe)
TAS2R38	rs10246939	TC	chip	C (+)	1	haplotype PAV (probe)

Sleeping · Petitar

LESS LIKELY **less probable**

6 score of 6 variants (signal effect alleles). the 22nd percentile (mean position) in the reference population (n=633); 23% of the reference population has exactly the same score.

Appetite and satiety variants (FTO, MC4R, LEP/LEPR, SH2B1, PER2/HES6, TCF7L2).

Example report: LESS LIKELY · Category: BODY FAT & APPETITE · Mechanisms: Dysfunction of adipose tissue; Circadian disorders

Alleles that pull to «MORE LIKELY» — in the opposite direction to the result (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TCF7L2	rs7903146	CT	chip	T (+)	1	TCF7L2
FTO	rs9939609	TA	chip	A (+)	1	FTO

No copies of the effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MC4R	rs2229616	CC	chip	T (—)	0	MC4R V103I (protector)
MC4R	rs17782313	TT	chip	C (+)	0	MC4R
LIN7C	rs925946	GG	chip	T (+)	0	GWAS: adiposity (Thorleifsson G; p=9e-10); unfavourable T allele
SH2B1	rs7498665	AA	chip	G (+)	0	SH2B1

Undefined effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
HES6	rs2304672	GG	chip	—	—	
LEP	rs791607	AA	imputation DR2=0.97	—	—	
LEPR	rs2025804	AA	chip	—	—	
LRR4	rs4577902	AG	imputation DR2=1.00	—	—	

ADRB2 (Body Composition) · ADRB2 (body composition)

WORSE GENETICS **genetics less favourable**

Genotype rule (as in the example report): Gln27Glu rs1042714 GC.

The β 2-adrenergic receptor participates in lipolysis. Gln27Glu (rs1042714 G = Glu27) is associated with higher adiposity in populations where it is less frequent.

Example report: WORSE GENETICS · Category: BODY FAT & APPETITE · Mechanisms: Decline of Thermogenesis

With the effect allele — in the direction of the result (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ADRB2	rs1042714	GC	imputation DR2=1.00	G (+)	1	Glu27: worst lipolytic response

UCP1 (Body Composition) · UCP1 (thermogenesis)

LOWER ACTIVITY lower activity

Genotype rule (as in the example report): -3826 rs1800592 TC.

UCP1 is the thermogenic protein of brown fat. -3826A>G (rs1800592 C in strand +) reduces expression.

Example report: LOWER ACTIVITY · Category: BODY FAT & APPETITE · Mechanisms: Decline of Thermogenesis

With the effect allele — contrary to the result (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
UCP1	rs1800592	TC	chip	C (+)	1	-3826G: minus UCP1

PPARG (Metabolism) · PPARG (metabolism)

TYPICAL ACTIVITY typical activity

Genotype rule (as in the example report): Pro12Ala rs1801282 CC.

Pro12Ala (rs1801282 G = Ala12) slightly reduces the activity of PPAR-γ and is associated with greater insulin sensitivity and less diabetes.

Example report: TYPICAL ACTIVITY · Category: BLOOD SUGAR CONTROL · Mechanisms: Insulin resistance; Dysfunction of adipose tissue

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
PPARG	rs1801282	CC	chip	G (+)	0	Pro12Ala: less activity of PPARγ2

FABP2 (Blood Sugar/ Cardiovascular) · FABP2 (blood glucose/cardiovascular)

HIGHER ACTIVITY higher activity

Genotype rule (as in the example report): Ala54Thr rs1799883 TC.

Ala54Thr (rs1799883 T = Thr54) increases the affinity for fatty acids and intestinal fat absorption; it is associated with more insulin resistance.

Example report: LOWER ACTIVITY · Category: BLOOD SUGAR CONTROL · Mechanisms: Insulin resistance

With the effect allele — in the direction of the result (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FABP2	rs1799883	TC	imputation DR2=1.00	T (+)	1	Thr54: higher affinity for fatty acids

UCP2 (Body Composition) · UCP2 (energy metabolism)

TYPICAL ACTIVITY typical activity

Genotype rule (as in the example report): -866 rs659366 CT.

UCP2 regulates mitochondrial decoupling and insulin secretion. -866G>A (rs659366 T in strand +) increases expression.

Example report: TYPICAL ACTIVITY · Category: BODY FAT & APPETITE · Mechanisms: Mitochondrial dysfunction

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
UCP2	rs659366	CT	chip	T (+)	1	-866A: more UCP2 expression

UCP3 (Body Composition) · UCP3 (energy metabolism)

TYPICAL ACTIVITY typical activity

Genotype rule (as in the example report): -55 rs1800849 GG.

UCP3 is expressed in the muscle. -55C>T (rs1800849 A in strand +) increases muscle expression.

Example report: TYPICAL ACTIVITY · Category: BODY FAT & APPETITE · Mechanisms: Mitochondrial dysfunction

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
UCP3	rs1800849	GG	chip	A (+)	0	-55T: more UCP3 expression

Undefined effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
UCP2	rs647126	GG	chip	—	—	
UCP2	rs2075577	AA	chip	—	—	

Tendency to Overeat · Trend to eat in excess

TYPICAL LIKELIHOOD typical probability

8 score of 8 variants (signal effect alleles). the 26th percentile (mean position) in the reference population (n=633); 14% of the reference population has exactly the same score.

Appetite, reward and satiety variants (FTO, MC4R, POMC, NPY, LEPR, DRD2/ANKK1, CNR1, FAAH).

Example report: TYPICAL LIKELIHOOD · Category: BODY FAT & APPETITE · Mechanisms: Dysfunction of adipose tissue

Alleles that pull to «MORE LIKELY» (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CNR1	rs1049353	CT	chip	T (+)	1	CNR1
FAAH	rs324420	CA	chip	A (+)	1	FAAH P129T
TCF7L2	rs7903146	CT	chip	C (+)	1	GWAS: adiposity (Huang J; p=2e-66); unfavourable C allele
FTO	rs9939609	TA	chip	A (+)	1	FTO

Alleles that pull to «LESS LIKELY» (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
COMT	rs4680	GG	chip	G (—)	2	GWAS: adiposity (Huang J; p=1e-11); favourable G allele

No copies of the effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ANKK1	rs1800497	GG	chip	A (+)	0	ANKK1 Taq1A (A1)
MC4R	rs17782313	TT	chip	C (+)	0	MC4R
MC4R	rs12970134	GG	chip	A (+)	0	MC4R

Undefined effect allele (13)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
POMC	rs6713532	TC	chip	—	—	
NPY	rs16147	TT	chip	—	—	
CLOCK	rs1801260	AG	chip	—	—	
LEPR	rs1805096	GG	chip	—	—	
LEPR	rs1137101	AG	chip	—	—	
TAS2R38	rs713598	CG	chip	—	—	
LEP	rs7799039	GA	imputation DR2=1.00	—	—	
TAS2R38	rs1726866	GA	chip	—	—	
LEPR	rs1137100	AG	chip	—	—	

SLC2A2	rs5400	GG	chip	—	—
FCER1G	rs5082	GA	chip	—	—
NPY	rs16139	TC	chip	—	—
POMC	rs1042571	GG	imputation DR2=0.97	—	—

SLC2A2 (Sugar Intake) · SLC2A2 (sugar consumption)

TYPICAL typical

Genotype rule (as in the example report): Thr110Ile rs5400 GG.

GLUT2 is a glucose sensor in the liver and brain. Thr110Ile (rs5400 A = Ile110) is associated with higher habitual consumption of sugar.

Example report: TYPICAL · Category: BLOOD SUGAR CONTROL · Mechanisms: Hyperglycemia and Intolerance to Glucose

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SLC2A2	rs5400	GG	chip	A (+)	0	Ile110: more sugar consumption

CLOCK (Sleep & Metabolism) · CLOCK (sleep and metabolism)

TYPICAL ACTIVITY typical activity

Genotype rule (as in the example report): 3111 rs1801260 AG.

CLOCK is a central regulator of the circadian clock. 3111T>C (rs1801260 G in strand +) is associated with later chronotype, less sleep and worse response to diets, especially in homozygous.

Example report: TYPICAL ACTIVITY · Category: BODY FAT & APPETITE · Mechanisms: Circadian disorders; Desynchronization of Biological Clocks

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CLOCK	rs1801260	AG	chip	G (+)	1	3111C

Excess Body Fat · Predisposition for excess body fat

TYPICAL LIKELIHOOD typical probability

PGS002679 (PGS Catalog): 965 801 of 987 879 variants with genotype (cover 98%). the 72nd percentile in the reference population (n=633); 0% of the reference population has the same score.

Polygenic score of adiposity. It measures only genetic predisposition; it does not say anything about the current body composition.

Example report: LESS LIKELY (p1) · Category: BODY FAT & APPETITE · Mechanisms: Dysfunction of adipose tissue; Metabolic Homeostasis Decline

Alleles that pull to «MORE LIKELY» (16)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
LEPR	rs1805096	GG	chip	G (+)	2	in PGS002679: G allele increases the score
MC4R	rs2229616	CC	chip	C (+)	2	GWAS: adiposity (Huang J; p=2e-78); allele C increases the trait
TCF7L2	rs7903146	CT	chip	C (+)	1	in PGS002679: C allele increases the score
NPY	rs16147	TT	chip	T (+)	2	in PGS002679: T allele increases the score
POMC	rs6713532	TC	chip	T (+)	1	in PGS002679: T allele increases the score
CLOCK	rs1801260	AG	chip	A (+)	1	in PGS002679: allele A increases the score
CDKAL1	rs2206734	CT	chip	C (+)	1	in PGS002679: C allele increases the score
GP2	rs12597579	CC	chip	C (+)	2	GWAS: adiposity (Wen W; p=1e-08); allele C increases the trait
RFC4	rs17300539	GG	chip	G (+)	2	in PGS002679: G allele increases the score

NPC1	rs1808579	CT	chip	C (+)	1	in PGS002679: C allele increases the score
MTCH2	rs10838738	AG	chip	G (+)	1	GWAS: adiposity (Willer CJ; p=5e-09); G allele increases the trait
NPC1	rs1805081	TT	chip	T (+)	2	in PGS002679: T allele increases the score
SH3YL1	rs6548238	CC	chip	C (+)	2	GWAS: adiposity (Huang J; p=3e-214); allele C increases the trait
LEPR	rs1137100	AG	chip	A (+)	1	in PGS002679: allele A increases the score
IL6R	rs4845623	AG	chip	A (+)	1	in PGS002679: allele A increases the score
IL1B	rs1143634	GG	chip	G (+)	2	in PGS002679: G allele increases the score

Alleles that pull to «LESS LIKELY» (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
NEGR1	rs3101336	TC	chip	T (−)	1	in PGS002679: T allele decreases the score
ADRB2	rs1042713	GA	chip	G (−)	1	in PGS002679: G allele decreases the score
ANKK1	rs1800497	GG	chip	G (−)	2	in PGS002679: G allele decreases the score
NEGR1	rs2815752	GA	chip	G (−)	1	in PGS002679: G allele decreases the score
LEP	rs2167270	GG	chip	G (−)	2	in PGS002679: G allele decreases the score
CRP	rs1205	CT	chip	C (−)	1	in PGS002679: C allele decreases the score
UCP2	rs660339	GA	chip	G (−)	1	in PGS002679: G allele decreases the score

No copies of the effect allele (10)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
SEC16B	rs591120	GG	chip	C (+)	0	GWAS: adiposity (Turcot V; p=5e-38); allele C increases the trait
PEX11A	rs894160	TT	chip	C (+)	0	in PGS002679: C allele increases the score
GIPR	rs2287019	CC	chip	T (−)	0	GWAS: adiposity (p=4e-25); T allele decreases the trait
KLF9	rs11142387	CC	chip	A (−)	0	in PGS002679: allele A decreases the score
KCTD15	rs29941	GG	chip	A (−)	0	in PGS002679: allele A decreases the score
LEPR	rs11208659	TT	chip	C (+)	0	GWAS: adiposity (Huang J; p=7e-17); allele C increases the trait
STEAP1B	rs10242595	AA	chip	G (+)	0	in PGS002679: G allele increases the score
STAT3	rs9891119	CC	chip	A (−)	0	in PGS002679: allele A decreases the score
UCP2	rs2075577	AA	chip	G (+)	0	in PGS002679: G allele increases the score
GNPDA2	rs16858082	CC	chip	T (+)	0	in PGS002679: T allele increases the score

Undefined effect allele (16)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
LPCAT2	rs2285053	CC	chip	—	—	
LEPR	rs1137101	AG	chip	—	—	
UCP1	rs1800592	TC	chip	—	—	
UCP2	rs659366	CT	chip	—	—	
TAS1R2	rs35874116	CC	chip	—	—	
ADRB2	rs1042714	GC	imputation DR2=1.00	—	—	
GNB3	rs5443	CT	chip	—	—	
IL6R	rs2228145	AC	chip	—	—	
LEP	rs3828942	GA	chip	—	—	
LEP	rs10244329	AT	chip	—	—	
SOCS3	rs9892622	AA	chip	—	—	
PYY	rs162431	GG	chip	—	—	
PGS1	rs4969170	GG	chip	—	—	

TMC8	rs4969168	GG	imputation DR2=0.81	—	—
UCP2	rs647126	GG	chip	—	—
RFC4	rs266729	CC	chip	—	—

SNP not found in the data (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
STMN4	rs140901272	—	SNP not found in the data (Imputed DR2=0.50, below the 0.8 threshold)		—	—

MC4R (Appetite / Blood Sugar) · MC4R (appetite / glycemia)

HIGHER ACTIVITY higher activity

Genotype rule (as in the example report): rs17782313 TT, rs12970134 GG: 0 allele(s) of risk.

MC4R is the satiety receptor of the hypothalamus. The rs17782313 C and rs12970134 A alleles are associated with lower signaling (more appetite, more adiposity).

Example report: HIGHER ACTIVITY · Category: BLOOD SUGAR CONTROL · Mechanisms: Dysfunction of adipose tissue

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MC4R	rs17782313	TT	chip	C (+)	0	Risk allele near MC4R
MC4R	rs12970134	GG	chip	A (+)	0	Risk allele near MC4R

Sugar Cravings · Sweets Desire

TYPICAL LIKELIHOOD typical probability

90 score of 92 variants (signal effect alleles). the 49th percentile (mean position) in the reference population (n=633); 0% of the reference population has exactly the same score. Complement: PGS000991 ("never eats sugar", inverted proxy) at the 93rd percentile, that is, a greater genetic tendency to avoid sugar.

Sweets receptors (TAS1R2/3), GLUT2 and GWAS loci of sweets taste and sugar consumption.

Example report: LESS LIKELY (p20) · Category: BLOOD SUGAR CONTROL · Mechanisms: Hyperglycemia and Intolerance to Glucose

Alleles that pull to «MORE LIKELY» (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
KISS1R	rs350132	AA	imputation DR2=0.96	A (+)	2	GWAS: Sweet taste preference (Kawafune K; p=8e-06); unfavourable A allele
ITGA4	rs13029040	CT	imputation DR2=1.00	T (+)	1	GWAS: Sweet taste preference (Kawafune K; p=5e-06); unfavourable T allele
DCC	rs55940710	CT	imputation DR2=0.99	T (+)	1	GWAS: Sweet taste preference (Kawafune K; p=2e-06); unfavourable T allele
MTDH	rs2448140	CT	chip	T (+)	1	GWAS: Sweet taste preference (Kawafune K; p=3e-06); unfavourable T allele

Alleles that pull to «LESS LIKELY» (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TAS1R2	rs35874116	CC	chip	C (—)	2	TAS1R2 Val191: less sugar consumption
FGF21	rs838133	GG	chip	G (—)	2	GWAS: F-sweet food like (derived food-like factor) (May-Wilson S; p=1e-36); G allele favourable
ATP10B	rs10037124	GG	imputation DR2=0.99	G (—)	2	GWAS: Sweet taste preference (Kawafune K; p=3e-06); G allele favourable
PNLIPRP3	rs2033397	CC	imputation DR2=0.88	C (—)	2	GWAS: Sweet taste preference (Kawafune K; p=9e-06); favourable C allele

No copies of the effect allele (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
—	rs56404116	AA	chip	G (—)	0	GWAS: Sweet taste preference (Kawafune K; p=5e-06); G allele favourable
ATP10B	rs4552669	GG	imputation DR2=1.00	A (+)	0	GWAS: Sweet taste preference (Kawafune K; p=5e-07); unfavourable A allele
AGMO	rs12699747	TT	imputation DR2=0.99	C (+)	0	GWAS: Sweet taste preference (Kawafune K; p=3e-06); unfavourable C allele
—	rs13347339	TT	imputation DR2=0.86	C (—)	0	GWAS: Sweet taste preference (Kawafune K; p=6e-07); favourable C allele
—	rs13182470	CC	chip	T (—)	0	GWAS: Sweet taste preference (Kawafune K; p=5e-06); favourable T allele
SLC2A2	rs5400	GG	chip	A (+)	0	SLC2A2 Ile110: more sugar
ALDH2	rs671	GG	chip	A (+)	0	GWAS: Sweet taste preference (Kawafune K; p=2e-70); unfavourable A allele

Undefined effect allele (15)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
TAS1R2	rs3935570	GG	chip	—	—	
SERPINA1	rs11568814	CT	imputation DR2=0.90	—	—	
ORC5	rs10953405	AA	imputation DR2=0.97	—	—	
WSCD1	rs35253088	TT	imputation DR2=0.91	—	—	
NOTCH1	rs710411	GA	chip	—	—	
DNAJA4	rs4887033	GT	imputation DR2=0.88	—	—	
GOLGA8B	rs2433267	AA	imputation DR2=0.99	—	—	
CLMP	rs17127163	AA	imputation DR2=0.83	—	—	
RELN	rs62485870	TT	imputation DR2=0.95	—	—	
CST7	rs62215296	GG	imputation DR2=0.96	—	—	
—	rs2815675	C	chip	—	—	
RARB	rs7619139	AA	chip	—	—	
CDH20	rs1497980	CC	imputation DR2=0.98	—	—	
HTR5A	rs7795216	AA	imputation DR2=0.98	—	—	
MTX3	rs139896719	TT	imputation DR2=1.00	—	—	

SNP not found in the data (12)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FSCB	rs537022264	—	SNP not found in the data	—	—	
TMEM63C	rs74340145	—	SNP not found in the data (Imputed DR2=0.29, below the 0.8 threshold)	A (+)	—	GWAS: Sweet taste preference (Kawafune K; p=2e-06); unfavourable A allele
BHLHE23	rs138884670	—	SNP not found in the data (Imputed DR2=0.09, below the 0.8 threshold)	—	—	
—	rs145221345	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—	
LRIT3	rs142911827	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—	
N6AMT1	rs139744648	—	SNP not found in the data (Imputed DR2=0.02, below the 0.8 threshold)	—	—	
HACE1	rs116683943	—		—	—	

			SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)		
OR51T1	rs142576658	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—
PTPRZ1	rs78017114	—	SNP not found in the data (Imputed DR2=0.04, below the 0.8 threshold)	—	—
SEC24D	rs569450439	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—
TMEM248	rs185288219	—	SNP not found in the data (Imputed DR2=0.01, below the 0.8 threshold)	—	—
DSEL	rs184797377	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—

More 82 variants of the GWAS Catalog (sweet preference) enter the score; 77 with genotype.

FTO (Body Composition) · FTO (body composition)

TYPICAL typical

Genotype rule (as in the example report): rs9939609 TA.

FTO rs9939609 A is the common allele with the greatest effect on adiposity, via IRX3/IRX5 and thermogenesis. Physical activity reduces the effect by ~30%.

Example report: BETTER · Category: BODY FAT & APPETITE · Mechanisms: Dysfunction of adipose tissue; Thermogenesis decline

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FTO	rs9939609	TA	chip	A (+)	1	FTO risk allele

Results in detail: Exercise and Fitness

Power · Power

TYPICAL typical

5 score of 5 variants (signal effect alleles). the 65th percentile (mean position) in the reference population (n=633); 26% of the reference population has exactly the same score.

ACTN3 R577X, ACE I/D (tag rs4343), AMPD1 Q12X, PPARGC1A, HIF1A and other genes linked to power and fast fibers.

Example report: TYPICAL · Category: FITNESS · Mechanisms: Sarcopenia; Loss of muscle strength

Alleles that pull to «HIGHER» (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ACTN3	rs1815739	TC	chip	C (+)	1	ACTN3 R (fast fibers)
ACE	rs4343	GA	chip	G (+)	1	ACE D
IL6	rs1800795	GG	chip	G (+)	2	IL6 -174G

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
HIF1A	rs11549465	CC	chip	T (+)	0	HIF1A P582S
AMPD1	rs17602729	GG	chip	A (–)	0	AMPD1 Q12X

Undefined effect allele (11)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ACVR1B	rs2854464	AG	chip	—	—	
CKM	rs8111989	TT	chip	—	—	
ADRB2	rs1042713	GA	chip	—	—	
PKDREJ	rs4253778	GC	chip	—	—	
ADRB2	rs1042714	GC	imputation DR2=1.00	—	—	
NOS3	rs2070744	CT	chip	—	—	
PPARGC1A	rs8192678	CC	chip	—	—	
VDR	rs1544410	CT	chip	—	—	
VDR	rs731236	AG	chip	—	—	
HIF1A	rs2301113	AA	chip	—	—	
AGT	rs699	AG	chip	—	—	

ACTN3 (Power) · ACTN3 (power)

TYPICAL ACTIVITY typical activity

Genotype rule (as in the example report): R577X rs1815739 TC.

The α-actinin-3 exists only in the fast fibers. The X allele (rs1815739 T, R577X) is a stop codon; XX does not produce the protein.

Example report: TYPICAL ACTIVITY · Category: FITNESS · Mechanisms: Sarcopenia; Loss of muscle strength

With the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ACTN3	rs1815739	TC	chip	T (+)	1	R577X: without α-actinin-3 in TT

Response Exercise (Metabolic) · Metabolic response to exercise

TYPICAL typical

7 score of 7 variants (signal effect alleles). the 57th percentile (mean position) in the reference population (n=633); 26% of the reference population has exactly the same score.

Variants (FTO, PPARG, ADRB2, ADRB3, LEPR, MC4R, CLOCK) associated with fat loss and metabolic improvement with training.

Example report: TYPICAL · Category: FITNESS · Mechanisms: Metabolic Homeostasis decline; Insulin resistance

Alleles that pull to «WORSE» (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FTO	rs9939609	TA	chip	A (+)	1	FTO
PPARG	rs1801282	CC	chip	C (+)	2	PPARG Pro12
ADRB2	rs1042713	GA	chip	A (+)	1	ADRB2 Arg16
ADRB2	rs1042714	GC	imputation DR2=1.00	G (+)	1	ADRB2 Glu27
LEPR	rs1137101	AG	chip	G (+)	1	LEPR Q223R

No copies of the effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
ADRB3	rs4994	AA	chip	G (+)	0	ADRB3 Arg64
MC4R	rs17782313	TT	chip	C (+)	0	MC4R

Undefined effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
LEPR	rs1137100	AG	chip	—	—	
CLOCK	rs1801260	AG	chip	—	—	
LEPR	rs1805094	GG	chip	—	—	

Muscle Cramps · Muscle cramps

LESS LIKELY less probable

2 score of 2 variants (GWAS Catalog coefficients). the 16th percentile (mean position) in the reference population (n=633); 23% of the reference population has exactly the same score. Very low reliability: the published associations are of rare variants, almost all missing from the data.

Weighted score with the associations of limb cramps and nocturnal leg cramps of GWAS Catalog (biobanks).

Example report: TYPICAL LIKELIHOOD (p59) · Category: FITNESS · Mechanisms: Neuromuscular Junction Disorder

Undefined effect allele (38)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
MSI2	rs116924898	GG	imputation DR2=0.97	—	—	
LMO7	rs73229904	TT	chip	—	—	
TBC1D4	rs76851570	TT	chip	—	—	
STK32A	rs35852718	AA	imputation DR2=1.00	—	—	
ST6GAL2	rs62155507	AA	imputation DR2=0.95	—	—	
RAB21	rs61754230	CT	chip	—	—	
MRPS34	rs11552434	CC	imputation DR2=0.99	—	—	
AGXT2	rs114286107	CC	chip	—	—	
CLEC18C	rs150702800	CC	imputation DR2=0.92	—	—	
PTK7	rs34021075	AA	imputation DR2=0.86	—	—	
DUOX2	rs151261408	GG	chip	—	—	
CHRNA5	rs2229961	GG	chip	—	—	
CC2D2A	rs144439937	AA	chip	—	—	
SLC36A2	rs77010315	CC	chip	—	—	
TMC3	rs150843673	GG	imputation DR2=0.92	—	—	

NEB	rs62167164	GG	chip	—	—
MCM10	rs35114749	CC	chip	—	—
PSCA	rs138377917	GG	chip	—	—
TENM2	rs114898031	AA	chip	—	—
GANC	rs35285091	CC	chip	—	—
AMPD1	rs17602729	GG	chip	—	—
CGRRF1	rs34839928	CC	imputation DR2=0.83	—	—
SPG11	rs78183930	CC	chip	—	—
EFHC1	rs1570624	GG	imputation DR2=0.87	—	—
ALDH5A1	rs62621664	GG	chip	—	—
CACNA2D3	rs112362995	CC	imputation DR2=0.92	—	—
OBSCN	rs62621832	TT	chip	—	—
XIRP2	rs16853333	GG	chip	—	—
CFHR2	rs41257904	GG	chip	—	—
GLCE	rs12440300	AA	chip	—	—
FAM193B	rs116293002	GG	chip	—	—
FPGS	rs35789560	CC	chip	—	—
RFX7	rs33984059	AA	chip	—	—
CYP4B1	rs45467195	AA	chip	—	—
BLVRA	rs1050916	AA	chip	—	—
IQCH	rs62014646	GG	imputation DR2=0.99	—	—
PDLIM5	rs75841704	AA	chip	—	—
THBS3	rs72704117	CC	chip	—	—

SNP not found in the data (12)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IKZF2	rs112988286	—	SNP not found in the data	—	—	
EPS8L2	rs201086107	—	SNP not found in the data	—	—	
TARS1	rs138936240	—	SNP not found in the data (Imputed DR2=0.73, below the 0.8 threshold)	—	—	
PCDHA3	rs141677026	—	SNP not found in the data	—	—	
KCTD18	rs565245989	—	SNP not found in the data	—	—	
SPATA22	rs150770000	—	SNP not found in the data	—	—	
—	rs35525189	—	SNP not found in the data	—	—	
—	rs1369355526	—	SNP not found in the data (Imputed DR2=0.71, below the 0.8 threshold)	—	—	
—	rs66906990	—	SNP not found in the data	—	—	
DCHS2	rs546143621	—	SNP not found in the data	—	—	
KRT6A	rs62617089	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	—	—	
—	rs78767524	—	SNP not found in the data	—	—	

Additional variants of the GWAS Catalog used in the score (8; 2 with genotype)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
—	rs146168804	—	SNP not found in the data	G (—)	—	GWAS: Cramp of limb (PheCode 771.2) (Verma A; p=3e-21)
—	rs542465761	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	G (—)	—	GWAS: Sleep related leg clamps (PheCode 327.72) (Verma A; p=3e-14)

—	rs564918597	—	SNP not found in the data	T (—)	—	GWAS: Sleep related leg clamps (PheCode 327.72) (Verma A; p=5e-13)
—	rs868269934	—	SNP not found in the data	T (—)	—	GWAS: Sleep related leg clamps (PheCode 327.72) (Verma A; p=2e-12)
—	rs571611582	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	G (—)	—	GWAS: Sleep related leg clamps (PheCode 327.72) (Verma A; p=2e-11)
—	rs10980545	CA	imputation DR2=1.00	C (+)	1	GWAS: Cramp of limb (PheCode 771.2) (Verma A; p=3e-11)
—	rs556436481	—	SNP not found in the data (Imputed DR2=0.00, below the 0.8 threshold)	T (—)	—	GWAS: Sleep related leg clamps (PheCode 327.72) (Verma A; p=3e-11)
—	rs113253936	CC	imputation DR2=0.98	C (—)	2	GWAS: Cramp of limb (PheCode 771.2) (Verma A; p=4e-11)

Injuries · Lesions (functional)

LESS LIKELY less probable

5 score of 5 variants (signal effect alleles). the 19th percentile (mean position) in the reference population (n=633); 18% of the reference population has exactly the same score.

Collagen variants (COL1A1, COL3A1, COL5A1, COL12A1), GDF5, VEGFA and TNF associated with tendon and ligament lesions.

Example report: TYPICAL LIKELIHOOD · Category: FITNESS · Mechanisms: Remodelling of the extracellular matrix; Stiffness of the Extracellular Matrix

Alleles that pull to «MORE LIKELY» — in the opposite direction to the result (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
COL5A1	rs12722	TT	chip	T (+)	2	COL5A1

Alleles that pull to «LESS LIKELY» — in the direction of the result (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
VEGFA	rs699947	AA	chip	A (—)	2	VEGFA -2578A

No copies of the effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CPNE1	rs143383	GG	chip	A (+)	0	GDF5 +104T
COL1A1	rs1800012	CC	chip	A (—)	0	COL1A1 Sp1 (protector)
TNF	rs1800629	GG	chip	A (+)	0	TNF -308A

Undefined effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
COL3A1	rs1800255	GG	chip	—	—	
COL12A1	rs970547	TT	chip	—	—	

COL5A1 (Collagen) · COL5A1 (collagen)

HIGHER ACTIVITY higher activity

Genotype rule (as in the example report): rs12722 TT.

Collagen V regulates the diameter of collagen I fibrils. Rs12722 T (3'UTR) is associated with greater stability of the mRNA, more rigid tendons, better running economy and, in some studies, more tendinopathy.

Example report: TYPICAL ACTIVITY · Category: FITNESS · Mechanisms: Remodelling of the extracellular matrix; Stiffness of the Extracellular Matrix

With the effect allele — in the direction of the result (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
COL5A1	rs12722	TT	chip	T (+)	2	3'UTR T: more stability of mRNA

Exercise Recovery · Recovery after exercise

TYPICAL typical

10 score of 10 variants (signal effect alleles). the 42nd percentile (mean position) in the reference population (n=633); 24% of the reference population has exactly the same score.

Inflammatory and antioxidant variants (IL6, IL6R, TNF, IL1B, CRP, CCL2/CCR2, SOD2, GPX1, CAT) and muscle variants (ACTN3, MYLK, IGF2, MCT1).

Example report: TYPICAL · Category: FITNESS · Mechanisms: Inflammaging; Sarcopenia

Alleles that pull to «WORSE» (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IL6R	rs4129267	CT	chip	T (+)	1	GWAS: Interleukin-6 subunit alpha levels (IL6R.4139.71.2) (Sun BB; p1e-300); unfavourable T allele
ACTN3	rs1815739	TC	chip	T (+)	1	ACTN3 (more muscle damage)
GPX1	rs1050450	GA	chip	A (+)	1	GPX1 Leu198
IL1B	rs16944	AG	chip	A (+)	1	IL1B -511T
SLC16A1	rs1049434	AT	chip	T (+)	1	MCT1 Glu490Asp: less lactate clearance
IL6	rs1800795	GG	chip	G (+)	2	IL6 -174G

Alleles that pull to «BETTER» (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IL6R	rs2228145	AC	chip	C (—)	1	IL6R Asp358Ala (less signaling)

No copies of the effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
IL1B	rs1143634	GG	chip	A (+)	0	IL1B +3954T
SOD2	rs4880	GG	chip	A (+)	0	SOD2 Val16
TNF	rs1800629	GG	chip	A (+)	0	TNF -308A

Undefined effect allele (25)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
FTO	rs9939609	TA	chip	—	—	
IGF2	rs7924316	TT	chip	—	—	
HGF	rs5745697	GG	chip	—	—	
HGF	rs5745678	GG	chip	—	—	
MYLK	rs28497577	GG	chip	—	—	
CCR2	rs1799865	TC	chip	—	—	
SLC30A8	rs13266634	CC	chip	—	—	
CRP	rs1205	CT	chip	—	—	
IGF2	rs680	TT	chip	—	—	
IL1A	rs4848306	GA	chip	—	—	
CYBA	rs4673	GG	chip	—	—	
IGF2	rs3213221	CC	chip	—	—	
CD28	rs3116496	TC	chip	—	—	
CCL8	rs2857656	GG	chip	—	—	
COL2A1	rs2070739	CC	chip	—	—	

COL5A1	rs12722	TT	chip	—	—
CAT	rs1001179	CC	chip	—	—
SLC16A1	rs7169	GA	chip	—	—
IGF2	rs4244808	TT	imputation DR2=0.95	—	—
CCR2	rs3918358	CA	chip	—	—
CCL2	rs3917878	CT	chip	—	—
C11ORF21	rs3842748	CG	chip	—	—
SPP1	rs28357094	TT	chip	—	—
MYLK	rs2700352	GG	imputation DR2=0.99	—	—
EEF1AKMT3	rs11613457	GG	chip	—	—

Endurance · Aerobic resistance

HIGHER higher

7 score of 7 variants (signal effect alleles). the 92nd percentile (mean position) in the reference population (n=633); 8% of the reference population has exactly the same score.

Variants associated with resistance athletes and VO2max (PPARGC1A, ACE I, ACTN3 X, HFE H63D, NFE2L2, GABPB1, COL5A1...).

Example report: HIGHER · Category: FITNESS · Mechanisms: Oxidative Phosphorylation Decline; Mitochondrial dysfunction

Alleles that pull to «HIGHER» — in the direction of the result (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
HFE	rs1799945	CG	chip	G (+)	1	HFE H63D (plus iron/hemoglobin)
ACTN3	rs1815739	TC	chip	T (+)	1	ACTN3 X
SLC16A1	rs1049434	AT	chip	A (+)	1	MCT1 Glu490
PPARGC1A	rs8192678	CC	chip	C (+)	2	PPARGC1A Gly482
COL5A1	rs12722	TT	chip	T (+)	2	COL5A1 T (running economy)
ACE	rs4343	GA	chip	A (+)	1	ACE I

No copies of the effect allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
HIF1A	rs11549465	CC	chip	T (—)	0	HIF1A 582S

Undefined effect allele (34)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
GABPB1	rs7181866	AA	chip	—	—	
GABPB1	rs8031031	CC	chip	—	—	
TTN	rs10497520	CC	chip	—	—	
KDR	rs1870377	TA	chip	—	—	
PRDM1	rs10499043	CC	chip	—	—	
ADRB3	rs4994	AA	chip	—	—	
CHRNA3	rs4950	GA	chip	—	—	
GSTP1	rs1695	AG	chip	—	—	
TSHR	rs7144481	TT	chip	—	—	
NFE2L2	rs35652124	TC	chip	—	—	
DES	rs7564856	AA	imputation DR2=0.96	—	—	
SATB1	rs4973706	TT	chip	—	—	
GALNTL6	rs558129	GA	chip	—	—	
ADRB2	rs1042713	GA	chip	—	—	
ADRB2	rs1042714	GC	imputation DR2=1.00	—	—	

GNB3	rs5443	CT	chip	—	—
BDKRB2	rs1799722	CC	chip	—	—
RBFOX1	rs7191721	AG	chip	—	—
CNDP2	rs6566810	TT	imputation DR2=0.96	—	—
PKDREJ	rs4253778	GC	chip	—	—
AGTR2	rs11091046	A	chip	—	—
CKM	rs8111989	TT	chip	—	—
DEF6	rs2016520	CT	chip	—	—
PPARGC1A	rs7665116	TC	chip	—	—
PPARGC1A	rs2970869	CC	imputation DR2=0.99	—	—
NR1H3	rs7120118	TT	chip	—	—
GABPB1	rs12594956	CA	chip	—	—
MYBPC3	rs1052373	CC	imputation DR2=1.00	—	—
NFIA	rs1572312	GG	chip	—	—
TTC23	rs1464430	CG	chip	—	—
KCNA4	rs1323860	AA	chip	—	—
PPARA	rs1800206	CC	chip	—	—
PPARGC1A	rs3774923	CC	chip	—	—
HIF1A	rs2301113	AA	chip	—	—

SNP not found in the data (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Note
CYFIP1	rs8029108	—	SNP not found in the data (Imputed DR2=0.24, below the 0.8 threshold)	—	—	

Recommendations in detail

All entries: suggestion (not applied).

1. Methylfolate (Methylolate; dose of the example: 400 mcg)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
MTHFR	LOWER ACTIVITY	5/5	5/5

Crossing with the v2 plan — already in the v2 stack: In the stack: 5-MTHF 800 µg (v2 plan: KEEP; without folic acid). The dose of the sample (400 mcg) is lower than the current one. Suggestion (not applied).

2. Strength training (Strength Training; dose of the example: 1 hour)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
ADRB2 (Body Composition)	WORSE GENETICS	5/5	5/5

Crossing with the v2 plan — already in the v2 stack: It is already part of the plan (with ~140 g/day protein and creatine 5 g). ADRB2 Glu27, UCP1 and FTO reinforce the priority. Suggestion (not applied).

3. Maintaining optimal levels of vitamin D (Maintain Optimal Vitamin D Levels; dose of the example: 1000 iu)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
Calcium	INCREASED NEED	4/5	5/5

Crossing with the v2 plan — already in the v2 stack: In the stack: D3 10 000 IU (coordination note: remains 10 000 IU). The 1000 IU of the example do not apply. Suggestion (not applied).

4. Mediterranean Diet (Mediterranean Diet; dose of the example: —)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
TCF7L2 (Blood Sugar/Diet)	WORSE GENETICS	4/5	4/5

Crossing with the v2 plan — already in the v2 stack: It coincides with the v2 plan: saturated fat <7% of energy, extra virgin olive oil 40–50 mL/day, 3×/week. Without wine: zero alcohol. Suggestion (not applied).

5. Green leaf vegetables (Leafy Green Vegetables; dose of the example: —)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
MTHFR	LOWER ACTIVITY	3/5	2/5
Calcium	INCREASED NEED	2/5	1/5
Methylation	HIGHER ABILITY	3/5	2/5

Crossing with the v2 plan — already in the v2 stack: Plan v2: daily green leaves (MTHFR, DHFR, NOS3). Suggestion (not applied).

6. Betaine (TMG) (Betaine (TMG); dose of example: 500 mg)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
MTHFR	LOWER ACTIVITY	3/5	2/5
Methylation	HIGHER ABILITY	4/5	3/5

Crossing with the v2 plan — already in the v2 stack: In the stack: TMG 1500 mg; v2 plan: up to 3 g/day in 2 tomas (maximum 3 g because of LDL). Suggestion (not applied).

7. Cinnamon (Cinnamon; dose of example: 1g)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
TCF7L2 (Blood Sugar/Diet)	WORSE GENETICS	3/5	3/5

Crossing with the v2 plan — caution (interaction or context): Cinnamon alone: cassava has coumarin, hepatotoxic at high doses. Effect added to acarbose (rare hypoglycemia). Suggestion (not applied).

8. Coffee (Caffeine; dose of the example: 100 mg)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
ADRB2 (Body Composition)	WORSE GENETICS	2/5	3/5

Crossing with the v2 plan — already in the v2 stack: Plan v2: caffeine ≤200 mg/day until 12h, filtered coffee (CYP1A2 CA, ADORA2A CC). 100 mg fits in this limit. **Suggestion (not applied).**

9. Zinc (Zinc; dose of the example: 15 mg)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
MTHFR	LOWER ACTIVITY	2/5	2/5
Methylation	HIGHER ABILITY	2/5	2/5

Crossing with the v2 plan — already in the v2 stack: Zinc in the stack (11 mg in trace minerals); plan v2: total 20–25 mg/day, ratio Zn:Cu 10–15:1, far from NAC, Ca-AKG and fiber. **Suggestion (not applied).**

10. Glutamine (Glutamine; dose of the example: 5g)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
TCF7L2 (Blood Sugar/Diet)	WORSE GENETICS	2/5	2/5

Crossing with the v2 plan — compatible with the stack: No known relevant interaction; weak evidence outside diabetes. **Suggestion (not applied).**

11. Beak grain (Chickpeas; dose of the example: —)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
Vitamin B6 (Pyridoxine)	TYPICAL NEED	2/5	1/5

Crossing with the v2 plan — already in the v2 stack: Plan v2: legumes 1–2 portions/day. **Suggestion (not applied).**

12. Quinoa (Quinoa; dose of the example: —)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
Riboflavin (Vitamin B2)	TYPICAL NEED	1/5	1/5

Crossing with the v2 plan — already in the v2 stack: Plan v2: quinoa and wheat germ in the diet **Suggestion (not applied).**

13. Fiber-rich foods (Eat Fiber-Rich Foods; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — already in the v2 stack: Plan v2: fiber ≥50 g/day of food and supplements; more fiber is always better (TCF7L2 CT, CETP, FUT2). **Suggestion (not applied).**

14. Berberine (Berberine; dose of the example: 1000 mg)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
TCF7L2 (Blood Sugar/Diet)	WORSE GENETICS	4/5	5/5

Crossing with the v2 plan — not recommended in this context: Plan v2: AVOID. It inhibits CYP3A4/P-gp and alters the exposure to rapamycin. TCF7L2 is already covered by fiber, diet and acarbose. **Suggestion (not applied).**

15. Alpha-lipoic acid (Alpha-Lipoic Acid; dose of the example: 100 mg)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — already in the v2 stack: In the stack: R-ALA 240 mg in fasting (v2 plan: KEEP, optional). **Suggestion (not applied).**

16. Psyllium (Psyllium; dose of the example: 5g)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — already in the v2 stack: In the stack (psyllium 10 g + PHGG 6 g); plan v2: rise to psyllium 15–20 g + PHGG 10 g. **Suggestion (not applied).**

17. Pine bark extract (Pine bark extract; dose of the example: 100 mg)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — already in the v2 stack: In the stack: pine bark extract 100 mg (Plan v2: KEEP, outside the rapamycin window). **Suggestion (not applied).**

18. Garlic (Garlic; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — already in the v2 stack: In the stack: aged garlic 1.2 g (Plan v2: KEEP). **Suggestion (not applied).**

19. Limit saturated fat (Limit Saturated Fat; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — already in the v2 stack: Plan v2: saturated fat <7% of energy (CETP GG, APOB, 9p21). **Suggestion (not applied).**

20. Lactobacillus rhamnosus (Lactobacillus Rhamnosus; dose of example: 10 billion CFU)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
Alcohol Sensitivity	TYPICAL	1/5	1/5

Crossing with the v2 plan — compatible with the stack: Compatible with the stack probiotic. The associated risk in the example (sensitivity to alcohol) does not apply: zero alcohol. **Suggestion (not applied).**

21. Sleeping 7 or more hours (Sleep for 7+ Hours; dose of the example: —)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
Methylation	HIGHER ABILITY	2/5	1/5

Crossing with the v2 plan — compatible with the stack: Compatible (CLOCK, MTNR1B); last meal ≥ 3 h before bedtime, as in the plan. **Suggestion (not applied).**

22. Yoga (Yoga; dose of the example: 30 minutes)

Risk of targeted DNA	Sérgio F result	Impact	Evidence
Methylation	HIGHER ABILITY	1/5	1/5

Crossing with the v2 plan — compatible with the stack: Compatible. **Suggestion (not applied).**

23. Plant-based integral food diet (Whole-Food Plant-Based Diet; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible with the v2 plan (fiber ≥ 50 g/day, legumes, green leaves). Keep the plan protein (~ 140 g/day) and the fatty fish. **Suggestion (not applied).**

24. High intensity interval training (HIIT) (High-Intensity Interval Training (HIIT); example dose: 30 minutes)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible; complements the plan strength training. **Suggestion (not applied).**

25. Glucomannan (Glucomannan; dose of the example: 3g)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible; sum of fiber (more is always better). With a lot of water and ≥ 2 h away from the medication and minerals. **Suggestion (not applied).**

26. Walk (Walking; dose of the example: 30 minutes)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible. **Suggestion (not applied).**

27. Exercise snacks (1 minute) (Practice Exercise Snacks; dose of the example: 1 minute)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible; useful after meals (TCF7L2). **Suggestion (not applied).**

28. Lactobacillus plantarum (Lactobacillus Plantarum; dose of the example: 10 billion CFU)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible with the secretory FUT2. **Suggestion (not applied).**

29. Okra (Okra; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible: more fiber. **Suggestion (not applied).**

30. Avoiding organochlorine pesticides (Avoid Organochlorine Pesticide Exposure; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Coherent with the v2 plan (probable null GSTM1, GSTA1 *B/*B, slow NAT2): reduce exposures and avoid processed, smoked or carbonized meat. **Suggestion (not applied).**

31. Avoid polycyclic aromatic hydrocarbons (PAH) (Avoid PAHs Exposure; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Coherent with the v2 plan (probable null GSTM1, GSTA1 *B/*B, slow NAT2): reduce exposures and avoid processed, smoked or carbonized meat. **Suggestion (not applied).**

32. Avoid PCB (Avoid PCBs; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Coherent with the v2 plan (probable null GSTM1, GSTA1 *B/*B, slow NAT2): reduce exposures and avoid processed, smoked or carbonized meat. **Suggestion (not applied).**

33. Vinegar apple cider (Apple Cider Vinegar; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible; diluted and to meals (dental enamel). Small effect. **Suggestion (not applied).**

34. Drinking at least 8 water cups per day (Drink at Least 8 Glasses of Water a Day; dose of example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible and useful with dapagliflozin (hydration). **Suggestion (not applied).**

35. Avoid phthalates (Avoid Phthalate Exposure; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Coherent with the v2 plan (probable null GSTM1, GSTA1 *B/*B, slow NAT2): reduce exposures and avoid processed, smoked or carbonized meat. **Suggestion (not applied).**

36. Whole body vibration (Whole-Body Vibration; dose of the example: 15 minutes)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible. **Suggestion (not applied).**

37. Mindfulness meditation (Mindfulness Meditation; dose of example: 30 minutes)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible. **Suggestion (not applied).**

38. Inositol (Inositol; dose of the example: 2g)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible; choosing one of the two (myo-inositol is the studied form). **Suggestion (not applied).**

39. Myo-inositol (Myo-inositol; dose of example: 4g)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Compatible; choosing one of the two (inositol or myo-inositol). **Suggestion (not applied).**

40. Avoid PFAS (Avoid PFAS Exposure; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — compatible with the stack: Coherent with the v2 plan (probable null GSTM1, GSTA1 *B/*B, slow NAT2): reduce exposures and avoid processed, smoked or carbonized meat. **Suggestion (not applied).**

41. Black cumin (Nigella sativa) (Black Seed (Black Cumin); example dose: 1000 mg)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — caution (interaction or context): Thanks to the use of CYP3A4 and CYP2C9 in vitro (rapamycin; CYP2C9 *1/*3) and there are rare reports of liver injury. No clear advantage over the current stack. **Suggestion (not applied).**

42. Green coffee extract (Green Coffee Extract; dose of the example: 400 mg)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — caution (interaction or context): Sums caffeine and concentrated chlorogenic acids; the plan limits caffeine to ≤200 mg/day only until 12h, and concentrated extracts have reports of hepatotoxicity. Filtered coffee arrives. **Suggestion (not applied).**

43. Seed extract of Irvingia gabonensis (dose of the example: 300 mg)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — caution (interaction or context): Poor evidence and low quality; the retatrutide already acts on appetite. No benefit expected. **Suggestion (not applied).**

44. Black cumin oil (Black Seed Oil; dose of example: 2 tsp)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — caution (interaction or context): Thanks to the use of CYP3A4 and CYP2C9 in vitro (rapamycin; CYP2C9 *1/*3) and there are rare reports of liver injury. No clear advantage over the current stack. **Suggestion (not applied).**

45. Whiteberry (White Mulberry; dose of example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — caution (interaction or context): Whiteberry inhibits alpha-glucosidases, such as acarbose: duplicate effect and more gastrointestinal symptoms. **Suggestion (not applied).**

46. Banaba (Banaba; dose of the example: 32 mg)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — caution (interaction or context): Corosolic acid lowers glycemia; added to acarbose, imeglimin, dapagliflozin and retatrutide can increase hypoglycemia. **Suggestion (not applied).**

47. Jiaogulan (Gynostemma pentaphyllum) (Jiaogulan (Gynostemma Pentaphyllum); dose of the example: 1350 mg)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — caution (interaction or context): It can reduce blood glucose and has an antiplatelet effect; there are few data of interaction with rapamycin. **Suggestion (not applied).**

48. Pheno-Greek (Fenugreek; dose of the example: 500 mg)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — caution (interaction or context): Low blood glucose (sum of acarbose and imeglimin) and may potentiate anticoagulants; small effect. **Suggestion (not applied).**

49. Ketogenic diet (Keto Diet; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — not recommended in this context: Conflict: ketogenic diet typically rises LDL and ApoB (the lower, the better), cuts the fiber and, with dapagliflozin, increases the risk of euglycemic ketoacidosis. The plan v2 uses hydrates 35–40% of low glycemic index. **Suggestion (not applied).**

50. Bergamot (Bergamot; dose of the example: —)

No DNA risks associated in the example report (general recommendation).

Crossing with the v2 plan — not recommended in this context: The bergamot contains furanocoumarins (bergamotin), inhibitors of CYP3A4 as grapefruit; the v2 plan avoids grapefruit and bitter orange because of rapamycin and tadalafil. **Suggestion (not applied).**

Comparison with the example report

The example report is from another person; the comparison shows where the results coincide or diverge. The methods are not identical: the percentiles of Sérgio F were calculated with public scores and the reference population, and the count traits use cured effect alleles.

Equal labels: 44 of 72.

Trait	Example report	Sérgio F	The same?
Methylation	LOWER ABILITY	HIGHER ABILITY (p7)	no
TCF7L2 (Blood Sugar/Diet)	WORSE GENETICS	WORSE GENETICS	yes
Triglycerides	TYPICAL LEVELS (p16)	TYPICAL LEVELS (p68)	yes
High Cholesterol	TYPICAL LIKELIHOOD (p46)	TYPICAL LIKELIHOOD (p39)	yes
LDL Cholesterol	TYPICAL LEVELS (p44)	LOWER LEVELS (p24)	no
HDL Cholesterol	TYPICAL LEVELS	LOWER LEVELS (p18)	no
Lipid Oxidation	TYPICAL	HIGHER (p77)	no
Oxidative Stress	TYPICAL	LOWER (p5)	no
Detox	TYPICAL ABILITY	HIGHER ABILITY (p21)	no
Phase I Detox	TYPICAL ABILITY	TYPICAL ABILITY (p49)	yes
Phase II Detox	TYPICAL ABILITY	TYPICAL ABILITY (p56)	yes
Inflammation (CRP)	TYPICAL LEVELS (p36)	TYPICAL LEVELS (p26)	yes
Bone Health (Functional)	TYPICAL	TYPICAL (p30)	yes
Type 2 Diabetes	TYPICAL LIKELIHOOD (p49)	TYPICAL LIKELIHOOD (p59)	yes
APOC3 (Blood Lipids/ Longevity)	TYPICAL ACTIVITY	LOWER ACTIVITY (p18)	no
IL6 Gene (Inflammation)	TYPICAL ACTIVITY	TYPICAL ACTIVITY	yes
TNF Gene (Inflammation)	TYPICAL ACTIVITY	LOWER ACTIVITY (p20)	no
CETP (Cholesterol/ Longevity)	LOWER ACTIVITY	HIGHER ACTIVITY (p11)	no
IL1B (Inflammation/ Fatigue)	LOWER ACTIVITY	TYPICAL ACTIVITY (p55)	no
MTHFR	LOWER ACTIVITY	LOWER ACTIVITY	yes
MUFA Response	WORSE	TYPICAL (p62)	no
Protein Metabolism	WORSE RESPONSE	WORSE RESPONSE (p82)	yes
Riboflavin (Vitamin B2)	INCREASED NEED	TYPICAL NEED	no
Vitamin B6 (Pyridoxine)	INCREASED NEED	TYPICAL NEED (p47)	no
Folate	INCREASED NEED	DECREASED NEED (p10)	no
Alcohol Sensitivity	HIGHER	TYPICAL	no
Calcium	INCREASED NEED	INCREASED NEED (p22)	yes
FUT2 (Gut/vitamin B12)	SECRETOR	SECRETOR	yes
Carbohydrate Metabolism	TYPICAL	TYPICAL (p61)	yes
Saturated Fat	TYPICAL RESPONSE	TYPICAL RESPONSE (p46)	yes
Unsaturated Fat	TYPICAL RESPONSE	TYPICAL RESPONSE (p68)	yes
Vitamin A	TYPICAL NEED	TYPICAL NEED (p46)	yes

Vitamin B12	TYPICAL NEED	TYPICAL NEED (p34)	yes
Vitamin C	TYPICAL NEED (p40)	TYPICAL NEED (p69)	yes
Vitamin D	TYPICAL NEED	DECREASED NEED (p84)	no
Choline	TYPICAL NEED	DECREASED NEED (p5)	no
Iron Overload	TYPICAL LIKELIHOOD	TYPICAL LIKELIHOOD	yes
HLA-DQ (Gluten)	TYPICAL GENETICS	TYPICAL GENETICS	yes
Coffee-Related Sleep Problems	TYPICAL LIKELIHOOD	LESS LIKELY (p13)	no
Salt Sensitivity	TYPICAL	TYPICAL (p49)	yes
MTRR (Methylation)	TYPICAL ACTIVITY	TYPICAL ACTIVITY	yes
VDR (Vitamin D)	TYPICAL ACTIVITY	TYPICAL ACTIVITY (p62)	yes
PUFA Response	TYPICAL	TYPICAL (p52)	yes
Fat Metabolism	BETTER	TYPICAL (p63)	no
Lactose Intolerance	LIKELY TOLERANT	LIKELY TOLERANT	yes
Caffeine Sensitivity	LOWER	LOWER (p18)	yes
Caffeine-Related Anxiety	LESS LIKELY	LESS LIKELY	yes
CYP1A2 (Detox)	HIGHER ACTIVITY	TYPICAL ACTIVITY	no
GC (Vitamin D)	HIGHER ACTIVITY	TYPICAL ACTIVITY (p27)	no
Bitter Taste Sensitivity	TASTER	TASTER	yes
Sleeping	LESS LIKELY	LESS LIKELY (p22)	yes
ADRB2 (Body Composition)	WORSE GENETICS	WORSE GENETICS	yes
UCP1 (Body Composition)	LOWER ACTIVITY	LOWER ACTIVITY	yes
PPARG (Metabolism)	TYPICAL ACTIVITY	TYPICAL ACTIVITY	yes
FABP2 (Blood Sugar/ Cardiovascular)	LOWER ACTIVITY	HIGHER ACTIVITY	no
UCP2 (Body Composition)	TYPICAL ACTIVITY	TYPICAL ACTIVITY	yes
UCP3 (Body Composition)	TYPICAL ACTIVITY	TYPICAL ACTIVITY	yes
Tendency to Overeat	TYPICAL LIKELIHOOD	TYPICAL LIKELIHOOD (p26)	yes
SLC2A2 (Sugar Intake)	TYPICAL	TYPICAL	yes
CLOCK (Sleep & Metabolism)	TYPICAL ACTIVITY	TYPICAL ACTIVITY	yes
Excess Body Fat	LESS LIKELY (p1)	TYPICAL LIKELIHOOD (p72)	no
MC4R (Appetite / Blood Sugar)	HIGHER ACTIVITY	HIGHER ACTIVITY	yes
Sugar Cravings	LESS LIKELY (p20)	TYPICAL LIKELIHOOD (p49)	no
FTO (Body Composition)	BETTER	TYPICAL	no
Power	TYPICAL	TYPICAL (p65)	yes
ACTN3 (Power)	TYPICAL ACTIVITY	TYPICAL ACTIVITY	yes
Response Exercise (Metabolic)	TYPICAL	TYPICAL (p57)	yes
Muscle Cramps	TYPICAL LIKELIHOOD (p59)	LESS LIKELY (p16)	no
Injuries	TYPICAL LIKELIHOOD	LESS LIKELY (p19)	no
COL5A1 (Collagen)	TYPICAL ACTIVITY	HIGHER ACTIVITY	no

Exercise Recovery	TYPICAL	TYPICAL (p42)	yes
Endurance	HIGHER	HIGHER (p92)	yes

Limitations

- Genotyping chip data plus local statistical imputation (full genome reference panel). The SNPs imputed with $DR2 < 0.8$ were treated as not found.
- The punctuation method of the example report is proprietary and not reproducible in detail. Here, counting traits use literature-cured effect alleles or the sense of the main association in GWAS Catalog; many SNPs of the example come from gene-diet interaction studies, approximated by main effects.
- Several count traits have few SNPs (B6 and VDR with 1, CETP and GC with 2): a single genotype decides the label, and the position in the reference population is coarse (many people with the same score).
- PGS explain a small fraction of the variation (typically 5–15% for lipids and less for other traits) and are not diagnosed. For LDL and ApoB, the target is always as low as possible, regardless of the score.
- Muscle Cramps: the published associations are almost all of rare variants; only 2 enter the score. Very low reliability result.
- Sugar Cravings: counting score with SNPs of preference for sweet; PGS000991 ("never eats sugar") is just a proxy.
- Indels, CNV and rare variants are almost never on the chip or attribute well: the deletion of GSTM1 and CNV of CYP2D6 are not visible, and rare alleles of HFE or CYP may go unnoticed.
- HLA-DQ: the tag SNPs (rs2187668, rs7454108) have high negative predictive value, but do not replace the HLA typing.
- The percentiles come from a reference population of 633 people; the example report uses a reference and its own method.
- Genetics is only one part: diet, training, sleep, medication and the current stack weigh more than almost all the results of this report.

Methods and sources

List of traits and SNPs: extracted from the example report (72 traits, 1362 lines, 931 unique rsIDs), with the same trait names, sections and labels.

Genotypes: first the chip (by rsID and, for cast rsIDs, by position), then the local imputation ($DR2 \geq 0.8$). Alleles in the strand + of dbSNP (GRCh38), confirmed against Ensembl. The raw genetic data were processed locally; to the public databases only rsIDs and gene names were sent.

Polygenic scores: PGS Catalog, harmonized for GRCh38: triglycerides PGS002719, total cholesterol PGS002700, LDL PGS002703, HDL PGS002695, diabetes type 2 PGS002720, vitamin D PGS002721, adiposity (Excess Body Fat) PGS002679 (Weissbrod 2022), CRP PGS002873, vitamin B12 PGS003158 (Ma 2022), calcium PGS001893 (Privé 2022) and PGS000991 as a proxy of preference for sugar. Percentages against the reference population ($n=633$), with the same SNPs and the same quality rule.

Counting traits: sum with signal of the effect alleles (strand +) of the SNPs of the example report, compared to the same sum in the reference population. Mean position $\geq 75 \rightarrow$ high label; $\leq 25 \rightarrow$ low label; rest $\rightarrow$ typical.

Traits by rule: genotype rules published for each gene (e.g. MTHFR 677/1298, HFE C282Y/H63D, LCT -13910, FTO rs9939609).

Main sources

- PGS Catalog (Lambert et al., Nat Genet 2021): harmonized scores for GRCh38.
- Weissbrod O et al. Leveraging fine-mapping and multipopulation training data to improve cross-population polygenic risk scores. Nat Genet 2022 (PGS002679, PGS002695, PGS002700, PGS002703, PGS002719, PGS002720, PGS002721).
- Ma Y et al. Genetic prediction of complex traits with Bayesian method (PGS002873, PGS003158), 2022.
- Privé F et al. Portability of 245 polygenic scores when derived from a large reference cohort. Am J Hum Genet 2022 (PGS001893).
- PGS000991 (large reference cohort, never eat sucking), used only as proxy.
- GWAS Catalog (NHGRI-EBI), full local version with alleles and coefficients.
- International genome reference panel (3 phase; reference population, $n=633$).
- Browning BL et al. Statistical genotype imputation and DR2.
- Ensembl REST and dbSNP (alleles in strand +, GRCh38).
- Frosst P et al. Nat Genet 1995; van der Put NM et al. Am J Hum Genet 1998 (MTHFR C677T and A1298C).
- McNulty H et al. Circulation 2006 (riboflavin and MTHFR 677TT).
- Grant SF et al. Nat Genet 2006 (TCF7L2).
- Feder JN et al. Nat Genet 1996 (HFE C282Y/H63D).
- Monsuur AJ et al. PLoS One 2008 (tags HLA-DQ2.5/DQ8).
- Enattah NS et al. Nat Genet 2002 (LCT -13910).
- Cornelis MC et al. JAMA 2006 (CYP1A2 and caffeine); Alsene K et al. Neuropsychopharmacology 2003 (ADORA2A and anxiety).
- Kim UK et al. Science 2003 (TAS2R38).
- Frailing TM et al. Science 2007 (FTO); Loos RJ et al. Nat Genet 2008 (MC4R).
- Yang N et al. Am J Hum Genet 2003 (ACTN3 R577X).
- September AV et al. Br J Sports Med 2009 (COL5A1 rs12722).
- Thompson A et al. JAMA 2008; Ference BA et al. JAMA 2017 (CETP and Mendelian randomization).
- Ference BA et al. 2017 (LDL causative in atherosclerosis).
- CPIC (Clinical Pharmacogenetics Implementation Consortium): CYP2C9, CYP2C19, CYP2D6, NAT2.

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